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| 2010-11-10 | First edition |
| 2016-11-08 | Date of change |

Volkswagen AG
D38436 Wolfsburg

# Function Catalogue - Navigation_SD
# Version 2.80
# Protocol version 3.0
# Data definition version 4.8

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# History

**Current Version:** 2.80

| Versionnumber | Date | Author | Description |
|---|---|---|---|
| 2.00 | 2011-06-22 | | Histroy clean-up for version 2.0 (P30DF40). |
| 2.00 | 2011-06-22 | | Incompatible change of 0x16. Data length of function x013, 0x14 changed |
| 2.00 | 2011-06-22 | | Parameter TotalNumListElements added to functions 0x1C, 0x1D (LastDestinations_Lis, FavoriteDestinations_Lis) |
| 2.00 | 2011-06-22 | | Fct. 0x15 (DistanceTimeToDestination) split into 2 functions (DistanceTODestination and TimeToDestination). |
| 2.00 | 2011-06-22 | | Fct. 0x19 (TMCInfo) extened. |
| 2.00 | 2011-06-22 | | Fct. 0x1F (PreferredDestinations_List) added. Order of function changed from 0x1F on. |
| 2.00 | 2011-06-22 | | Fct. 0x22 (VoiceGuidance) added. |
| 2.00 | 2011-06-22 | | Fcf. 0x23 (FunctionSynchronsation), 0x24 (InfoStates), 0x25 (ActiveRgType) added. |
| 2.00 | 2011-06-22 | | Fct. 026 (TrafficBlock_Indication) added. |
| 2.00 | 2011-06-22 | | LSG name shortened from Navigation_SecondaryDisplay to Navigation_SD. |
| 2.00 | 2011-06-22 | | Fct 0x29 ("GetNextListPos") and fct. 0x2A ("NbSpeller") added. |
| 2.01 | 2011-06-22 | | Correction of functions 0x24, 0x25. |
| 2.01 | 2011-06-22 | | removed parameter label ID in function 0x1D, 0x1E, 0x21, 0x26 |
| 2.01 | 2011-06-22 | | Parameter "Reserve" renamed to "Reserve1" for all occurences. |
| 2.01 | 2011-06-22 | | Missing NMax for array functions set. |
| 2.01 | 2011-06-22 | | Function 0x29: Parameter range for "Offset" added. |

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| | | | |
|---|---|---|---|
| 2.01a | 2011-06-22 | 2009-08-21, M. Wuschke | correction of bit14 in function list |
| 2.01b | 2011-06-22 | 2009-11-20, M. Wuschke | corrections to fct. 0x18, 0x20, 0x21. |
| 2.02 | 2011-06-22 | 2010-02-23, M: Wuschke | Fct. 0x17: Reference to "MOST API 1.11" removed |
| 2.02 | 2011-06-22 | 2010-03-16, M. Wuschke | Fct. 0x18: 0x1D, 0x1E, 0x20, 0x21, 0x2A: error code 0x45 added. |
| 2.02 | 2011-06-22 | 2010-04-07, M. Kellermeier, M. Wuschke | Fct. 0x17: ENUM for MainElement changed |
| 2.02 | 2011-06-22 | 2010-04-08, M. Wuschke | Fct. 0x21: RecordAddress 0x6 added |
| 2.02 | 2011-06-22 | 2010-04-13, M.Kellermeier | Fct. 0x17: changed maneuvers starting from 0x20 |
| 2.02 | 2011-06-22 | 2010-04-13, M.Kellermeier | Fct. 0x18 changed LaneGuidance.Lane.LaneType 0x11 |
| 2.03 | 2011-06-22 | 2010-04-30, M: Wuschke | Fct. 0x22: Result code(s) added. |
| 2.03 | 2011-06-22 | 2010-05-18, M. Wuschke | Fct. 0x1D, 0x1E, 0x20, 0x21: Bits within MODE-Nibble aligned to definitions for LSG Telephone. |
| 2.03 | 2011-06-22 | 2010-06-02, M. Wuschke, M. Kellermeier | Fct. 0x12, 0x15: New distance unit 1/4mi added |
| 2.04 | 2011-06-22 | 2010-10-05, M. Wuschke | Fct. 0x2A: Length of Processing message changed |
| | 2011-06-22 | | P30DF41 |
| 2.10 | 2011-06-22 | 2011-06-22, C. Sander | Functions 0x2B..0x2F added |
| 2.10 | 2011-06-22 | 2011-06-28, M. Wuschke | Fct. 0x16: Comment changed. |
| 2.10 | 2011-06-22 | 2011-07-21, M. Kellermeier | Fct. 0x30..0x32: Functions added |
| 2.10 | 2011-06-22 | 2011-07-07, M. Wuschke | Fct. 0x21: ParameterOtherListType extended by HomeAddress |
| 2.10 | 2011-06-22 | 2011-08-11, M. Wuschke | Fct. 0x21: ParameterOtherListType extended by completeList |
| 2.10 | 2011-06-22 | 2011-09-01, C. Sander | Fct. 0x24: Parameter extended |
| 2.10 | 2011-06-22 | 2011-09-20, C. Sander | Fct. 0x22: Parameter ControlType extended |

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| | | | |
|---|---|---|---|
| 2.10 | 2011-06-22 | 2011-09-20, C. Sander | Fct. 0x11: Parameter extended |
| 2.10 | 2011-06-22 | 2011-09-23, M. Wuschke | Fct. 0x16: Bugfix in ValidityInformation |
| 2.10 | 2011-06-22 | 2011-10-11, C. Sander | Fct. 0x27: Parameter RGType extended |
| 2.11 | 2011-06-22 | 2012-01-12, M. Wuschke | Fct. 0x22: Result value extended |
| 2.11 | 2011-06-22 | 2012-01-18, M. Wuschke | Fct. 0x21: Bugfix of RA1 definition |
| 2.11 | 2011-06-22 | 2012-03-07, M. Wuschke | Fct. 0x2F: Bugfix in parameter set of SetGet |
| 2.11 | 2011-06-22 | 2012-03-22, C. Sander | Fct. 0x26: Parameter States extended |
| 2.12 | 2011-06-22 | 2012-05-24, C. Sander | Fct. 0x33: Function added |
| 2.12 | 2011-06-22 | 2012-06-14, M. Wuschke | Fct. 0x16: Sub-Parameter TimeInfoType extended |
| 2.12 | 2011-06-22 | 2012-07-25, M. Kellermeier | Fct. 0x34 Function added |
| 2.12 | 2011-06-22 | 2012-07-26, M. Kellermeier | Fct. 0x22: Parameter CI_Type extended |
| 2.12 | 2011-06-22 | 2012-07-26, M. Kellermeier | Fct. 0x35: MapSizeSetup added |
| 2.12 | 2011-06-22 | 2012-08-26, M. Wuschke | Fct. 0x2F: Parameter AutoZoom added |
| 2.12 | 2011-06-22 | 2012-09-13, A. Grenzendorf | Fct. 0x35, 0x36, 0x37 added |
| 2.12 | 2011-06-22 | 2012-09-14, M. Kellermeier | Fct. 0x2F; MapScale, Autozoom: on for intersection added |
| 2.12 | 2011-06-22 | 2012-09-14, M. Kellermeier | Fct. 0x37; Bitfield for autozoom added |
| 2.12 | 2011-06-22 | 2012-09-14, M. Kellermeier | Fct. 0x2D; MapView; extended parameter for MapVisibility and supplementary map view |
| 2.12 | 2011-06-22 | 2012-09-14, M. Kellermeier | Fct. 0x2C; MapType; extended setting for primary and secondary map |
| 2.12 | 2011-06-22 | 2012-09-17, M Kellermeier | Fct. 0x36; Function "Map_Presentation" added |
| 2.12 | 2011-06-22 | 2012-09-20, A. Grenzendorf | Fct. 0x2B, 0x2C combined, 0x2D, 0x2E combined, remaining function IDs adapted |
| 2.12 | 2011-06-22 | 2012-09-20, A. Grenzendorf | Fct. 0x33, 0x34, 0x35 completly reworked |
| 2.12 | 2011- | 2012-09-20, A. | Fct. 0x1D, 0x1E, 0x21: Parameter POI_Type |

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| | | | |
|---|---|---|---|
| | 06-22 | Grenzendorf | extended |
| 2.12 | 2011-06-22 | 2012-09-20, A. Grenzendorf | Fct. 0x29: Paramter LisTyoe extended |
| 2.12 | 2011-06-22 | 2012-10-12, A. Grenzendorf | Fct. 0x33: POI_Type unused values set to reserved |
| 2.12 | 2011-06-22 | 2012-10-15, M. Wuschke | Fct. 0x28: Paramter TMC_Symbol extended |
| 2.12 | 2011-06-22 | 2012-10-17, M. Wuschke | Fct. 0x13: OpType HB-Status added |
| 2.13 | 2011-06-22 | 2012-10-26, A. Grenzendorf | Fct. 0x32: Parameter NewTimeToDestination added to OpTypes parameter list |
| 2.13 | 2011-06-22 | 2012-10-31, A. Grenzendorf | Fct. 0x2B Parameter SupportedMapTypes, 0x33 Parameter SearchType spelling error corrected |
| 2.13 | 2011-06-22 | 2012-10-31, A. Grenzendorf | Fct. 0x2B Value 0x04 added to parameter ActiveMapType |
| 2.13 | 2011-06-22 | 2012-10-31, A. Grenzendorf | Fct. 0x2B ModificationReason and ModifcationState combined to the parameter Modification, value added for Google map |
| 2.14 | 2011-06-22 | 2013-01-29, M. Wuschke | Fct. 0x34: Comments changed. Bugfix for maxElements. |
| 2.15 | 2011-06-22 | 2013-04-26, A. Grenzendorf | Fct. 0x30: Parameter State extended by value 0x03 |
| 2.20 | 2011-06-22 | | P30DF42 |
| 2.20 | 2011-06-22 | 2013-09-25, S. Stefanowitsch | Fct. 0x26 extend: Initializing MOST Map |
| 2.20 | 2011-06-22 | 2013-09-25, S. Stefanowitsch | Fct. 0x15 changed ValidityInformation splitted into two nibbles |
| 2.20 | 2011-06-22 | 2013-09-27, S. Stefanowitsch | Fct. 0x18 ASG_ID extended 0x3 |
| 2.20 | 2011-06-22 | 2013-09-27, S. Stefanowitsch | Fct. 0x1D, 0x1E, 0x20, 0x21, 0x34 ASG_ID extended 0x3, 0xB |
| | 2011-06-22 | | P30FD43 |
| 2.30 | 2011-06-22 | 2014-03-15, G. Roll | Fct. 0x37 added |
| 2.30 | 2011-06-22 | 2014-04-01, M. Prasser | Fct. 0x27: RGType extended by LVDS - KDK |
| 2.30 | 2011-06-22 | 2014-04-04, M. Prasser | Fct. 0x2D: Comment changed in parameter Scale |
| 2.30 | 2011-06-22 | 2014-05-04, C. Franz | Fct. 0x26 extend: Navigation in mobile device active |

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| | | | |
|---|---|---|---|
| 2.30 | 2011-06-22 | 2014-05-07, M. Prasser | Fct. 0x35: Description in Bitfield VoiceGuidance changed for combatibility to code generator |
| 2.31 | 2011-06-22 | 2014-08-29, M. Prasser | Fct. 0x12, 0x16, 0x17, 0x2E, 0x32: ParamPos corrected |
| 2.40 | 2011-06-22 | - | P30DF44 |
| 2.40 | 2011-06-22 | 2014-08-29, M. Prasser | Fct. 0x18: Parameter LaneType extended |
| 2.40 | 2011-06-22 | 2014-08-29, M. Prasser | Fct. 0x38 added |
| 2.40 | 2011-06-22 | 2014-09-26, C. Franz | Fct. 0x35: Parameter FunctionSupport added |
| 2.40 | 2011-06-22 | 2014-11-05, C. Franz | Fct. 0x18: ASG_ID extended 0x9, 0xA, 0xB |
| | 2015-09-30 | | P30DF45 |
| 2.50 | 2015-09-30 | S. Stefanowitsch | Fct. 0x22: Parameter ControlType, CI_Type and RG_ActDeact_Result extended |
| 2.50 | 2015-09-30 | S. Stefanowitsch | Fct. 0x35: Parameter FunctionSupport extended |
| 2.60 | 2016-01-13 | | P30DF46 |
| 2.60 | 2016-01-13 | S. Stefanowitsch | Fct. 0x39 added |
| 2.60 | 2016-01-13 | S. Stefanowitsch | Fct. 0x1D, 0x1E, 0x2E, 0x34: POI_Type extended |
| 2.60 | 2016-01-13 | S. Stefanowitsch | Secondary Updates in Fct. 0x03, 0x25 for the other changes |
| 2.60 | 2016-02-15 | C. Franz | Fct 0x26: Parameter states extended |
| 2.61 | 2016-03-03 | S. Stefanowitsch | Fct. 0x21 OtherListType extended and Desription of Strings changed |
| 2.70 | 2016-04-07 | | P30DF47 |
| 2.70 | 2016-04-08 | C. Franz | Fct. 0x17: Parameter MainElement extended |
| 2.70 | 2016-04-12 | C. Franz | Fct. 0x3A, 0x3B, 0x3C, 0x3D, 3E added |
| 2.70 | 2016-04-13 | C. Franz | Fct. 0x35: Parameter FunctionSupport extended |
| 2.70 | 2016-05-04 | C. Franz | Fct. 0x3D: Parameter FctList_3 extended |

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| | | | |
|---|---|---|---|
| 2.70 | 2016-05-11 | C. Franz | Fct. 0x17: Parameter "Z-LevelGuidance" renamed in "Z_LevelGuidance" |
| 2.70 | 2016-05-12 | C. Franz | Fct. 0x19, 0x1A, 0x1B, 0x22, 0x24, 0x27, 0x29, 0x2B, 0x2C, 0x2D, 0x33, 0x39: Errorcode 0x46 - 'AppErr_InvalidArgument' added ' |
| 2.71 | 2016-06-15 | C. Franz | Fct. 0x3E: correction in parameter TotalNumListElements (base datatype word -> byte) |
| 2.72 | 2016-06-24 | C. Franz | Fct. 0x3E: correction in parameter RecordAddress |
| 2.73 | 2016-09-05 | C. Franz | Fct. 0x3A, 0x3E: FcID description extended accoding to arbitrary id definitions |
| 2.8 | 2016-10-06 | | P30DF48 |
| 2.8 | 2016-10-06 | S. Stefanowitsch | Fct. 0x3C: Parameter "E-RangeWarning" added |
| 2.8 | 2016-10-06 | S. Stefanowitsch | Fct. 0x0F Parameter "Op_State" extended |

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# References

| Ref.No | Document | Version |
|---|---|---|
| [1] | BAP-LH_Protokollspezifikation | 1.90 |
| [2] | BAP-LH_Datentypen | 1.70 |
| [3] | BAP-LH_Grundverhalten_der_Applikation | 1.06 |
| [4] | BAP-MSCs_NAV-SD_P30DF47 | 2.70 |
| [4] | MSCs_GeneralArrayHandling_MQB | 1.00 |

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# Purpose of this Document

This document is a description of data format respectively of the BAP functions for a specific generation of control units (Function Catalog).

The BAP protocol is specified in [1].

For the order of parameters in function-calls the order described in the header of each function shall be used.

Example:

| LSG | Function | OPType | Parameter |
|---|---|---|---|
| LSG-Name (LSG-ID) | Functionname (Fct-ID) | Get | - |
| | | Status | Param1, Param2, Param3 |
| | | HeartbeatStatus | Param1, Param2, Param3 |
| | | Error | ErrorCode |

For Function/OpType combinations with simple type parameter (uint8, unit16, uint32) the respective typed function (e.g. BAP_RequestInt16, BAP_IndicationInt32) shall be used.

For Function/OpType combinations without parameter the void function call (e.g. BAP_RequestVoid, BAP_IndicationVoid) shall be used.

For Function/OpType combinations with more than one parameter or structured parameter the byteSequence function call with a corresponding required datalength (e.g. BAP_RequestByteSequence, BAP_IndicationByteSequence) shall be used.

Application specific dataypes and their evaluation are described in [2]

All parameters transmitted via BAP shall be stored in non-volatile memory on the FSG to keep information during loss of power.

Function-IDs 1 to 4 shall not be evaluated by application if not otherwise specified

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# LCU Properties Navigation_SD ( 50, 0x32 )

| Attribute | Value |
|---|---|
| ID | 50 |
| Name | Navigation_SD |
| Class | 50 |
| SubClass | 0 |

**Description:** The LSG "Navigation - Secondary Display" is used for navigation systems as well as for standalone compass modules.
This LSG is used for transfer of navigation information (RGI, compass, TMC, destination lists) and navigation setup/control (e.g. activation/deactivation of a route guidance, control of voice guidance/navigation announcements) etc.

Technical contact person VW: Franz, Christian (EECF/1)
Technical contact person Audi: Prasser, Moritz (I/EE-52)
Technical contact person Porsche: Rheindt, Ralf, (EEI1)
Technical contact person Skoda: Kemr, Jakub (TMI/6)
Technical contact person Seat: Rovira, Jaume (EE-52)

Exception:
In case a car configuration requires an installation of a compass module in parallel to a navigation system, the compass module shall use the LSG-ID 0x15 (LSG-Class:0x32).

Navigation_SD:.. Navigation - Secondary Display.

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# Function List

| ID (dez) | ID (hex) | Name | Type |
|---|---|---|---|
| 1 | 0x1 | GetAll | Cache |
| 2 | 0x2 | BAP_Config | Property |
| 3 | 0x3 | FunctionList | Property |
| 4 | 0x4 | HeartBeat | Property |
| 15 | 0xF | FSG_OperationState | Property |
| 16 | 0x10 | CompassInfo | Property |
| 17 | 0x11 | RG_Status | Property |
| 18 | 0x12 | DistanceToNextManeuver | Property |
| 19 | 0x13 | CurrentPositionInfo | Property |
| 20 | 0x14 | TurnToInfo | Property |
| 21 | 0x15 | DistanceToDestination | Property |
| 22 | 0x16 | TimeToDestination | Property |
| 23 | 0x17 | ManeuverDescriptor | Property |
| 24 | 0x18 | LaneGuidance | Array |
| 25 | 0x19 | TMCinfo | Property |
| 26 | 0x1A | MagnetFieldZone | Property |
| 27 | 0x1B | Calibration | Property |
| 28 | 0x1C | ASG_Capabilities | Property |
| 29 | 0x1D | LastDest_List | Array |
| 30 | 0x1E | FavoriteDest_List | Array |
| 31 | 0x1F | PreferredDest_List | Property |
| 32 | 0x20 | NavBook | Array |
| 33 | 0x21 | Address_List | Array |
| 34 | 0x22 | RG_ActDeact | Method |
| 35 | 0x23 | RepeatLastNavAnnouncement | Method |
| 36 | 0x24 | VoiceGuidance | Property |
| 37 | 0x25 | FunctionSynchronisation | Property |
| 38 | 0x26 | InfoStates | Property |
| 39 | 0x27 | ActiveRgType | Property |
| 40 | 0x28 | TrafficBlock_Indication | Property |
| 41 | 0x29 | GetNextListPos | Method |
| 42 | 0x2A | NbSpeller | Method |
| 43 | 0x2B | MapColorAndType | Property |
| 44 | 0x2C | MapViewAndOrientation | Property |
| 45 | 0x2D | MapScale | Property |
| 46 | 0x2E | DestinationInfo | Property |

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| | | | |
|---|---|---|---|
| 47 | 0x2F | Altitude | Property |
| 48 | 0x30 | OnlineNavigationState | Property |
| 49 | 0x31 | Exitview | Property |
| 50 | 0x32 | SemidynamicRouteGuidance | Property |
| 51 | 0x33 | POI_Search | Method |
| 52 | 0x34 | POI_List | Array |
| 53 | 0x35 | FSG_Setup | Property |
| 54 | 0x36 | Map_Presentation | Property |
| 55 | 0x37 | ManeuverState | Property |
| 56 | 0x38 | ETC_Status | Property |
| 57 | 0x39 | MapContentSettings | Property |
| 58 | 0x3A | VideoStreams | Array |
| 59 | 0x3B | RequestSync_Video | Method |
| 60 | 0x3C | DistanceToDestinationExtended | Property |
| 61 | 0x3D | LaneGuidance2 | Array |
| 62 | 0x3E | Picture | Array |

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# 1.) GetAll ( 1, 0x1 )

**Type:** Cache

**Description:** Get all properties.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Data | Void | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** GetAll (1, 0x1 )

| OpCode | Associated Parameters |
|---|---|
| GetAll | NoParam, |
| StatusAll | Data, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 1.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** GetAll,

**Base Datatype:** Empty

## 1.2.) Data [Void]

**Type:** Void

**Description:** The parameter "Data" contains all property function values successively in a segmented message.
See BAP specification [1] for "GetAll" functionality.

**OpCodes:** StatusAll,

**Base Datatype:** Empty

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## 1.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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## 2.) BAP_Config ( 2, 0x2 )

**Type:** Property

**Description:** Version information

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| BAP_Version_major | Number | Byte |
| BAP_Version_minor | Number | Byte |
| LSG_Class | Number | Byte |
| LSG_Sub_Class | Number | Byte |
| LSG_Version_major | Number | Byte |
| LSG_Version_minor | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** BAP_Config (2, 0x2 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | BAP_Version_major, BAP_Version_minor, LSG_Class, LSG_Sub_Class, LSG_Version_major, LSG_Version_minor, |
| Reset | BAP_Version_major, BAP_Version_minor, LSG_Class, LSG_Sub_Class, LSG_Version_major, LSG_Version_minor, |
| HeartbeatStatus | BAP_Version_major, BAP_Version_minor, LSG_Class, LSG_Sub_Class, LSG_Version_major, LSG_Version_minor, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

#### 2.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 2.2.) BAP_Version_major [Number]

**Type:** Number

**Description:** Major Protocol Version = 3.
BAP-Version is 3.0

**OpCodes:** Status, Reset, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 3

**Range Max:** 3

**Step:** 1

**Unit:** unit_const

## 2.3.) BAP_Version_minor [Number]

**Type:** Number

**Description:** Minor Protocol Version = 0.
BAP-Version is 3.0

**OpCodes:** Status, Reset, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 2.4.) LSG_Class [Number]

**Type:** Number

**Description:** LSG_Class_ID
The LSG_Class_ID classifies the function catalogue uniquely because a function catalogue can occur several times with different LSG-IDs (e.g. 0x15, 0x32).

**OpCodes:** Status, Reset, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 50

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**Range Max:** 50

**Step:** 1

**Unit:** unit_const

## 2.5.) LSG_Sub_Class [Number]

**Type:** Number

**Description:** LSG_Sub_Class_ID.
Not used; for this ECU (LSG) is the parameter set to 0x00.

**OpCodes:** Status, Reset, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 2.6.) LSG_Version_major [Number]

**Type:** Number

**Description:** Major Version of LSG-Definition = 4.
LSG-Version is 4.8.

**OpCodes:** Status, Reset, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 4

**Range Max:** 4

**Step:** 1

**Unit:** unit_const

## 2.7.) LSG_Version_minor [Number]

**Type:** Number

**Description:** Minor Version of LSG-Definition = 8.
LSG-Version is 4.8.

**OpCodes:** Status, Reset, HeartbeatStatus,

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**Base Datatype:** Byte

**Range Min:** 8

**Range Max:** 8

**Step:** 1

**Unit:** unit_const

## 2.8.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 3.) FunctionList ( 3, 0x3 )

**Type:** Property

**Description:** Bitfield of available functions.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| FctList | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** FunctionList (3, 0x3 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | FctList, |
| HeartbeatStatus | FctList, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 3.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 3.2.) FctList [Bitfield]

**Type:** Bitfield

**Description:** This bitfield indicates the support of a certain function during runtime.

For the encoding of this function the reader is referred to BAP_Datentypen [2].

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** ByteStream

**Length:**8

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | reserved | reserved |
| 1 | Fct "GetAll" available | Fct "GetAll" not available |
| 2 | Fct "BAP_Config" available | Fct "BAP_Config" not available |
| 3 | Fct "FunctionList" available | Fct "FunctionList" not available |
| 4 | Fct "HeartBeat" available | Fct "HeartBeat" not available |
| 5..14 | reserved | reserved |
| 15 | Fct "FSG_OperationState" available | Fct "FSG_OperationState" not available |
| 16 | Fct "CompassInfo" available | Fct "CompassInfo" not available |
| 17 | Fct "RG_Status" available | Fct "RG_Status" not available |
| 18 | Fct "DistanceToNextManeuver" available | Fct "DistanceToNextManeuver" not available |
| 19 | Fct "CurrentPositionInfo" available | Fct "CurrentPositionInfo" not available |
| 20 | Fct "TurnToInfo" available | Fct "TurnToInfo" not available |
| 21 | Fct "DistanceToDestination" available | Fct "DistanceToDestination" not available |
| 22 | Fct "TimeToDestination" available | Fct "TimeToDestination" not available |
| 23 | Fct "ManeuverDescriptor" available | Fct "ManeuverDescriptor" not available |
| 24 | Fct "LaneGuidance" available | Fct "LaneGuidance" not available |
| 25 | Fct "TMCinfo" available | Fct "TMCinfo" not available |
| 26 | Fct "MagnetFieldZone" available | Fct "MagnetFieldZone" not available |
| 27 | Fct "Calibration" available | Fct "Calibration" not available |
| 28 | Fct "ASG_Capabilities" available | Fct "ASG_Capabilities" not available |
| 29 | Fct "LastDest_List" available | Fct "LastDest_List" not available |
| 30 | Fct "FavoriteDest_List" available | Fct "FavoriteDest_List" not available |
| 31 | Fct "PreferredDest_List" available | Fct "PreferredDest_List" not available |
| 32 | Fct "NavBook" available | Fct "NavBook" not available |
| 33 | Fct "Address_List" available | Fct "Address_List" not available |
| 34 | Fct "RG_ActDeact" available | Fct "RG_ActDeact" not available |
| 35 | Fct "RepeatLastNavAnnouncement" available | Fct "RepeatLastNavAnnouncement" not available |
| 36 | Fct "VoiceGuidance" available | Fct "VoiceGuidance" not available |
| 37 | Fct "FunctionSynchronisation" available | Fct "FunctionSynchronisation" not available |
| 38 | Fct "InfoStates" available | Fct "InfoStates" not available |

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| | | |
|---|---|---|
| 39 | Fct "ActiveRgType" available | Fct "ActiveRgType" not available |
| 40 | Fct "TrafficBlock_Indication" available | Fct "TrafficBlock_Indication" not available |
| 41 | Fct "GetNextListPos" available | Fct "GetNextListPos" not available |
| 42 | Fct "NbSpeller" available | Fct "NbSpeller" not available |
| 43 | Fct "MapColorAndType" available | Fct "MapColorAndType" not available |
| 44 | Fct "MapViewAndOrientation" available | Fct "MapViewAndOrientation" not available |
| 45 | Fct "MapScale" available | Fct "MapScale" not available |
| 46 | Fct "DestinationInfo" available | Fct "DestinationInfo" not available |
| 47 | Fct "Altitude" available | Fct "Altitude" not available |
| 48 | Fct "OnlineNavigationState" available | Fct "OnlineNavigationState" not available |
| 49 | Fct "Exitview" available | Fct "Exitview" not available |
| 50 | Fct "SemidynamicRouteGuidance" available | Fct "SemidynamicRouteGuidance" not availalbe |
| 51 | Fct "POI_Search" available | Fct "POI_Search" not available |
| 52 | Fct "POI_List" available | Fct "POI_List" not available |
| 53 | Fct "FSG_Setup" available | Fct "FSG_Setup" not available |
| 54 | Fct "Map_Presentation" available | Fct "Map_Presentation" not available |
| 55 | Fct "ManeuverState" available | Fct "ManeuverState" not available |
| 56 | Fct "ETC_Status" available | Fct "ETC_Status" not available |
| 57 | Fct "MapContentSettings" available | Fct "MapContentSettings" not available |
| 58 | Fct "VideoStreams" available | Fct "VideoStreams" not available |
| 59 | Fct "RequestSync_Video" available | Fct "RequestSync_Video" not available |
| 60 | Fct "DistanceToDestinationExtended" available | Fct "DistanceToDestinationExtended" not available |
| 61 | Fct "LaneGuidance2" available | Fct "LaneGuidance2" not available |
| 62 | Fct "Picture" available | Fct "Picture" not available |
| 63 | reserved | reserved |

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## 3.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 4.) HeartBeat ( 4, 0x4 )

**Type:** Property

**Description:** FSG sends out the time between two HeartBeat messages. If a heart beat error occurs for a BAP-connection, it is reported on function HeartBeat (0x04).

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| HeartBeatTime | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** HeartBeat (4, 0x4 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | HeartBeatTime, |
| HeartbeatStatus | HeartBeatTime, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 4.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 4.2.) HeartBeatTime [Number]

**Type:** Number

**Description:** Time between two Heartbeat-messages.
The transmitted value is a multiple of 100ms. For the HeartBeatTime = 1000ms the value 10 has to be transmitted.
Values greater than the specified maximum value are reserved.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 25500

**Step:** 100

**Unit:** unit_ms

## 4.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 5.) FSG_OperationState ( 15, 0xF )

**Type:** Property

**Description:** Operation state of the FSG.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| OP_State | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** FSG_OperationState (15, 0xF )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | OP_State, |
| HeartbeatStatus | OP_State, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 5.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 5.2.) OP_State [Enum]

**Type:** Enum

**Description:** Operation state.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | normal operation |
| 0x01 | OFF / Stand-by |
| 0x02 | reserved |
| 0x03 | initializing |
| 0x04..0x0D | reserved |
| 0x0E | function inactive (Function on Demand)(DF4.8) |
| 0x0F | defective |
| 0x10..0xFF | reserved |

## 5.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 6.) CompassInfo ( 16, 0x10 )

**Type:** Property

**Description:** This function delivers the numeric (in degrees) and the symbolic driving direction.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Direction_Symbolic | Enum | Byte |
| Direction_Angle | Number | Word |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** CompassInfo (16, 0x10 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | Direction_Symbolic, Direction_Angle, |
| HeartbeatStatus | Direction_Symbolic, Direction_Angle, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 6.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 6.2.) Direction_Symbolic [Enum]

**Type:** Enum

**Description:** The Direction_Symbolic shall be the same as shown on the head unit HMI (navigation device primary display).
If the symbolic direction is not shown on the primary display at all, the FSG shall set Direction_Symbolic to 0xFF (not supported). In this case, the FSG has to supply the parameter Direction_Angle with valid values.

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The FSG is requested to support at least one of both parameters Direction_Symbolic or Direction_Angle (in the sense of a minimum requirement).

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | N |
| 0x01 | NNO |
| 0x02 | NO |
| 0x03 | ONO |
| 0x04 | O |
| 0x05 | OSO |
| 0x06 | SO |
| 0x07 | ssO |
| 0x08 | S |
| 0x09 | ssW |
| 0x0A | SW |
| 0x0B | WSW |
| 0x0C | W |
| 0x0D | WNW |
| 0x0E | NW |
| 0x0F | NNW |
| 0x10..0xFE | reserved |
| 0xFF | not supported |

## 6.3.) Direction_Angle [Number]

**Type:** Number

**Description:** If the FSG does not support the parameter Direction_Angle, it shall send the value 0xFFFF in this parameter.
The angle in degrees shall be given in mathematic positive orientation in order to have the same meaning for Direction_Angle and ManeuverDescriptor.Maneuver_(n).Direction.

0Â° North,
90Â° West,
180Â° South,
270Â° East.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Word

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**Range Min:** 0

**Range Max:** 360

**Step:** 1

**Unit:** unit_none

**EnumValues:**

| Code | Description |
|---|---|
| 0xFFFF | not supported |

## 6.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 7.) RG_Status ( 17, 0x11 )

**Type:** Property

**Description:** This function indicates if the route guidance is currently active or inactive.

RG_Status:.. RouteGuidance_Status

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| RG_Status | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** RG_Status (17, 0x11 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | RG_Status, |
| HeartbeatStatus | RG_Status, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 7.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 7.2.) RG_Status [Enum]

**Type:** Enum

**Description:** Contains RG_Status

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | RG not active |
| 0x01 | RG active |
| 0x02 | RG suspended (DF4.1) |
| 0x03..0xFE | reserved |
| 0xFF | not supported |

## 7.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 8.) DistanceToNextManeuver ( 18, 0x12 )

**Type:** Property

**Description:** This function returns absolute and relative distances to the next maneuver.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| DistanceToNextManeuver | Record | Empty |
| BargraphInfo | Record | Empty |
| ValidityInformation | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** DistanceToNextManeuver (18, 0x12 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | DistanceToNextManeuver, BargraphInfo, ValidityInformation, |
| HeartbeatStatus | DistanceToNextManeuver, BargraphInfo, ValidityInformation, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 8.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 8.2.) DistanceToNextManeuver [Record]

**Type:** Record

**Description:** Absolute distance to the next maneuver.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_8_2

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Distance | Number | Long |
| Unit | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 8.2.1.) Distance [Number]

**Type:** Number

**Description:** [transmitted distance value]=[displayed distance value]*10 (e.g. [displayed distance value]=0.9 -> [transmitted distance value]=9).
In some situations there may be no distance available. In this case, the FSG shall set the ValidityInformation.Bit0 := FALSE.

If the Unit == 0x05 (quarter mile), then Distance shall be always an integer, e.g. 30*0.1=3. In that case, the ASG may display 3/4mile (displayed value := Distance * 1/4mi, e.g. 3.0*1/4 = 3/4mi.)

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** 0

**Range Max:** 429496729

**Step:** 0.1

**Unit:** unit_none

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## 8.2.2.) Unit [Enum]

**Type:** Enum

**Description:** Unit belonging to the Distance value.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (UK and US (statute mile)) |
| 0x05 | quarter mile(s) (n* 1/4mile) |
| 0x06..0xFE | reserved |
| 0xFF | not supported |

## 8.3.) BargraphInfo [Record]

**Type:** Record

**Description:** Relative distance information to the next maneuver, which can be used to display a bargraph, or similar.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_8_3

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| BargraphOnOff | Enum | Byte |
| Bargraph | Number | Byte |

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## *Detailed Sub Parameter Overview*

### 8.3.1.) BargraphOnOff [Enum]

**Type:** Enum

**Description:** The route guidance application within the navigation unit decides, when counting the relative distance (bargraph) starts.
If not specified in another requirement document, the ASG shall handle received reserved values as 0x00 ("bargraph shall not be displayed").

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | bargraph shall not be displayed |
| 0x01 | display bargraph |
| 0x02..0xFE | reserved |
| 0xFF | not supported |

### 8.3.2.) Bargraph [Number]

**Type:** Number

**Description:** Relative fill of Bargraph: 0%..100%.
The ASG shall map values > 100% to 100%.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 100

**Step:** 1

**Unit:** unit_percent

### 8.4.) ValidityInformation [Bitfield]

**Type:** Bitfield

**Description:** This paramters contains information about the validity of parameter(s). The ASG shall ignore reserved values. The FSG shall set reserved bits of the bitfield to FALSE.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | DistanceToNextManeuver valid | DistanceToNextManeuver invalid |
| 1..7 | reserved | reserved |

## 8.5.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 9.) CurrentPositionInfo ( 19, 0x13 )

**Type:** Property

**Description:** This function returns the textual description of the current vehicle position (e.g. street name if this is available in the navigation device).

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| PositionInfo | String | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** CurrentPositionInfo (19, 0x13 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | PositionInfo, |
| HeartbeatStatus | PositionInfo, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 9.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 9.2.) PositionInfo [String]

**Type:** String

**Description:** String describing the current position. The string contains UTF8 encoded unicode3.0 characters. The string may contain more than one element, separated by the Unicode control code "LineFeed" (0x000A). It may contain street name, street-number and additional description.

If the PositionInfo is not known to a specific moment, empty string shall be transmitted.

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The maximum number of characters in this string is limited by the string length only. The encoded length must not exceed 96 bytes plus an additional byte for the string length information.

For the definition of "empty string" the reader is referred to [2].

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:** 97

## 9.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 10.) TurnToInfo ( 20, 0x14 )

**Type:** Property

**Description:** This function returns the textual description of the street to turn to in the next maneuver (e.g. street-name if this is available in the navigation device).

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| TurnToInfo | String | ByteStream |
| SignPost | String | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** TurnToInfo (20, 0x14 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | TurnToInfo, SignPost, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 10.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 10.2.) TurnToInfo [String]

**Type:** String

**Description:** String describing the next street to turn to. The string contains UTF8 encoded unicode3.0 characters. The string may contain more than one element, separated by the Unicode control code "LineFeed" (0x000A). It may contain streetname, street-number and additional description.

If the TurnToInfo is not known at a specific moment, empty string shall be transmitted.

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The maximum number of characters in this string is limited by the string length only. The encoded length must not exceed 75 bytes plus an additional byte for the string length information.

For the defnition of "empty string" the reader is referred to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 76

## 10.3.) SignPost [String]

**Type:** String

**Description:** String containing the sign post information. The string contains UTF8 encoded unicode3.0 characters. If there is no sign post available, the FSG shall set this parameter to empty string.
The meaning is "in the direction to < SignPost >", e.g. TurnToInfo = "A10", SignPost = "Hamburg" means: "turn to A10, direction to Hamburg".

The maximum number of characters in this string is limited by the string length only. The encoded length must not exceed 30 bytes plus an additional byte for the string length information.

For the defnition of "empty string" the reader is referred to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 31

## 10.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 11.) DistanceToDestination ( 21, 0x15 )

**Type:** Property

**Description:** This function provides information regarding remaining travel distance.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| DistanceToDestination | Record | Empty |
| DistanceToDestinationType | Bitfield | Nibble |
| ValidityInformation | Bitfield | Nibble |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** DistanceToDestination (21, 0x15 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | DistanceToDestination, DistanceToDestinationType, ValidityInformation, |
| HeartbeatStatus | DistanceToDestination, DistanceToDestinationType, ValidityInformation, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 11.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 11.2.) DistanceToDestination [Record]

**Type:** Record

**Description:** This parameter contains the absolute distance to the next or to the final destination along the planned route, depending on the settings in the navigation unit.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_11_2

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Distance | Number | Long |
| Unit | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 11.2.1.) Distance [Number]

**Type:** Number

**Description:** [transmitted distance value]=[displayed distance value]*10 (e.g. [displayed distance value]=0.9 -> [transmitted distance value]=9).

In some situations there may be no distance available. Therefore the following coding is defined. If the distance is not known to a specific moment, the bit0 in parameter ValidityInformation is set to 0.

If the Unit == 0x05 (quarter mile), then Distance shall be always an integer, e.g. 30*0.1=3. In that case, the ASG may display 3/4mile (displayed value := Distance * 1/4mi, e.g. 3.0*1/4 = 3/4mi.)

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** 0

**Range Max:** 429496729

**Step:** 0.1

**Unit:** unit_none

## 11.2.2.) Unit [Enum]

**Type:** Enum

**Description:** Unit belonging to the distance value.

**OpCodes:** Empty,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (UK and US (statute mile)) |
| 0x05 | quarter mile(s) (n* 1/4mile) |
| 0x06..0xFE | reserved |
| 0xFF | not supported / no information about unit available |

## 11.3.) DistanceToDestinationType [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble (DF4.2))

This parameter gives the information what type of distance to destination is transmitted in parameter â€• DistanceToDestinationâ€• along the planned route.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Distance to Stopover | Distance to Destination |
| 1..3 | reserved | reserved |

## 11.4.) ValidityInformation [Bitfield]

**Type:** Bitfield

**Description:** (lower nibble (DF4.2))

This parameter contains information about the validity of parameter(s).

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Nibble

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | DistanceToDestination valid | DistanceToDestination invalid |
| 1..3 | reserved | reserved |

## 11.5.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 12.) TimeToDestination ( 22, 0x16 )

**Type:** Property

**Description:** This function provides information regarding travel time or arrival time at destination.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| TimeInfo | Record | Empty |
| ValidityInformation | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** TimeToDestination (22, 0x16 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | TimeInfo, ValidityInformation, |
| HeartbeatStatus | TimeInfo, ValidityInformation, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 12.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 12.2.) TimeInfo [Record]

**Type:** Record

**Description:** Provides the estimated driving or arrival time.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

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**Count:** 1

**Subparameter von:** 1_12_2

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| TimeInfoType | Enum | Nibble |
| NavigationTimeFormat | Enum | Nibble |
| Minute | Number | Byte |
| Hour | Number | Byte |
| Day | Number | Byte |
| Month | Number | Byte |
| Year | Number | Byte |

### *Detailed Sub Parameter Overview*

## 12.2.1.) TimeInfoType [Enum]

**Type:** Enum

**Description:** (upper nibble)

Type of information is given by the following parameters. The user setting within the navigation unit decides which type is used.

TimeToDestination: Estimated driving time to the next or to the final destination.
ArrivalTime: Estimated arrival time for the next or the final destination.

If not specified in another requirement document, the ASG shall handle a reserved value as value 0xF.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | TimeToDestination |
| 0x1 | ArrivalTime |
| 0x2 | ArrivalTime is local time in another time zone (DF4.1) |
| 0x3..0xE | reserved |
| 0xF | not supported (= any time info type) |

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## 12.2.2.) NavigationTimeFormat [Enum]

**Type:** Enum

**Description:** (lower nibble)

Information about the used format for the arrival time in the navigation unit. The Secondary Display may or may not use the same time format, depending on the HMI definitions.
The format of the transmitted time value (Hour, Minute) is always the 24h mode, independent from the parameter NavigationTimeFormat.
If the ASG receives a reserved value, it shall apply the ASG-internal default value for the NavigationTimeFormat.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | 24 hour format |
| 0x1 | 12 hour format |
| 0x2..0xF | reserved |

## 12.2.3.) Minute [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit1 in parameter ValidityInformation is set to 0.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 60

**Step:** 1

**Unit:** unit_min

## 12.2.4.) Hour [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit2 in parameter ValidityInformation is set to 0.

**OpCodes:** Empty,

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**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_h

## 12.2.5.) Day [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit3 in parameter ValidityInformation is set to FALSE (0).
This parameter is only be used if TimeInfoType=0x1 "ArrivalTime" or 0x2 "ArrivalTime in local time another time zone".

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 31

**Step:** 1

**Unit:** unit_d

## 12.2.6.) Month [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit4 in parameter ValidityInformation is set to FALSE (0).
This parameter is only be used if TimeInfoType=0x1 "ArrivalTime" or 0x2 "ArrivalTime in local time another time zone".

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 12

**Step:** 1

**Unit:** unit_mon

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## 12.2.7.) Year [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit5 in parameter ValidityInformation is set to FALSE (0).
Parameter "Year" is only be used if TimeInfoType=0x1 "ArrivalTime" or 0x2 "ArrivalTime in local time another time zone".
Offset is 2000. I.e. for Year=2014 the value 14 is to be sent.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 2000

**Range Max:** 2099

**Step:** 1

**Unit:** unit_a

## 12.3.) ValidityInformation [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains information about the validity of parameters and sub-parameters.

The ASG shall ignore reserved bits of the bit field. The FSG shall set them to FALSE.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | reserved | reserved |
| 1 | TimeInfo.Minute is valid | TimeInfo.Minute is invalid |
| 2 | TimeInfo.Hour is valid | TimeInfo.Hour is invalid |
| 3 | TimeInfo.Day is available / to be displayed | TimeInfo.Day is not available / not to be displayed |
| 4 | TimeInfo.Month is available / to be displayed | TimeInfo.Month is not available / not to be displayed |
| 5 | TimeInfo.Year is available / to be displayed | TimeInfo.Year is not available / not to be displayed |
| 6..7 | reserved | reserved |

## 12.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 13.) ManeuverDescriptor ( 23, 0x17 )

**Type:** Property

**Description:** This function returns the symbolic description of the next maneuver relative to the current vehicle position.
This function may contain up to 3 maneuvers.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Maneuver_1 | Record | Empty |
| Maneuver_2 | Record | Empty |
| Maneuver_3 | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** ManeuverDescriptor (23, 0x17 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | Maneuver_1, Maneuver_2, Maneuver_3, |
| HeartbeatStatus | Maneuver_1, Maneuver_2, Maneuver_3, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 13.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 13.2.) Maneuver_1 [Record]

**Type:** Record

**Description:** Contains Maneuver_1.

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**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_13_2

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| MainElement | Enum | Byte |
| Direction | Number | Byte |
| Z_LevelGuidance | Enum | Byte |
| Sidestreets | String | ByteStream |

### *Detailed Sub Parameter Overview*

## 13.2.1.) MainElement [Enum]

**Type:** Enum

**Description:** Enumeration type to describe the type of the main element within a maneuver description.
If not denoted otherwise, each main element type can be combined with sidestreets within one maneuver description (based on "BAP-Navigationsdarstellung").
For exit maneuvers with ramp/Z-level, the values 0x0F and 0x10 shall be used and combined with z-Level.
The values 0x20, 0x21, 0x24 and 0x25 are reserved for projects not supporting Z-Level as a separate parameter and shall not be used on this interface.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | NoSymbol |
| 0x01 | NoInfo |
| 0x02 | DirectionToDestination |
| 0x03 | Arrived |
| 0x04 | NearDestination |
| 0x05 | ArrivedDestinationOffmap |
| 0x06 | OffRoad |
| 0x07 | OffMap |

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| | |
|---|---|
| 0x08 | NoRoute |
| 0x09 | CalcRoute |
| 0x0A | RecalcRoute |
| 0x0B | FollowStreet |
| 0x0C | ChangeLane |
| 0x0D | Turn |
| 0x0E | TurnOnMainroad |
| 0x0F | ExitRight |
| 0x10 | ExitLeft |
| 0x11 | ServiceRoadRight |
| 0x12 | ServiceRoadLeft |
| 0x13 | Fork-2 |
| 0x14 | Fork-3 |
| 0x15 | RoundaboutTrsRight |
| 0x16 | RoundaboutTrsLeft |
| 0x17 | SquareTrsRight |
| 0x18 | SquareTrsLeft |
| 0x19 | Uturn |
| 0x1A | ExitRoundaboutTrsRight |
| 0x1B | ExitRoundaboutTrsLeft |
| 0x1C | PrepareTurn |
| 0x1D | PrepareRoundabout |
| 0x1E | PrepareSquare |
| 0x1F | PrepareUTurn |
| 0x20 | reserved - ExitRight-RampUp |
| 0x21 | reserved - ExitRight-RampDown |
| 0x22 | MichigangTurn |
| 0x23 | DoubleTurn |
| 0x24 | reserved - ExitLeft-RampUp |
| 0x25 | reserved - ExitLeft-RampDown |
| 0x26 | DirectionToWaypoint |
| 0x27 | ReducedGuidance (DF4.7) |
| 0x28 | LeaveParking (DF4.7) |
| 0x29 | MainRoadRight (DF4.7) |
| 0x2A | MainRoadLeft (DF4.7) |
| 0x2B | DirectionToEntrypoint (DF4.7) |
| 0x2C | Take Ferry (DF4.7) |
| 0x2D | Take CarTrain (DF4.7) |
| 0x2E | UnderpassRight (DF4.7) |

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| | |
|---|---|
| 0x2F | UnderpassLeft (DF4.7) |
| 0x30 | OverpassRight (DF4.7) |
| 0x31 | OverpassLeft (DF4.7) |
| 0x32 | Toll Gate (DF4.7) |
| 0x33 | Toll Gate - ETC (DF4.7) |
| 0x34..0xFF | reserved |

## 13.2.2.) Direction [Number]

**Type:** Number

**Description:** Direction of the street to turn into relative to the current driving direction. Resolution: 360Â°/256 (approx. 1.4Â°). The exact meaning of the direction depends on the MainElement.

0x00 -> straight (0Â°)
0x40 -> left (90Â°)
0x80 -> back (180Â°)
0xC0 -> right (270Â°)

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 13.2.3.) Z_LevelGuidance [Enum]

**Type:** Enum

**Description:** Parameter provides Z-LevelGuidance information associated with the Direction.

If not specified in another requirement document, the ASG shall handle a reserved value as if 0x00 ("no Z-LevelGuidance") is received.

**OpCodes:** Empty,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no Z-LevelGuidance |
| 0x01 | up |
| 0x02 | down |
| 0x03..0xFE | reserved |
| 0xFF | not supported |

## 13.2.4.) Sidestreets [String]

**Type:** String

**Description:** The string contains the complete set of available sidestreets as data stream (string). The data stream doesn't contain any UTF8 encoded unicode3.0 characters. String contains information about the (existing) sidestreets, which shall be combined with the current MainElement.

The first byte provides information about the string length. One byte is used for the description of each sidestreet combined to the MainElement.

Byte1 -> Length
Byte2 -> Sidestreet1
Byte3 -> Sidestreet2
.
.
.
ByteN -> SidestreetN-1

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 17

## 13.3.) Maneuver_2 [Record]

**Type:** Record

**Description:** Contains Maneuver_2.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_13_3

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### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| MainElement | Enum | Byte |
| Direction | Number | Byte |
| Z_LevelGuidance | Enum | Byte |
| Sidestreets | String | ByteStream |

### *Detailed Sub Parameter Overview*

## 13.3.1.) MainElement [Enum]

**Type:** Enum

**Description:** Enumeration type to describe the type of the main element within a maneuver description.
If not denoted otherwise, each main element type can be combined with sidestreets within one maneuver description.
For exit maneuvers with ramp/Z-level, the values 0x0F and 0x10 shall be used and combined with z-Level.
The values 0x20, 0x21, 0x24 and 0x25 are reserved for projects not supporting Z-Level as a separate parameter and shall not be used on this interface.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | NoSymbol |
| 0x01 | NoInfo |
| 0x02 | DirectionToDestination |
| 0x03 | Arrived |
| 0x04 | NearDestination |
| 0x05 | ArrivedDestinationOffmap |
| 0x06 | OffRoad |
| 0x07 | OffMap |
| 0x08 | NoRoute |
| 0x09 | CalcRoute |
| 0x0A | RecalcRoute |
| 0x0B | FollowStreet |
| 0x0C | ChangeLane |
| 0x0D | Turn |
| 0x0E | TurnOnMainroad |

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| | |
|---|---|
| 0x0F | ExitRight |
| 0x10 | ExitLeft |
| 0x11 | ServiceRoadRight |
| 0x12 | ServiceRoadLeft |
| 0x13 | Fork-2 |
| 0x14 | Fork-3 |
| 0x15 | RoundaboutTrsRight |
| 0x16 | RoundaboutTrsLeft |
| 0x17 | SquareTrsRight |
| 0x18 | SquareTrsLeft |
| 0x19 | Uturn |
| 0x1A | ExitRoundaboutTrsRight |
| 0x1B | ExitRoundaboutTrsLeft |
| 0x1C | PrepareTurn |
| 0x1D | PrepareRoundabout |
| 0x1E | PrepareSquare |
| 0x1F | PrepareUTurn |
| 0x20 | reserved - ExitRight-RampUp |
| 0x21 | reserved - ExitRight-RampDown |
| 0x22 | MichigangTurn |
| 0x23 | DoubleTurn |
| 0x24 | reserved - ExitLeftRampUp |
| 0x25 | reserved - ExitLeftRampDown |
| 0x26 | DirectionToWaypoint |
| 0x27 | ReducedGuidance (DF4.7) |
| 0x28 | LeaveParking (DF4.7) |
| 0x29 | MainRoadRight (DF4.7) |
| 0x2A | MainRoadLeft (DF4.7) |
| 0x2B | DirectionToEntrypoint (DF4.7) |
| 0x2C | Take Ferry (DF4.7 |
| 0x2D | Take CarTrain (DF4.7) |
| 0x2E | UnderpassRight (DF4.7) |
| 0x2F | UnderpassLeft (DF4.7) |
| 0x30 | OverpassRight (DF4.7) |
| 0x31 | OverpassLeft (DF4.7) |
| 0x32 | Toll Gate (DF4.7) |
| 0x33 | Toll Gate - ETC (DF4.7) |
| 0x34..0xFF | reserved |

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## 13.3.2.) Direction [Number]

**Type:** Number

**Description:** Direction of the street to turn into relative to the current driving direction. Resolution: 360Â°/256 (approx. 1.4Â°). The exact meaning of the direction depends on the MainElement.

0x00 -> straight (0Â°)
0x40 -> left (90Â°)
0x80 -> back (180Â°)
0xC0 -> right (270Â°)

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 13.3.3.) Z_LevelGuidance [Enum]

**Type:** Enum

**Description:** Parameter provides Z-LevelGuidance information associated with the Direction.

If not specified in another requirement document, the ASG shall handle a reserved value as if 0x00 ("no Z-LevelGuidance") is received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no Z-LevelGuidance |
| 0x01 | up |
| 0x02 | down |
| 0x03..0xFE | reserved |
| 0xFF | not supported |

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## 13.3.4.) Sidestreets [String]

**Type:** String

**Description:** The string contains the complete set of available sidestreets as data stream (string). The data stream doesn't contain any UTF8 encoded unicode3.0 characters. String contains information about the (existing) sidestreets, which shall be combined with the current MainElement.

The first byte provides information about the string length. One byte is used for the description of each sidestreet combined to the MainElement.

Byte1 -> Length
Byte2 -> Sidestreet1
Byte3 -> Sidestreet2
.
.
.
ByteN -> SidestreetN-1

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 17

## 13.4.) Maneuver_3 [Record]

**Type:** Record

**Description:** Contains Maneuver_3.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_13_4

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| MainElement | Enum | Byte |
| Direction | Number | Byte |
| Z_LevelGuidance | Enum | Byte |
| Sidestreets | String | ByteStream |

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## *Detailed Sub Parameter Overview*

### 13.4.1.) MainElement [Enum]

**Type:** Enum

**Description:** Enumeration type to describe the type of the main element within a maneuver description.
If not denoted otherwise, each main element type can be combined with sidestreets within one maneuver description.
For exit maneuvers with ramp/Z-level, the values 0x0F and 0x10 shall be used and combined with z-Level.
The values 0x20, 0x21, 0x24 and 0x25 are reserved for projects not supporting Z-Level as a separate parameter and shall not be used on this interface.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | NoSymbol |
| 0x01 | NoInfo |
| 0x02 | DirectionToDestination |
| 0x03 | Arrived |
| 0x04 | NearDestination |
| 0x05 | ArrivedDestinationOffmap |
| 0x06 | OffRoad |
| 0x07 | OffMap |
| 0x08 | NoRoute |
| 0x09 | CalcRoute |
| 0x0A | RecalcRoute |
| 0x0B | FollowStreet |
| 0x0C | ChangeLane |
| 0x0D | Turn |
| 0x0E | TurnOnMainroad |
| 0x0F | ExitRight |
| 0x10 | ExitLeft |
| 0x11 | ServiceRoadRight |
| 0x12 | ServiceRoadLeft |
| 0x13 | Fork-2 |
| 0x14 | Fork-3 |
| 0x15 | RoundaboutTrsRight |
| 0x16 | RoundaboutTrsLeft |

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| | |
|---|---|
| 0x17 | SquareTrsRight |
| 0x18 | SquareTrsLeft |
| 0x19 | Uturn |
| 0x1A | ExitRoundaboutTrsRight |
| 0x1B | ExitRoundaboutTrsLeft |
| 0x1C | PrepareTurn |
| 0x1D | PrepareRoundabout |
| 0x1E | PrepareSquare |
| 0x1F | PrepareUTurn |
| 0x20 | reserved - ExitRight-RampUp |
| 0x21 | reserved - ExitRight-RampDown |
| 0x22 | MichigangTurn |
| 0x23 | DoubleTurn |
| 0x24 | reserved - ExitLeftRampUp |
| 0x25 | reserved - ExitLeftRampDown |
| 0x26 | DirectionToWaypoint |
| 0x27 | ReducedGuidance (DF4.7) |
| 0x28 | LeaveParking (DF4.7) |
| 0x29 | MainRoadRight (DF4.7) |
| 0x2A | MainRoadLeft (DF4.7) |
| 0x2B | DirectionToEntrypoint (DF4.7) |
| 0x2C | Take Ferry (DF4.7) |
| 0x2D | Take CarTrain (DF4.7) |
| 0x2E | UnderpassRight (DF4.7) |
| 0x2F | UnderpassLeft (DF4.7) |
| 0x30 | OverpassRight (DF4.7) |
| 0x31 | OverpassLeft (DF4.7) |
| 0x32 | Toll Gate (DF4.7) |
| 0x33 | Toll Gate - ETC (DF4.7) |
| 0x34..0xFF | reserved |

## 13.4.2.) Direction [Number]

**Type:** Number

**Description:** Direction of the street to turn into relative to the current driving direction. Resolution: 360Â°/256 (approx. 1.4Â°). The exact meaning of the direction depends on the MainElement.

0x00 -> straight (0Â°)
0x40 -> left (90Â°)
0x80 -> back (180Â°)
0xC0 -> right (270Â°)

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**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 13.4.3.) Z_LevelGuidance [Enum]

**Type:** Enum

**Description:** Parameter provides Z-LevelGuidance information associated with the Direction.

If not specified in another requirement document, the ASG shall handle a reserved value as if 0x00 ("no Z-LevelGuidance") is received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no Z-LevelGuidance |
| 0x01 | up |
| 0x02 | down |
| 0x03..0xFE | reserved |
| 0xFF | not supported |

## 13.4.4.) Sidestreets [String]

**Type:** String

**Description:** The string contains the complete set of available sidestreets as data stream (string). The data stream doesn't contain any UTF8 encoded unicode3.0 characters. String contains information about the (existing) sidestreets, which shall be combined with the current MainElement.

The first byte provides information about the string length. One byte is used for the description of each sidestreet combined to the MainElement.

Byte1 -> Length
Byte2 -> Sidestreet1
Byte3 -> Sidestreet2
.

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.
.
ByteN -> SidestreetN-1

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 17

## 13.5.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 14.) LaneGuidance ( 24, 0x18 )

**Type:** Array

**Description:** Function provides information needed for lane guidance. Lane guidance may be displayed additionally to the maneuver of the route guidance.

Lane guidance is modelled as array of consecutive IDs. Each array entry corresponds to a single lane. For the definition of "consecutive IDs" and the usage for BAP array functions the reader is referred to [2].

NOTE: In case of a ChangedArray message the parameter LaneGuidance is either empty (not sent) or contains only Pos information.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| LaneGuidanceOnOff | Enum | Byte |
| ArrayHeader | Record | Empty |
| LaneGuidance | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** LaneGuidance (24, 0x18 )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, LaneGuidanceOnOff, ArrayHeader, LaneGuidance, |
| ChangedArray | LaneGuidanceOnOff, ArrayHeader, LaneGuidance, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 14.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

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If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaenous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear (DF4.2) |
| 0x4..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs (DF4.4) |
| 0xA | head-up display, to be evaluated by all ASGs (DF4.4) |
| 0xB | operating unit rear, to be evaluated by all ASGs (DF4.4) |
| 0xC..0xF | reserved |

## 14.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceeding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

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**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 14.3.) LaneGuidanceOnOff [Enum]

**Type:** Enum

**Description:** Contains on / off trigger for lane guidance.

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | lane guidance shall not be displayed |
| 0x01 | display lane guidance |
| 0x02..0xFE | reserved |
| 0xFF | not supported |

## 14.4.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_14_4

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

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### *Detailed Sub Parameter Overview*

## 14.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
The IndexSize shall be set to 8bit always (Bit3 := FALSE). I.e., only 8-bit values are allowed for Start, Elements and Pos.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Array direction is backward | Array direction is forward |
| 2 | Array position is transmitted | Array position is not transmitted |
| 3 | reserved | IndexSize 8 Bit for Start/Elements |

## 14.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records [2].

RecordAddress 0x0 delivers all record elements.
RecordAddress 0xF delivers Pos only.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | LaneDirection, LaneSidestreets, LaneType, LaneMarking_left, LaneMarking_right, LaneDescription, GuidanceInfo |
| 0x1 | LaneDirection, LaneSidestreets, LaneType, LaneDescription, GuidanceInfo |
| 0x2 | LaneDirection, LaneDescription, GuidanceInfo |

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| | |
|---|---|
| 0x3 | LaneDescription, GuidanceInfo |
| 0x4..0xE | reserved |
| 0xF | Pos |

### 14.4.3.) Start [Number]

**Type:** Number

**Description:** For detailed specification refer to [2].
Parameter Start is the first required array element.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 8

**Step:** 1

**Unit:** unit_none

### 14.4.4.) Elements [Number]

**Type:** Number

**Description:** For detailed specification refer to [2].
Parameter Elements contains the number of required array elements.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 14.5.) LaneGuidance [Record]

**Type:** Record

**Description:** LaneGuidance may contain up to 8 lanes together with the respective lane and guidance information.
MaxElements = 8. Please refer to [2], Â§3.10 for the meaning of MaxElements.

**OpCodes:** StatusArray, ChangedArray,

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**Base Datatype:** Empty

**Count:** 255

**Subparameter von:** 1_14_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| LaneDirection | Number | Byte |
| LaneSidestreets | String | ByteStream |
| LaneType | Enum | Byte |
| LaneMarking_left | Enum | Nibble |
| LaneMarking_right | Enum | Nibble |
| LaneDescription | Enum | Nibble |
| GuidanceInfo | Enum | Nibble |

### *Detailed Sub Parameter Overview*

## 14.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos = 0 denotes the left most lane.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 8

**Step:** 1

**Unit:** unit_none

## 14.5.2.) LaneDirection [Number]

**Type:** Number

**Description:** The encoded lane direction represents the traffic direction the lane will most likely be used for, in the context of the upcoming maneuver. The encoding of the LaneDirection is the same as for parameter "Maneuver_< x > .Direction" of function "ManeuverDescriptor" (0x17). The encoding and meaning of "LaneDirection" is similar to the encoding of "Direction" in maneuver "turn on mainroad". The details are given the functional requirement specification.

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The combination of LaneDirection with LaneSidestreets indicates which of the sidestreets of the lane the driver shall take (in the sense of a reduced RGI). Usually, the direction (sidestreet) contained in LaneDirection is not contained additionally in parameter LaneSidestreets. Further details are given in the functional requirement specification.

Parameter LaneDirection is not intended to provide a faithful representation of the symbol painted on the road surface.
NOTE: It makes no sense to combine an inner lane of a street with sidestreets; only outer lanes may be combined with LaneSidestreets.

The default value of LaneDirection is "straight" (value: 0x00).

LaneDirection = 0xFF denotes an invalid LaneDirection.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 254

**Step:** 1

**Unit:** unit_none

**EnumValues:**

| Code | Description |
|---|---|
| 0xFF | invalid LaneDirection |

## 14.5.3.) LaneSidestreets [String]

**Type:** String

**Description:** The string contains the complete set of available sidestreets of the lane as data stream (string). The data stream doesn't contain any UTF8 encoded Unicode3.0 characters.
This string contains information about the (existing) sidestreets, which shall be combined with the current lane.

The first byte provides information about the string length.
The encoding and meaning is similar to the encoding of "sidestreets" in maneuver "turn on mainroad". Further details are given in functional requirement specification.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 17

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## 14.5.4.) LaneType [Enum]

**Type:** Enum

**Description:** This parameter contains the lane type information, e.g. bus lane, taxi lane, bicycle lane.

If the ASG receives a reserved value, it shall act as if value 0x00 ("unspecific lane type") had been received.

Value 0x17 ("further lanes on the left hand side") is applicable for the left most lane only.
Value 0x18 ("further lanes on the right hand side") is applicable for the rightmost lane only.
The leftmost lane is the one with the smallest numeric Pos value.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | unspecific lane type |
| 0x01 | normal lane |
| 0x02 | lane blocked |
| 0x03 | lane forbidden |
| 0x04 | lane restricted |
| 0x05 | breakdown lane |
| 0x06 | dynamically controlled |
| 0x07 | car pools |
| 0x08 | bus lane |
| 0x09 | bicycle lane |
| 0x0A | taxi lane |
| 0x0B | truck lane |
| 0x0C | bus & bicycle lane |
| 0x0D | bus & taxi lane |
| 0x0E | taxi & bicycle lane |
| 0x0F | bus, taxi & bicycle lane |
| 0x10 | green strip |
| 0x11 | motorway/highway lane |
| 0x12 | tramway on normal road |
| 0x13 | separated tramway |
| 0x14 | exit lane |
| 0x15 | enter lane |
| 0x16 | toll lane |
| 0x17 | further lanes on the left hand side (not shown) |

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| | |
|---|---|
| 0x18 | further lanes on the right hand side (not shown) |
| 0x19 | hide lane |
| 0x1A | ETC / non ETC mixed lane (DF4.4) |
| 0x1B..0xFF | reserved |

## 14.5.5.) LaneMarking_left [Enum]

**Type:** Enum

**Description:** (upper nibble)

This parameter contains the lane marking (at the left border of the lane). Lane markings of adjacent lanes may be combined. E.g., if the lane marking is a double solid line, lane "n-1" reports LaneMarking_right = 0x1 and lane "n" reports LaneMarking_left = 0x1.
If the ASG receives a reserved value, it shall act as if value 0x0 ("no line") had been received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no lane marking at the left border of the lane (no line) |
| 0x1 | solid divider line |
| 0x2 | dashed divider line |
| 0x3..0xF | reserved |

## 14.5.6.) LaneMarking_right [Enum]

**Type:** Enum

**Description:** (lower nibble)

This parameter contains the lane marking (at the right border of the lane). Lane markings of adjacent lanes may be combined. E.g., if the lane marking is a double solid line, lane "n-1" reports LaneMarking_right = 0x1 and lane "n" reports LaneMarking_left = 0x1.
If the ASG receives a reserved value, it shall act as if value 0x0 ("no line") had been received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

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**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no lane marking at the right border of the lane (no line) |
| 0x1 | solid divider line |
| 0x2 | dashed divider line |
| 0x3..0xF | reserved |

## 14.5.7.) LaneDescription [Enum]

**Type:** Enum

**Description:** (upper nibble)

Provides information of lane availability further down the route. This information can be used to provide suitable graphical representations for additional lanes, which are not yet available at the present vehicle position (expanding lanes) or lanes which will end/merge ahead of the present vehicle position (merging lanes).

If the ASG receives a reserved value, it shall act as if 0x00 ("lane available") is received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | lane available |
| 0x1 | lane merging to left |
| 0x2 | lane merging to right |
| 0x3 | lane ending later |
| 0x4 | forbidden lane |
| 0x5 | lane expanding to left |
| 0x6 | lane expanding to right |
| 0x7 | lane available later |
| 0x8 | multiple additional lane begin |
| 0x9 | multiple continuation with lane ends |
| 0xA | lane begins in the middle |
| 0xB..0xE | reserved |
| 0xF | not supported - no description available |

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## 14.5.8.) GuidanceInfo [Enum]

**Type:** Enum

**Description:** (lower nibble)

This parameter describes the lane recommendation.

If the ASG receives a reserved value, it shall act if the value 0x0 ("not recommended lane") is received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | not recommended lane |
| 0x1 | recommended lane |
| 0x2 | best recommendation |
| 0x3..0xF | reserved |

## 14.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved |

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# 15.) TMCinfo ( 25, 0x19 )

**Type:** Property

**Description:** This function is used to transfer TMC messages.
TMCinfo should only be used for X-Urgent messages. It is not intended to display all TMC messages, since these are available at the navigation system itself.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| MessageID | Number | Nibble |
| MessageStatus | Enum | Nibble |
| MessageWaitingIndication | Enum | Byte |
| Reserve1 | Number | Byte |
| Reserve2 | String | ByteStream |
| TMCinfo | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** TMCinfo (25, 0x19 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | MessageID, MessageStatus, Reserve1, Reserve2, |
| Status | MessageID, MessageStatus, MessageWaitingIndication, TMCinfo, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 15.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 15.2.) MessageID [Number]

**Type:** Number

**Description:** (upper nibble)

Contains the TMC message ID. This parameter is used by the ASG to confirm the presentation of a certain TMC message. This message ID is not necessarily the TMC message ID received over air.

**OpCodes:** SetGet, Status,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 15.3.) MessageStatus [Enum]

**Type:** Enum

**Description:** (lower nibble)

This parameter is used to indicate new messages and to confirm messages.

**OpCodes:** SetGet, Status,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no message |
| 0x1 | presentation request for new message |
| 0x2 | reserved |
| 0x3 | message presentation confirmed (to be set by ASG) |
| 0x4..0xF | reserved |

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## 15.4.) MessageWaitingIndication [Enum]

**Type:** Enum

**Description:** This parameter contains an indication about messages still to be presented to the driver.
If the ASG receives a reserved value, it shall act as if 0x02 ("several messages are waiting") is received.

**OpCodes:** Status,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no messages waiting |
| 0x01 | 1 message waiting |
| 0x02 | several messages are waiting |
| 0x03..0xFF | reserved |

## 15.5.) Reserve1 [Number]

**Type:** Number

**Description:** This parameter is just for symmetric reasons. It shall be set to 0 by the ASG. The FSG shall ignore this parameter.

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 15.6.) Reserve2 [String]

**Type:** String

**Description:** This parameter is just for symmetric reasons.
The FSG shall set this parameter to 0x00 (empty string, refer to [2] for its definition).

**OpCodes:** SetGet,

**Base Datatype:** ByteStream

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**Length:** 250

## 15.7.) TMCinfo [Record]

**Type:** Record

**Description:** Provides some abstract TMC information.

**OpCodes:** Status,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_15_7

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Priority | Number | Byte |
| StreetName | String | ByteStream |
| Location | String | ByteStream |
| Infotext | String | ByteStream |
| Length_Validity | Bitfield | ByteStream |
| Length_Value | Number | Long |
| Length_Unit | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 15.7.1.) Priority [Number]

**Type:** Number

**Description:** This parameter (record element) contains the priority of the corresponding TMC message.

Logical lowest priority: 255
Logical highest priority: 0

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

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**Unit:** unit_none

## 15.7.2.) StreetName [String]

**Type:** String

**Description:** StreetName includes also the direction if available (e.g. "A2 Hannover Braunschweig").
The string may contain more than one element (e.g. A2; Hannover; Braunschweig), separated by the Unicode control code "LineFeed" (0x000A).

The maximum number of characters in this string is limited by the string length only. The encoded length must not exceed 60 bytes plus an additional byte for the string length information.

"Empty string" if unknown.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 15.7.3.) Location [String]

**Type:** String

**Description:** The location is marked by the names of two exits bordering the traffic event.
The string may contain more than one element (e.g. AS Braunschweig Ost; AS Wolfsburg/Koenigslutter), separated by the Unicode control code "LineFeed" (0x000A).

The maximum number of characters in this string is limited by the string length only. The encoded length must not exceed 60 bytes plus an additional byte for the string length information.

"Empty string" if unknown.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 15.7.4.) Infotext [String]

**Type:** String

**Description:** Infotext contents a short description of the traffic event.
The string may contain more than one element, separated by the Unicode control code "LineFeed" (0x000A).

The maximum number of characters in this string is limited by the string length only. The encoded length must not exceed 120 bytes plus an additional byte for the string length information.

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"Empty string" if unknown. The definition of "empty string" is given in [2].

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 121

## 15.7.5.) Length_Validity [Bitfield]

**Type:** Bitfield

**Description:** This element contains the validity information of the length field. The ASG shall ignore reserved bits of this bit field. The FSG shall set them to FALSE.

Length provides information about the length of the congestion.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Length valid | Length invalid |
| 1..7 | reserved | reserved |

## 15.7.6.) Length_Value [Number]

**Type:** Number

**Description:** Length_Value provides information about the length of the congestion / traffic event.

[transmitted length value]=[displayed length value]*10 (e.g. [displayed length value]=0.9 -> [transmitted length value]=9).

In some situations there may be no length available. Therefore the following coding is defined: If the length is not known to a specific moment, the bit0 in parameter Length_Validity.Bit0 is set to 0.

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** 0

**Range Max:** 429496729

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**Step:** 0.1

**Unit:** unit_none

## 15.7.7.) Length_Unit [Enum]

**Type:** Enum

**Description:** Unit belonging to the Length_Value.

The ASG shall handle a reserved value the same way as value 0xFF ("no information about unit available").

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (US / UK (british statute mile)) |
| 0x05..0xFE | reserved |
| 0xFF | not supported / no information about unit available |

## 15.8.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |

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| | |
|---|---|
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 16.) MagnetFieldZone ( 26, 0x1A )

**Type:** Property

**Description:** Function is only relevant for stand alone compass units. With this function it is possible to change the magnetic field zone.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Zone | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** MagnetFieldZone (26, 0x1A )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | Zone, |
| Status | Zone, |
| HeartbeatStatus | Zone, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 16.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 16.2.) Zone [Enum]

**Type:** Enum

**Description:** Zone

**OpCodes:** SetGet, Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | reserved |
| 0x01 | zone 1 |
| 0x02 | zone 2 |
| 0x03 | zone 3 |
| 0x04 | zone 4 |
| 0x05 | zone 5 |
| 0x06 | zone 6 |
| 0x07 | zone 7 |
| 0x08 | zone 8 |
| 0x09 | zone 9 |
| 0x0A | zone 10 |
| 0x0B | zone 11 |
| 0x0C | zone 12 |
| 0x0D | zone 13 |
| 0x0E | zone 14 |
| 0x0F | zone 15 |
| 0x10..0xFF | reserved |

## 16.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 17.) Calibration ( 27, 0x1B )

**Type:** Property

**Description:** Function is only relevant for stand alone compass units. Provides information about and sets the calibration state of the compass control unit.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| CalibrationState | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** Calibration (27, 0x1B )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | CalibrationState, |
| Status | CalibrationState, |
| HeartbeatStatus | CalibrationState, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 17.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 17.2.) CalibrationState [Enum]

**Type:** Enum

**Description:** The calibration state "calibration active" may also used to display driving instructions.

**OpCodes:** SetGet, Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | calibrated |
| 0x01 | calibration active |
| 0x02 | interfering field |
| 0x03..0xFE | reserved |
| 0xFF | not supported |

## 17.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 18.) ASG_Capabilities ( 28, 0x1C )

**Type:** Property

**Description:** This function indicates the ASG display capabilities to the FSG.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| PresentationCapabilities | Bitfield | Nibble |
| FurtherCapabilities | Bitfield | Nibble |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** ASG_Capabilities (28, 0x1C )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | PresentationCapabilities, FurtherCapabilities, |
| Status | PresentationCapabilities, FurtherCapabilities, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 18.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 18.2.) PresentationCapabilities [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

Parameter indicates the ASG display or presentation capabilities for the compass and maneuver presentation.

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The sender shall set reserved bits to FALSE (0).
The receiver shall ignore reserved bits.

**OpCodes:** SetGet, Status,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | 2.5D symbol presentation | 2D symbol presentation |
| 1..3 | reserved | reserved |

## 18.3.) FurtherCapabilities [Bitfield]

**Type:** Bitfield

**Description:** (lower nibble)

Parameter reserved for further ASG capabilities of interest for the FSG.

The FSG shall ignore reserved values if received in a SetGet messages.
The ASG shall ignore reserved values if received. The FSG shall set reserved values to FALSE in any Status message.

**OpCodes:** SetGet, Status,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | reserved | reserved |
| 1 | reserved | reserved |
| 2 | reserved | reserved |
| 3 | reserved | reserved |

## 18.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software.
Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

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Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0xFF | reserved, refer to annex |

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# 19.) LastDest_List ( 29, 0x1D )

**Type:** Array

**Description:** This function contains the list of last destinations stored in the FSG. The Pos value is used by the ASG as destination information if a route guidance shall be started.

This list is referred to as "arbitrary list". For the meaning of "arbitrary IDs" the reader is referred to [2].

If the ASG requests an invalid array element in the Start parameter, the FSG returns the required START element, ELEMENTS = 0 and no DATA because an invalid element has no successor or predecessor. Please note the special rules for START=0, refer to [2].
If the list contains fewer elements in the given direction than required, the FSG returns the existing elements.
The value of the START byte in a StatusArray message has to be equal to the one from the GetArray message. The value of the Transmit_Pos bit has to be taken into account.

For the general specification of the array function class please refer to [2]. For the general description of the sliding window principle the reader is referred to [3].

NOTE: In case of a ChangedArray message the parameter Data is either empty or contains only Pos information.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| TotalNumListElements | Number | Byte |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** LastDest_List (29, 0x1D )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

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## 19.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaneous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear (DF4.2) |
| 0x4..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB | operating unit rear, to be evaluated by all ASGs (DF4.2) |
| 0xC..0xF | reserved |

## 19.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID

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Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 19.3.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of list elements. In most cases this number is greater than the number of elements stored in the ASG.
This information may be used for dimensioning of scroll bars etc.

Value 0xFF means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 19.4.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

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**Subparameter von:** 1_19_4

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

### *Detailed Sub Parameter Overview*

## 19.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
Both, the ASG and the FSG shall always use 8-bit addressing for Pos. This means, bit3 shall always be set to 0 by the ASG and the FSG.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Direction = backward | Direction = forward |
| 2 | Transmit_Pos = Pos is transmitted | Transmit_Pos = Pos is not transmitted |
| 3 | reserved | IndexSize = 8 Bit for Start, Elements, Pos |

## 19.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is part of the ModeByte of the ArrayHeader (see also "BAP Datentypen" [2]!).
The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records.

**OpCodes:** Empty,

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**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | POI_Type, Description |
| 0x1 | Description |
| 0x2 | POI_Type |
| 0x3..0xE | reserved |
| 0xF | Pos |

## 19.4.3.) Start [Number]

**Type:** Number

**Description:** Parameter Start is the first array element the messages relates to. For usage of this parameter the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 19.4.4.) Elements [Number]

**Type:** Number

**Description:** Parameter Elements contains the number of elements the message relates to. For usage of this parameter the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

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## 19.5.) Data [Record]

**Type:** Record

**Description:** "Destination" contains a description of the destination.
MaxElements = 4. This means, a StatusArray message is limited to 4 complete elements.
Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 254 elements. The FSG is not allowed to generate 0x00 as a valid Pos value.

The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 254

**Subparameter von:** 1_19_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| POI_Type | Enum | Byte |
| Description | String | ByteStream |

### *Detailed Sub Parameter Overview*

## 19.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

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## 19.5.2.) POI_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates, if the destination is a POI (Point Of Interest) and the POI type.
If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no POI |
| 0x01 | Accommodation |
| 0x02 | Airport |
| 0x03 | all restaurants |
| 0x04 | Amusement Park |
| 0x05 | Apparel |
| 0x06 | ATM (cashpoint) |
| 0x07 | Attraction |
| 0x08 | Automobile Club |
| 0x09 | Banking and Shopping |
| 0x0A | Banking |
| 0x0B | Banquet-Halls |
| 0x0C | Bars and Lounge |
| 0x0D | Boating |
| 0x0E | Bookstore |
| 0x0F | Border Cross |
| 0x10 | Bowling |
| 0x11 | Broadcasting |
| 0x12 | Business and Service |
| 0x13 | Business |
| 0x14 | Busstation |
| 0x15 | Camping |
| 0x16 | Car and Travel |
| 0x17 | Car Track |
| 0x18 | Car Train |
| 0x19 | Car Service |
| 0x1A | Car Wash |
| 0x1B | Casino |

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| | |
|---|---|
| 0x1C | Cemetry |
| 0x1D | Cityhall |
| 0x1E | coffee |
| 0x1F | Commuter Railstaion |
| 0x20 | Cleaning Launch |
| 0x21 | Clothing Store |
| 0x22 | Communication |
| 0x23 | Community Center |
| 0x24 | Computer Hardware-Software |
| 0x25 | Convenience Store |
| 0x26 | Convention Center |
| 0x27 | Corporations |
| 0x28 | Corporations Public Places |
| 0x29 | Culture Facility |
| 0x2A | Court |
| 0x2B | Department Store |
| 0x2C | Dining and Entertainment |
| 0x2D | Dinning Shopping |
| 0x2E | Discount Store |
| 0x2F | Education |
| 0x30 | Electronics |
| 0x31 | Embassy |
| 0x32 | Emergency |
| 0x33 | Emergency and Public |
| 0x34 | Entertainment |
| 0x35 | Fast Food |
| 0x36 | Ferry |
| 0x37 | Fire Department |
| 0x38 | Flooring and Carpeting |
| 0x39 | Flower and Jewelry |
| 0x3A | Food and Beverage |
| 0x3B | Funeral Director |
| 0x3C | Furniture |
| 0x3D | Games, Music and Video |
| 0x3E | Garden and other |
| 0x3F | Gasstation |
| 0x40 | Gasstation DieselGasstation LPG |
| 0x41 | General |
| 0x42 | Gifts and Leisure |

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| | |
|---|---|
| 0x43 | Gifts- Antiques and Arts |
| 0x44 | Glass and Window |
| 0x45 | Golfing |
| 0x46 | Goverment |
| 0x47 | Grocery Stores |
| 0x48 | Guesthouse |
| 0x49 | Hair and Beauty |
| 0x4A | Harbor |
| 0x4B | Hardware and Lumber |
| 0x4C | Health and Fitness |
| 0x4D | Health- Care |
| 0x4E | Highrise- BLDG |
| 0x4F | Home and Improvment |
| 0x50 | Home Center |
| 0x51 | Home Service |
| 0x52 | Hospital |
| 0x53 | Hotel / Motel |
| 0x54 | Iceskating |
| 0x55 | Insurance |
| 0x56 | Karaokay |
| 0x57 | Library |
| 0x58 | LIQ-Gas-Station |
| 0x59 | Major-Appliance |
| 0x5A | Mens-Apparel |
| 0x5B | Monument |
| 0x5C | Motorway Exit |
| 0x5D | Motorway Cross |
| 0x5E | Motorway Service |
| 0x5F | Movie-Theater |
| 0x60 | Moving and Storage |
| 0x61 | Museum |
| 0x62 | Nightclubs |
| 0x63 | Optical |
| 0x64 | Restaurant |
| 0x65 | Park and Fitness |
| 0x66 | Park and Ride |
| 0x67 | Parking |
| 0x68 | Parking Garage |
| 0x69 | Parks |

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| | |
|---|---|
| 0x6A | Performing-Arts |
| 0x6B | Personal- Service |
| 0x6C | Pharmacies |
| 0x6D | Photographer |
| 0x6E | Place or workship |
| 0x6F | Police Station |
| 0x70 | Post Office |
| 0x71 | Public Places |
| 0x72 | Publisher |
| 0x73 | Railway Station |
| 0x74 | Recreation |
| 0x75 | Rentel Service |
| 0x76 | Repair |
| 0x77 | Rest-Area |
| 0x78 | Retirement Nursing |
| 0x79 | Search |
| 0x7A | Service |
| 0x7B | Shoe Stores |
| 0x7C | Shopping Center |
| 0x7D | Skiing |
| 0x7E | Social Service |
| 0x7F | Speciality-Clothing |
| 0x80 | Speciality-Store |
| 0x81 | Speciality-Food |
| 0x82 | Sport-Goods |
| 0x83 | Sports |
| 0x84 | Sport Airport |
| 0x85 | Sports-Complex |
| 0x86 | Tailor |
| 0x87 | Tax-Service |
| 0x88 | Tool-Booth |
| 0x89 | Tourist Attraction |
| 0x8A | Tourist Information |
| 0x8B | Train Station |
| 0x8C | Transportation |
| 0x8D | Travel |
| 0x8E | Travel-Agent |
| 0x8F | University |
| 0x90 | Utilities |

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| | |
|---|---|
| 0x91 | Variety-Store |
| 0x92 | Wine and Liquor |
| 0x93 | Womens Apparel |
| 0x94 | Winery |
| 0x95 | car repair shop - VW |
| 0x96 | car repair shop - Audi |
| 0x97 | car repair shop - Seat |
| 0x98 | car repair shop - Skoda |
| 0x99 | car repair shop - VW Nutzfahrzeuge |
| 0x9A | car repair shop - Bentley |
| 0x9B | car repair shop - Bugatti |
| 0x9C | car repair shop - Lamborghini |
| 0x9D | car repair shop - Scania |
| 0x9E | car repair shop - MAN |
| 0x9F | car repair shop - Porsche |
| 0xA0 | electrical service station (DF4.1) |
| 0xA1 | Diesel fuel station (DF4.1) |
| 0xA2 | CNG fuel station (DF4.1) |
| 0xA3 | rest area ("Autohof") (DF4.1) |
| 0xA4 | rest area with WC (DF4.1) |
| 0xA5 | motorway service area ("Raststaette") (DF4.1) |
| 0xA6..0xAF | reserved |
| 0xB0 | Predictive NAV (DF4.6) |
| 0xB1..0xFE | reserved |
| 0xFF | unknown |

## 19.5.3.) Description [String]

**Type:** String

**Description:** String contains the string to be shown for the listed destination.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 96 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 97

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## 19.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved |

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# 20.) FavoriteDest_List ( 30, 0x1E )

**Type:** Array

**Description:** This function contains the list of favourite destinations stored in the FSG. The Pos value is used by the ASG as destination information if a route guidance shall be started.

This list is referred to as "arbitrary list". For the meaning of "arbitrary IDs" the reader is referred to [2].

If the ASG requests an invalid array element in the Start parameter, the FSG returns the required START element, ELEMENTS = 0 and no DATA because an invalid element has no successor or predecessor. Please note the special rules for START=0, refer to [2].
If the list contains fewer elements in the given direction than required, the FSG returns the existing elements.
The value of the START byte in a StatusArray message has to be equal to the one from the GetArray message. The value of the Transmit_Pos bit has to be taken into account.

For the general specification of the array function class please refer to [2]. For the general description of the sliding window principle the reader is referred to [3].

NOTE: In case of a ChangedArray message the parameter Data is either empty or contains only Pos information.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| TotalNumListElements | Number | Byte |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** FavoriteDest_List (30, 0x1E )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

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## 20.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaneous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear (DF4.2) |
| 0x4..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB | operating unit rear, to be evaluated by all ASGs (DF4.2) |
| 0xC..0xF | reserved |

## 20.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

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TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 20.3.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of current list elements. In most cases this number is greater than the number of elements stored in the ASG.
This information may be used for dimensioning of scroll bars etc.

Value 0xFF means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 20.4.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

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**Count:** 1

**Subparameter von:** 1_20_4

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

### *Detailed Sub Parameter Overview*

## 20.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
Both, the ASG and the FSG shall always use 8-bit addressing for Pos. This means, Bit3 shall always be set to 0 by the ASG and the FSG.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Direction = backward | Direction = forward |
| 2 | Transmit_Pos = Pos is transmitted | Transmit_Pos = Pos is not transmitted |
| 3 | reserved | IndexSize = 8 Bit for Start, Elements, Pos |

## 20.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is part of the ModeByte of the ArrayHeader (see also "BAP Datentypen" [2]).

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The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | POI_Type, Description |
| 0x1 | Description |
| 0x2 | POI_Type |
| 0x3..0xE | reserved |
| 0xF | Pos |

## 20.4.3.) Start [Number]

**Type:** Number

**Description:** Parameter Start is the first array element the message relates to. For detailed specification refer to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 20.4.4.) Elements [Number]

**Type:** Number

**Description:** Parameter Elements contains the number of array elements the message relates to. For detailed specification refer to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

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**Step:** 1

**Unit:** unit_none

## 20.5.) Data [Record]

**Type:** Record

**Description:** "Destination" contains a description of the destination.
MaxElements = 4. This means, a StatusArray message is limited to 4 complete elements.
Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 254 elements. The FSG is not allowed to generate 0x00 as a valid Pos value.
The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 254

**Subparameter von:** 1_20_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| POI_Type | Enum | Byte |
| Description | String | ByteStream |

### *Detailed Sub Parameter Overview*

## 20.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

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**Unit:** unit_none

## 20.5.2.) POI_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates, if the destination is a POI (Point Of Interest) and the POI type.
If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no POI |
| 0x01 | Accommodation |
| 0x02 | Airport |
| 0x03 | all restaurants |
| 0x04 | Amusement Park |
| 0x05 | Apparel |
| 0x06 | ATM (cashpoint) |
| 0x07 | Attraction |
| 0x08 | Automobile Club |
| 0x09 | Banking and Shopping |
| 0x0A | Banking |
| 0x0B | Banquet-Halls |
| 0x0C | Bars and Lounge |
| 0x0D | Boating |
| 0x0E | Bookstore |
| 0x0F | Border Cross |
| 0x10 | Bowling |
| 0x11 | Broadcasting |
| 0x12 | Business and Service |
| 0x13 | Business |
| 0x14 | Busstation |
| 0x15 | Camping |
| 0x16 | Car and Travel |
| 0x17 | Car Track |
| 0x18 | Car Train |

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| | |
|---|---|
| 0x19 | Car Service |
| 0x1A | Car Wash |
| 0x1B | Casino |
| 0x1C | Cemetry |
| 0x1D | Cityhall |
| 0x1E | coffee |
| 0x1F | Commuter Railstaion |
| 0x20 | Cleaning Launch |
| 0x21 | Clothing Store |
| 0x22 | Communication |
| 0x23 | Community Center |
| 0x24 | Computer Hardware-Software |
| 0x25 | Convenience Store |
| 0x26 | Convention Center |
| 0x27 | Corporations |
| 0x28 | Corporations Public Places |
| 0x29 | Culture Facility |
| 0x2A | Court |
| 0x2B | Department Store |
| 0x2C | Dining and Entertainment |
| 0x2D | Dinning Shopping |
| 0x2E | Discount Store |
| 0x2F | Education |
| 0x30 | Electronics |
| 0x31 | Embassy |
| 0x32 | Emergency |
| 0x33 | Emergency and Public |
| 0x34 | Entertainment |
| 0x35 | Fast Food |
| 0x36 | Ferry |
| 0x37 | Fire Department |
| 0x38 | Flooring and Carpeting |
| 0x39 | Flower and Jewelry |
| 0x3A | Food and Beverage |
| 0x3B | Funeral Director |
| 0x3C | Furniture |
| 0x3D | Games, Music and Video |
| 0x3E | Garden and other |
| 0x3F | Gasstation |

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| | |
|---|---|
| 0x40 | Gasstation DieselGasstation LPG |
| 0x41 | General |
| 0x42 | Gifts and Leisure |
| 0x43 | Gifts- Antiques and Arts |
| 0x44 | Glass and Window |
| 0x45 | Golfing |
| 0x46 | Goverment |
| 0x47 | Grocery Stores |
| 0x48 | Guesthouse |
| 0x49 | Hair and Beauty |
| 0x4A | Harbor |
| 0x4B | Hardware and Lumber |
| 0x4C | Health and Fitness |
| 0x4D | Health- Care |
| 0x4E | Highrise- BLDG |
| 0x4F | Home and Improvment |
| 0x50 | Home Center |
| 0x51 | Home Service |
| 0x52 | Hospital |
| 0x53 | Hotel / Motel |
| 0x54 | Iceskating |
| 0x55 | Insurance |
| 0x56 | Karaokay |
| 0x57 | Library |
| 0x58 | LIQ-Gas-Station |
| 0x59 | Major-Appliance |
| 0x5A | Mens-Apparel |
| 0x5B | Monument |
| 0x5C | Motorway Exit |
| 0x5D | Motorway Cross |
| 0x5E | Motorway Service |
| 0x5F | Movie-Theater |
| 0x60 | Moving and Storage |
| 0x61 | Museum |
| 0x62 | Nightclubs |
| 0x63 | Optical |
| 0x64 | Restaurant |
| 0x65 | Park and Fitness |
| 0x66 | Park and Ride |

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| | |
|---|---|
| 0x67 | Parking |
| 0x68 | Parking Garage |
| 0x69 | Parks |
| 0x6A | Performing-Arts |
| 0x6B | Personal- Service |
| 0x6C | Pharmacies |
| 0x6D | Photographer |
| 0x6E | Place or workship |
| 0x6F | Police Station |
| 0x70 | Post Office |
| 0x71 | Public Places |
| 0x72 | Publisher |
| 0x73 | Railway Station |
| 0x74 | Recreation |
| 0x75 | Rentel Service |
| 0x76 | Repair |
| 0x77 | Rest-Area |
| 0x78 | Retirement Nursing |
| 0x79 | Search |
| 0x7A | Service |
| 0x7B | Shoe Stores |
| 0x7C | Shopping Center |
| 0x7D | Skiing |
| 0x7E | Social Service |
| 0x7F | Speciality-Clothing |
| 0x80 | Speciality-Store |
| 0x81 | Speciality-Food |
| 0x82 | Sport-Goods |
| 0x83 | Sports |
| 0x84 | Sport Airport |
| 0x85 | Sports-Complex |
| 0x86 | Tailor |
| 0x87 | Tax-Service |
| 0x88 | Tool-Booth |
| 0x89 | Tourist Attraction |
| 0x8A | Tourist Information |
| 0x8B | Train Station |
| 0x8C | Transportation |
| 0x8D | Travel |

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| | |
|---|---|
| 0x8E | Travel-Agent |
| 0x8F | University |
| 0x90 | Utilities |
| 0x91 | Variety-Store |
| 0x92 | Wine and Liquor |
| 0x93 | Womens Apparel |
| 0x94 | Winery |
| 0x95 | car repair shop - VW |
| 0x96 | car repair shop - Audi |
| 0x97 | car repair shop - Seat |
| 0x98 | car repair shop - Skoda |
| 0x99 | car repair shop - VW Nutzfahrzeuge |
| 0x9A | car repair shop - Bentley |
| 0x9B | car repair shop - Bugatti |
| 0x9C | car repair shop - Lamborghini |
| 0x9D | car repair shop - Scania |
| 0x9E | car repair shop - MAN |
| 0x9F | car repair shop - Porsche |
| 0xA0 | electrical service station (DF4.1) |
| 0xA1 | Diesel fuel station (DF4.1) |
| 0xA2 | CNG fuel station (DF4.1) |
| 0xA3 | rest area ("Autohof") (DF4.1) |
| 0xA4 | rest area with WC (DF4.1) |
| 0xA5 | motorway service area ("Raststaette") (DF4.1) |
| 0xA6..0xAF | reserved |
| 0xB0 | Predictive NAV (DF4.6) |
| 0xB1..0xFE | reserved |
| 0xFF | unknown |

### 20.5.3.) Description [String]

**Type:** String

**Description:** String contains the string to be shown for the listed destination.

String may contain up to 32 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 97 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

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**Length:** 97

## 20.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved, refer to annex |

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# 21.) PreferredDest_List ( 31, 0x1F )

**Type:** Property

**Description:** This function is used to change the list to be shown in the ASG (e.g. instrument cluster) if only one list is available or only one list can be presented by the ASG.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| ListType | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** PreferredDest_List (31, 0x1F )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | ListType, |
| HeartbeatStatus | ListType, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 21.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 21.2.) ListType [Enum]

**Type:** Enum

**Description:** If parameter "List" is set to a reserved value, the ASG shall act as if 0x00 ("no list preferred") is received.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no list preferred / undefined |
| 0x01 | LastDestinations_List (fct 0x1D) |
| 0x02 | FavoriteDestinations_List (fct. 0x1E) |
| 0x03 | NavBook (fct. 0x20) |
| 0x04..0xFF | reserved |

## 21.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 22.) NavBook ( 32, 0x20 )

**Type:** Array

**Description:** This function contains a list of name entries. Each entry has at least a relation to one destination, which is suitable for navigation. The address details for one or more addresses of that entry are contained in function Address_List (0x21).

NOTE: This list is referred to as "arbitrary list". Please refer to [2] for definition and usage of that list type.

If the ASG requests an invalid array element in the Start parameter, the FSG returns the required START element, ELEMENTS = 0 and no DATA because an invalid element has no successor or predecessor. Please note the special rules for START=0, refer to [2].

If the list contains fewer elements in the given direction than required, the FSG returns the existing elements.

NOTE: In case of a ChangedArray message the parameter Data is either empty or contains only POS information.

For the general specification of the array function class please refer to [2]. For the general description of the sliding window principle the reader is referred to [3].

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| TotalNumListElements | Number | Word |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** NavBook (32, 0x20 )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

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## *Detailed Parameter Overview*

### 22.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

This parameter identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaneous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear (DF4.2) |
| 0x4..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB | operating unit rear, to be evaluated by all ASGs (DF4.2) |
| 0xC..0xF | reserved |

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## 22.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 22.3.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of list elements. In most cases this number is greater than the number of elements stored in the ASG.
This information may be used for dimensioning of scroll bars etc.

Value 0xFFFF (65535) means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

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## 22.4.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_22_4

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Word |
| Elements | Number | Word |

### *Detailed Sub Parameter Overview*

## 22.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to "BAP Datentypen" [2]. The meaning of the bits depends on the OpType. Refer to [2].

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Direction = backward | Direction = forward |
| 2 | Transmit_Pos = Pos transmitted | Transmit_Pos = Pos is not transmitted |
| 3 | IndexSize 16 Bit for Start/Elements | IndexSize = 8 Bit for Start, Elements, Pos |

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## 22.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is part of the ModeByte of the ArrayHeader (refer to [2]). The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | LastName, FirstName |
| 0x1 | LastName |
| 0x2 | FirstName |
| 0x3..0xE | reserved |
| 0xF | Pos |

## 22.4.3.) Start [Number]

**Type:** Number

**Description:** For detailed specification the reader is referred to [2].
Parameter Start is the first required array element. Parameter length depends on the Bit IndexSize in Mode:
IndexSize=0 -> 8 Bit, IndexSize=1 -> 16 Bit.

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

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## 22.4.4.) Elements [Number]

**Type:** Number

**Description:** For detailed specification see [2].
Parameter Elements contains the number of required array elements. Parameter length is depending on the Bit IndexSize in Mode:
IndexSize=0 -> 8 Bit, IndexSize=1 -> 16 Bit.

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 22.5.) Data [Record]

**Type:** Record

**Description:** "Destination" contains a description of the destination.
MaxElements = 4. This means, a StatusArray message is limited to 4 complete elements.
Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 65534 elements. The FSG is not allowed value generate 0x00/0x0000 as a valid POS value.
The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 65534

**Subparameter von:** 1_22_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Word |
| LastName | String | ByteStream |
| FirstName | String | ByteStream |

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### *Detailed Sub Parameter Overview*

## 22.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 22.5.2.) LastName [String]

**Type:** String

**Description:** This parameter contains the "last name" field.

This string may contain up to 20 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 61 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 22.5.3.) FirstName [String]

**Type:** String

**Description:** String contains the string to be shown for the listed destination.

String may contain up to 20 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 61 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

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## 22.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved, refer to annex |

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# 23.) Address_List ( 33, 0x21 )

**Type:** Array

**Description:** This function contains a (detailed) list of addresses. The parameter ListType_Reference links other lists to the Address_List. I.e., using function Address_List the ASG may ask for address details of a phonebook entry, a last destinations list entry, a favourite destinations list entry etc.
Each of those list entries (e.g. of the phonebook) may contain one or several addresses. These addresses belonging to another logical list are conveyed in function "Address_List".

The Pos value of Address_List may used by the ASG as destination information if a route guidance shall be started.

This list is referred to as "arbitrary list". Please refer to [2] for definition and usage of that list type.

If the ASG requests an invalid array element in the Start parameter, the FSG returns the required START element, ELEMENTS = 0 and no DATA because an invalid element has no successor or predecessor. Please note the special rules for START=0, refer to [2].

If the list contains fewer elements in the given direction than required, the FSG returns the existing elements.

NOTE: In case of a ChangedArray message the parameter Data is either empty or contains only Pos information.

For the general specification of the array function class please refer to [2]. For the general description of the sliding window principle the reader is referred to [3].

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| OtherListType | Enum | Byte |
| OtherList_Reference | Number | Word |
| TotalNumListElements | Number | Word |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

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### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** Address_List (33, 0x21 )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, OtherListType, OtherList_Reference, ArrayHeader, |
| StatusArray | ASG_ID, TAID, OtherListType, OtherList_Reference, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 23.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

This parameter identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaneous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear (DF4.2) |

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| | |
|---|---|
| 0x4..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB | operating unit rear, to be evaluated by all ASGs (DF4.2) |
| 0xC..0xF | reserved |

## 23.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 23.3.) OtherListType [Enum]

**Type:** Enum

**Description:** OtherListType indicates the list to which the list entries are related to. OtherList_Reference refers to the corresponding list entry of OtherListType.

If the ASG wants to load details of the HomeAddress, it shall set OtherListType to 0x04. If only one home address is defined, then OtherListReference := 0x0000 and ArrayHeader.Elements := 1 for this use case.

If the ASG wants to load the complete list (consisting of all logical (partial) lists), it shall set OtherListType to 0x0F.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | LastDestinations_List (function 0x1D) |
| 0x01 | FavoriteDestinations_List (function 0x1E) |
| 0x02 | NavBook (function 0x20) |
| 0x03 | PhoneBook (BAP function catalogue Phone) |
| 0x04 | HomeAddress (DF4.1) |
| 0x05 | BusinessAddress (DF4.6) |
| 0x06..0x0E | reserved |
| 0x0F | completeList (DF4.1) |
| 0x10..0xFF | reserved |

## 23.4.) OtherList_Reference [Number]

**Type:** Number

**Description:** The OtherList_Reference refers to a list entry (POS value) of OtherListType.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 23.5.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of current list elements. In most cases this number is greater than the number of elements stored in the ASG.
This information may be used for dimensioning of scroll bars etc.

Value 0xFFFF (65535) means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

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**Step:** 1

**Unit:** unit_none

## 23.6.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_23_6

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Word |
| Elements | Number | Word |

### *Detailed Sub Parameter Overview*

## 23.6.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Direction = backward | Direction = forward |
| 2 | Transmit_Pos = Pos is transmitted | Transmit_Pos = Pos is not transmitted |
| 3 | IndexSize = 16 Bit for Start, Elements, Pos | IndexSize = 8 Bit for Start, Elements, Pos |

## 23.6.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records.
For detailed specification see [2].

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | LastName, FirstName, Street, City, Region, PostalCode, Country, Coordinates, POI_Description, POI_Type, Address_Type |
| 0x1 | Street, City, PostalCode, Country, Coordinates, Address_Type |
| 0x2 | reserved |
| 0x3 | Street, City, Region, PostalCode, Country, Coordinates, POI_Description, POI_Type, Address_Type |
| 0x4 | Street, City, PostalCode, Coordinates, Address_Type |
| 0x5 | POI_Description, POI_Type |
| 0x6 | Street, City |
| 0x7..0xE | reserved |
| 0xF | Pos |

## 23.6.3.) Start [Number]

**Type:** Number

**Description:** For detailed specification see [2].
Parameter Start is the first array element the message relates to. Parameter length depends

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on the Mode.IndexSize (Bit3):
IndexSize=0 -> 8 Bit, IndexSize=1 -> 16 Bit.

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 23.6.4.) Elements [Number]

**Type:** Number

**Description:** For detailed specification see document "BAP Datentypen" [2]!
Parameter Elements contains the number of required array elements. Parameter length is depending on the Bit IndexSize in Mode: IndexSize=0 -> 8 Bit, IndexSize=1 -> 16 Bit.

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 23.7.) Data [Record]

**Type:** Record

**Description:** "Destination" contains a description of the destination.
MaxElements = 2. This means, a StatusArray message is limited to 2 complete elements.
Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 65534 elements. The FSG is not allowed to generate value 0x00 or 0x0000 as a valid POS value.
The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 65534

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**Subparameter von:** 1_23_7

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Word |
| LastName | String | ByteStream |
| FirstName | String | ByteStream |
| Street | String | ByteStream |
| City | String | ByteStream |
| Region | String | ByteStream |
| PostalCode | String | ByteStream |
| Country | String | ByteStream |
| Coordinates | String | ByteStream |
| POI_Description | String | ByteStream |
| POI_Type | Enum | Byte |
| Address_Type | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 23.7.1.) Pos [Number]

**Type:** Number

**Description:** Pos

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 23.7.2.) LastName [String]

**Type:** String

**Description:** String contains the last name of the contact.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length

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information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.3.) FirstName [String]

**Type:** String

**Description:** String contains the first name of the contact.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.4.) Street [String]

**Type:** String

**Description:** String contains the street.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.5.) City [String]

**Type:** String

**Description:** String contains the city.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length

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information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.6.) Region [String]

**Type:** String

**Description:** String contains the region or state.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.7.) PostalCode [String]

**Type:** String

**Description:** String contains the Postal code.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 21 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 22

## 23.7.8.) Country [String]

**Type:** String

**Description:** String contains the country.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length

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information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.9.) Coordinates [String]

**Type:** String

**Description:** String contains the geographic (nautical) coordinates.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.10.) POI_Description [String]

**Type:** String

**Description:** String contains the POI descriptive string (textual description of the POI).

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 60 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 61

## 23.7.11.) POI_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates, if the destination is a POI (Point Of Interest) and the POI type.
If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Empty,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no POI |
| 0x01 | Accommodation |
| 0x02 | Airport |
| 0x03 | all restaurants |
| 0x04 | Amusement Park |
| 0x05 | Apparel |
| 0x06 | ATM (cashpoint) |
| 0x07 | Attraction |
| 0x08 | Automobile Club |
| 0x09 | Banking and Shopping |
| 0x0A | Banking |
| 0x0B | Banquet-Halls |
| 0x0C | Bars and Lounge |
| 0x0D | Boating |
| 0x0E | Bookstore |
| 0x0F | Border Cross |
| 0x10 | Bowling |
| 0x11 | Broadcasting |
| 0x12 | Business and Service |
| 0x13 | Business |
| 0x14 | Busstation |
| 0x15 | Camping |
| 0x16 | Car and Travel |
| 0x17 | Car Track |
| 0x18 | Car Train |
| 0x19 | Car Service |
| 0x1A | Car Wash |
| 0x1B | Casino |
| 0x1C | Cemetry |
| 0x1D | Cityhall |
| 0x1E | coffee |
| 0x1F | Commuter Railstaion |
| 0x20 | Cleaning Launch |
| 0x21 | Clothing Store |
| 0x22 | Communication |

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| | |
|---|---|
| 0x23 | Community Center |
| 0x24 | Computer Hardware-Software |
| 0x25 | Convenience Store |
| 0x26 | Convention Center |
| 0x27 | Corporations |
| 0x28 | Corporations Public Places |
| 0x29 | Culture Facility |
| 0x2A | Court |
| 0x2B | Department Store |
| 0x2C | Dining and Entertainment |
| 0x2D | Dinning Shopping |
| 0x2E | Discount Store |
| 0x2F | Education |
| 0x30 | Electronics |
| 0x31 | Embassy |
| 0x32 | Emergency |
| 0x33 | Emergency and Public |
| 0x34 | Entertainment |
| 0x35 | Fast Food |
| 0x36 | Ferry |
| 0x37 | Fire Department |
| 0x38 | Flooring and Carpeting |
| 0x39 | Flower and Jewelry |
| 0x3A | Food and Beverage |
| 0x3B | Funeral Director |
| 0x3C | Furniture |
| 0x3D | Games, Music and Video |
| 0x3E | Garden and other |
| 0x3F | Gasstation |
| 0x40 | Gasstation DieselGasstation LPG |
| 0x41 | General |
| 0x42 | Gifts and Leisure |
| 0x43 | Gifts- Antiques and Arts |
| 0x44 | Glass and Window |
| 0x45 | Golfing |
| 0x46 | Goverment |
| 0x47 | Grocery Stores |
| 0x48 | Guesthouse |
| 0x49 | Hair and Beauty |

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| | |
|---|---|
| 0x4A | Harbor |
| 0x4B | Hardware and Lumber |
| 0x4C | Health and Fitness |
| 0x4D | Health- Care |
| 0x4E | Highrise- BLDG |
| 0x4F | Home and Improvment |
| 0x50 | Home Center |
| 0x51 | Home Service |
| 0x52 | Hospital |
| 0x53 | Hotel / Motel |
| 0x54 | Iceskating |
| 0x55 | Insurance |
| 0x56 | Karaokay |
| 0x57 | Library |
| 0x58 | LIQ-Gas-Station |
| 0x59 | Major-Appliance |
| 0x5A | Mens-Apparel |
| 0x5B | Monument |
| 0x5C | Motorway Exit |
| 0x5D | Motorway Cross |
| 0x5E | Motorway Service |
| 0x5F | Movie-Theater |
| 0x60 | Moving and Storage |
| 0x61 | Museum |
| 0x62 | Nightclubs |
| 0x63 | Optical |
| 0x64 | Restaurant |
| 0x65 | Park and Fitness |
| 0x66 | Park and Ride |
| 0x67 | Parking |
| 0x68 | Parking Garage |
| 0x69 | Parks |
| 0x6A | Performing-Arts |
| 0x6B | Personal- Service |
| 0x6C | Pharmacies |
| 0x6D | Photographer |
| 0x6E | Place or workship |
| 0x6F | Police Station |
| 0x70 | Post Office |

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| | |
|---|---|
| 0x71 | Public Places |
| 0x72 | Publisher |
| 0x73 | Railway Station |
| 0x74 | Recreation |
| 0x75 | Rentel Service |
| 0x76 | Repair |
| 0x77 | Rest-Area |
| 0x78 | Retirement Nursing |
| 0x79 | Search |
| 0x7A | Service |
| 0x7B | Shoe Stores |
| 0x7C | Shopping Center |
| 0x7D | Skiing |
| 0x7E | Social Service |
| 0x7F | Speciality-Clothing |
| 0x80 | Speciality-Store |
| 0x81 | Speciality-Food |
| 0x82 | Sport-Goods |
| 0x83 | Sports |
| 0x84 | Sport Airport |
| 0x85 | Sports-Complex |
| 0x86 | Tailor |
| 0x87 | Tax-Service |
| 0x88 | Tool-Booth |
| 0x89 | Tourist Attraction |
| 0x8A | Tourist Information |
| 0x8B | Train Station |
| 0x8C | Transportation |
| 0x8D | Travel |
| 0x8E | Travel-Agent |
| 0x8F | University |
| 0x90 | Utilities |
| 0x91 | Variety-Store |
| 0x92 | Wine and Liquor |
| 0x93 | Womens Apparel |
| 0x94 | Winery |
| 0x95 | car repair shop - VW |
| 0x96 | car repair shop - Audi |
| 0x97 | car repair shop - Seat |

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| | |
|---|---|
| 0x98 | car repair shop - Skoda |
| 0x99 | car repair shop - VW Nutzfahrzeuge |
| 0x9A | car repair shop - Bentley |
| 0x9B | car repair shop - Bugatti |
| 0x9C | car repair shop - Lamborghini |
| 0x9D | car repair shop - Scania |
| 0x9E | car repair shop - MAN |
| 0x9F | car repair shop - Porsche |
| 0xA0 | electrical service station |
| 0xA1 | Diesel fuel station (DF4.1) |
| 0xA2 | CNG fuel station (DF4.1) |
| 0xA3 | rest area ("Autohof") (DF4.1) |
| 0xA4 | rest area with WC (DF4.1) |
| 0xA5 | motorway service area ("Raststaette") (DF4.1) |
| 0xA6..0xAF | reserved |
| 0xB0 | Predictive NAV (DF4.6) |
| 0xB1..0xFE | reserved |
| 0xFF | unknown |

## 23.7.12.) Address_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates the address type.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | private |
| 0x01 | business |
| 0x02 | business - parcel address |
| 0x03..0xFE | reserved |
| 0xFF | unknown |

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## 23.8.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved, refer to annex |

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# 24.) RG_ActDeact ( 34, 0x22 )

**Type:** Method

**Description:** This function is used to start and stop/abort a route guidance.
If a route guidance is started, the wanted destination is linked by the Pos information either from the LastDestinations_List or from the FavoriteDestinations_List.
Please note RouteGuidance_ActDeact affects the function RG_Status (0x11).

RG_ActDeact:.. RouteGuidance_ActivationDeactivation

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| ControlType | Enum | Nibble |
| RG_ActDeact_Result | Enum | Byte |
| Reserve1 | Number | Byte |
| CI_Type | Enum | Nibble |
| ControlInformation | Number | Word |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** RG_ActDeact (34, 0x22 )

| OpCode | Associated Parameters |
|---|---|
| Abort | NoParam, |
| StartResult | ControlType, CI_Type, ControlInformation, |
| Result | RG_ActDeact_Result, |
| Processing | Reserve1, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 24.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Abort,

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**Base Datatype:** Empty

## 24.2.) ControlType [Enum]

**Type:** Enum

**Description:** (upper nibble)

Indicates if the route guidance shall be started or stopped.
The FSG shall handle reserved values like value 0x1 ("stop route guidance").

**OpCodes:** StartResult,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | start route guidance |
| 0x1 | stop route guidance |
| 0x2 | suspend route guidance (DF4.1) |
| 0x3 | resume route guidance (DF4.1) |
| 0x4 | add stopover to active RG (DF4.5) |
| 0x5 | replace current destination (DF4.5) |
| 0x6..0xF | reserved |

## 24.3.) RG_ActDeact_Result [Enum]

**Type:** Enum

**Description:** Indicates the result of the method.
If RouteGuidance cannot be started because the ControlInformation does not match to a stored destination in the FSG, the method ends immediately. (FSG sends result message.)
If a RouteGuidance cannot be stoped because no RG is active, the method ends immediately.
If RG is deactivated when no RG is active, the FSG shall return "RG not active" (0x04).
If RG shall be activated when the RG is already active, the FSG shall return "RG already active" (0x05). The already active RG shall not be influenced in this case.

If the ControlType is set to a reserved value in a StartResult message, the FSG shall send an Error message with ErrorCode set to 0x41 (out of range).

**OpCodes:** Result,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | successful |
| 0x01 | not successful |
| 0x02 | abort successful |
| 0x03 | abort not successful |
| 0x04 | RG not active |
| 0x05 | RG already active |
| 0x06 | not successful - ControlInformation does not match to FSG internal destination (RG not started!) |
| 0x07 | not successful - destination address needs to be refined/detailed at head unit |
| 0x08 | not successful - home address not specified in FSG |
| 0x09 | not successful - select Route at FSG |
| 0x0A | not successful - select home address at FSG (several home addresses available) (DF4.1) |
| 0x0B | not successful - no navigation data (map data) available (DF4.1) |
| 0x0C | not successful - no semi-dynamic route available (DF4.1) |
| 0x0D | not successful - select semi-dynamic route at FSG (several semi-dynamic routes available) (DF4.1) |
| 0x0E | not successful - business address not specified in FSG (DF4.5) |
| 0x0F..0xFF | reserved |

## 24.4.) Reserve1 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value. The FSG shall set this parameter to "0".

**OpCodes:** Processing,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

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## 24.5.) CI_Type [Enum]

**Type:** Enum

**Description:** (lower nibble)

This parameter contains information about type of parameter ControlInformation.
CI_Type:.. ControlInformation Type

If the current route guidance is to be stopped, suspended or resumed, parameter CI_Type shall be set to 0x00 "invalid" by the ASG.

If a route guidance to the home address shall be started, the CI_Type shall be set to 0x4 ("home") and parameter "ControlInformation" shall be set to 0x0000. If more than one home address is defined, ControlInformation is used to select a particular address.

**OpCodes:** StartResult,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | invalid |
| 0x1 | LastDestinations_List (function 0x1D) |
| 0x2 | FavoriteDestinations_List (function 0x1E) |
| 0x3 | TEL addressbook (default address) |
| 0x4 | home (home address) |
| 0x5 | NavBook (function 0x20) |
| 0x6 | semidynamicRoute (DF4.1) |
| 0x7 | POI_List (DF4.1) |
| 0x8 | business (business adress) (DF4.5) |
| 0x9..0xF | reserved |

## 24.6.) ControlInformation [Number]

**Type:** Number

**Description:** If a route guidance is started, this parameter is to be set to the Pos value taken from function LastDest_List, FavoriteDest_List, Address_List or the PhoneBook. Please refer to description of parameter CI_Type for the route guidance to the home address.
If a route guidance is stopped, suspended, started as a semidaynamic route or resumed, this parameter is to be set to 0x0000.

**OpCodes:** StartResult,

**Base Datatype:** Word

**Range Min:** 0

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**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 24.7.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0x4F | reserved |
| 0x50 | AppErr_MethodAborted |
| 0x51..0xFF | reserved |

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# 25.) RepeatLastNavAnnouncement ( 35, 0x23 )

**Type:** Method

**Description:** This function is used for repeating the last navigation announcement.
The FSG shall send the Result message as soon as the announcement repetition has finished.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| RepeatLNA_Result | Enum | Byte |
| Reserve1 | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** RepeatLastNavAnnouncement (35, 0x23 )

| OpCode | Associated Parameters |
|---|---|
| StartResult | NoParam, |
| Abort | NoParam, |
| Result | RepeatLNA_Result, |
| Processing | Reserve1, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 25.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** StartResult, Abort,

**Base Datatype:** Empty

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## 25.2.) RepeatLNA_Result [Enum]

**Type:** Enum

**Description:** Indicates the result of the method.

**OpCodes:** Result,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | successful |
| 0x01 | not successful |
| 0x02 | abort successful |
| 0x03 | abort not successful |
| 0x04..0xFF | reserved |

## 25.3.) Reserve1 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value. The FSG shall set this parameter to "0".

**OpCodes:** Processing,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 25.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

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**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x4F | reserved, refer to annex |
| 0x50 | AppErr_MethodAborted |
| 0x51..0xFF | reserved |

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# 26.) VoiceGuidance ( 36, 0x24 )

**Type:** Property

**Description:** This function is used for enabling and disabling the navigation announcements (so-called "voice guidance").

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| VoiceGuidance_State | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** VoiceGuidance (36, 0x24 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | VoiceGuidance_State, |
| Status | VoiceGuidance_State, |
| HeartbeatStatus | VoiceGuidance_State, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 26.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 26.2.) VoiceGuidance_State [Enum]

**Type:** Enum

**Description:** Contains the voice guidance state.

**OpCodes:** SetGet, Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | VoiceGuidance ON (full / complete announcements) |
| 0x01 | VoiceGuidance OFF |
| 0x02 | VoiceGuidance ON (reduced announcements / reduced mode) |
| 0x03 | VoiceGuidance ON (reduced announcement traffic) (DF4.1) |
| 0x04..0xFF | reserved |

## 26.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 27.) FunctionSynchronisation ( 37, 0x25 )

**Type:** Property

**Description:** This function this used to indicate the need of synchronous HMI update of certain functions. This function is brand specific.
The content of a function should not be evaluated by the ASG, if the corresponding bit is set to "1". If the bit associated to a certain function is 0", the function shall be evaluated immediately.

The encoding of this function is the same as used for function FunctionList (0x03).
If this function is not contained in the FunctionList function (0x03), the ASG shall act as if the value (0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00) has been received (all bits of this parameter set to 0).

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| FctList_1 | Bitfield | ByteStream |
| FctList_2 | Bitfield | ByteStream |
| FctList_3 | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** FunctionSynchronisation (37, 0x25 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | FctList_1, FctList_2, FctList_3, |
| HeartbeatStatus | FctList_1, FctList_2, FctList_3, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 27.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 27.2.) FctList_1 [Bitfield]

**Type:** Bitfield

**Description:** This bit field indicates the synchronisation need of a certain function for functions 0x0..0x2F.

For the encoding of this function the reader is referred to BAP_Datentypen [2]. The encoding complies to teh same rules as function FunctionList.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**6

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0..15 | reserved | reserved |
| 16 | Fct CompassInfo is to be synchronised | Fct CompassInfo is not to be synchronised |
| 17 | Fct RG_Status is to be synchronised | Fct RG_Status is not to be synchronised |
| 18 | Fct DistanceToNextManeuver is to be synchronised | Fct DistanceToNextManeuver is not to be synchronised |
| 19 | Fct CurrentPositionInfo is to be synchronised | Fct CurrentPositionInfo is not to be synchronised |
| 20 | Fct TurnToInfo is to be synchronised | Fct TurnToInfo is not to be synchronised |
| 21 | Fct DistanceToDestination is to be synchronised | Fct DistanceToDestination is not to be synchronised |
| 22 | Fct TimeToDestination is to be synchronised | Fct TimeToDestination is not to be synchronised |
| 23 | Fct ManeuverDescriptor is to be synchronised | Fct ManeuverDescriptor is not to be synchronised |
| 24 | reserved | Fct LaneGuidance is not to be synchronised |
| 25 | Fct TMCinfo is to be synchronised | Fct TMCinfo is not to be synchronised |
| 26 | Fct MagnetFieldZone is to be synchronised | Fct MagnetFieldZone is not to be synchronised |
| 27 | Fct Calibration is to be synchronised | Fct Calibration is not to be synchronised |
| 28 | Fct ASG_Capabilities is to be synchronised | Fct ASG_Capabilities is not to be synchronised |
| 29 | reserved | Fct LastDest_List is not to be synchronised |
| 30 | reserved | Fct FavoriteDest_List is not to be |

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| | | |
|---|---|---|
| | | synchronised |
| 31 | Fct PreferredDest_List is to be synchronised | Fct PreferredDest_List is not to be synchronised |
| 32 | reserved | Fct NavBook is not to be synchronised |
| 33 | reserved | Fct Address_List is not to be synchronised |
| 34 | reserved | Fct RG_ActDeact is not to be synchronised |
| 35 | reserved | Fct RepeatLastNavAnnouncement is not to be synchronised |
| 36 | Fct VoiceGuidance is to be synchronised | Fct VoiceGuidance is not to be synchronised |
| 37 | reserved | Fct FunctionSynchronisation is not to be synchronised |
| 38 | Fct InfoStates is to be synchronised | Fct InfoStates is not to be synchronised |
| 39 | Fct ActiveRgType is to be synchronised | Fct ActiveRgType is not to be synchronised |
| 40 | Fct TrafficBlock_Indication is to be synchronised | Fct TrafficBlock_Indication is not to be synchronised |
| 41 | reserved | Fct GetNextListPos is not to be synchronised |
| 42 | reserved | Fct NbSpeller is not to be synchronised |
| 43 | Fct MapColorAndType is to be synchronised | Fct MapColorAndType is not to be synchronised |
| 44 | Fct MapViewAndOrientation is to be synchronised | Fct MapViewAndOrientation is not to be synchronised |
| 45 | Fct MapScale is to be synchronised | Fct MapScale is not to be synchronised |
| 46 | Fct DestinationInfo is to be synchronised | Fct DestinationInfo is not to be synchronised |
| 47 | Fct Altitude is to be synchronised | Fct Altitude is not to be synchronised |

## 27.3.) FctList_2 [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains information about function synchronisation for functions 48 (0x30) to 55 (0x37). Reserved bis shall be set to FALSE (0) by the FSG.

The binary encoding is according to the same rules as for function FunctionList (0x03), refer to [2].

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Fct OnlineNavigationState is to be synchronised | Fct OnlineNavigationState is not to be synchronised |
| 1 | Fct ExitView is to be synchronised | Fct ExitView is not to be synchronised |
| 2 | Fct SemidynamicRouteGuidance is to be synchronised | Fct SemidynamicRouteGuidance is not to be synchronised |
| 3 | reserved | Fct POI_Search is not to be synchronised |
| 4 | reserved | Fct POI_List is not to be synchronised |
| 5 | Fct FSG_Setup is to be synchronised | Fct FSG_Setup is not to be synchronised |
| 6 | Fct MapPresentation is to be synchronised | Fct MapPresentation is not to be synchronised |
| 7 | Fct ManeuverState is to be synchronised | Fct ManeuverState is not to be synchronised |

## 27.4.) FctList_3 [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains information about function synchronisation for functions 56 (0x38) to 63 (0x3F). In this version of the specifiaction, the FSG shall set all bits of this bitfield to FALSE (0).

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Fct ETC_Status is to be synchronised | Fct ETC_Status is not to be synchronised |
| 1 | Fct MapContentSettings is to be synchronised | Fct MapContentSettings is not to be synchronised |
| 2 | reserved | Fct VideoStreams is not to be synchronised |
| 3 | reserved | Fct RequestSync_Video is not to be synchronised |
| 4 | Fct DistanceToDestinationExtended is to be synchronised | Fct DistanceToDestinationExtended is not to be synchronised |
| 5 | reserved | Fct LaneGuidance2 is not to be |

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| | | |
|---|---|---|
| | | synchronised |
| 6 | reserved | Fct "Picture" is not to be synchronised |
| 7 | reserved | reserved |

## 27.5.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 28.) InfoStates ( 38, 0x26 )

**Type:** Property

**Description:** This function delivers additional information which may be displayed (project dependent).
If this function is not supported by the FSG (indicated in function FunctionList (0x03)) the ASG shall act as if States = "no error/no information" was received.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| States | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** InfoStates (38, 0x26 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | States, |
| HeartbeatStatus | States, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 28.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 28.2.) States [Enum]

**Type:** Enum

**Description:** If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no error/no information |
| 0x01 | no navigation data medium inserted |
| 0x02 | navigation database corrupted |
| 0x03 | no GPS signal available |
| 0x04 | navigation database update ongoing/in progress (DF4.1) |
| 0x05 | Initializing MOST Map (DF4.2) |
| 0x06 | Navigation in mobile device active (DF 4.3) |
| 0x07 | No navigation database available (DF4.6) |
| 0x08..0xFE | reserved |
| 0xFF | unknown |

## 28.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 29.) ActiveRgType ( 39, 0x27 )

**Type:** Property

**Description:** This function contains the kind of route guidance (to be) used by the ASG. Depending on the ASG capabilities and the system setup different route guidance presentations are possible.

The STATUS message shall contain the route guidance being used by the ASG. The FSG may alter this value using Status.

If this function is not supported by the FSG, the ASG shall apply route guidance presentation according to the ASG HMI definitions.

## *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| RGType | Enum | Byte |
| ErrorCode | Enum | Byte |

## *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** ActiveRgType (39, 0x27 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | RGType, |
| Status | RGType, |
| HeartbeatStatus | RGType, |
| Error | ErrorCode, |

## *Detailed Parameter Overview*

### 29.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 29.2.) RGType [Enum]

**Type:** Enum

**Description:** This parameter contains the source of route guidance presentation / the way of route guidance presentation to be used by the ASG.

If a reserved value is received by the ASG, it shall act as if it has received the value 0xFF.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | RGI (from BAP function ManeuverDescriptior) |
| 0x01 | MOST - KDK (data stream for JZ/EV received via MOST FBLOCK) |
| 0x02 | compass (from BAP function CompassInfo) |
| 0x03 | MOST - Map (data stream received via MOST FBLOCK) (DF4.1) |
| 0x04 | LVDS - Map (video stream received via LVDS) (DF4.1) |
| 0x05 | LVDS - KDK (video stream received via LVDS) (DF4.3) |
| 0x06..0xFE | reserved |
| 0xFF | apply ASG default route guidance presentation |

## 29.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 30.) TrafficBlock_Indication ( 40, 0x28 )

**Type:** Property

**Description:** This function is used to indicate a traffic disturbance (congestion, accident etc.) on the current street or route.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| TMC_Symbol | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** TrafficBlock_Indication (40, 0x28 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | TMC_Symbol, |
| HeartbeatStatus | TMC_Symbol, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 30.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 30.2.) TMC_Symbol [Enum]

**Type:** Enum

**Description:** This parameter contains a TMC symbol to be used as congestion or danger indication for the driver.
If the ASG receives a reserved value, it shall act as if the value 0x00 has been received.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no TMC symbol |
| 0x01 | general danger (part 1) |
| 0x02 | general danger (part 2) |
| 0x03 | general danger (part 3) |
| 0x04 | road works |
| 0x05 | stationary traffic / jam |
| 0x06 | slow / heavy traffic |
| 0x07 | road narrowed (road width reduced) (on both sides) |
| 0x08 | snowfall |
| 0x09 | slippery road |
| 0x0A | heavy winds / storm |
| 0x0B | rockfall, landslip |
| 0x0C | people on the road |
| 0x0D | biker on the road |
| 0x0E | animals on road |
| 0x0F | smog |
| 0x10 | closed road |
| 0x11 | overtake restriction |
| 0x12 | end of overtake restriction |
| 0x13 | end of all restrictions |
| 0x14 | snow chains mandatory |
| 0x15 | delay |
| 0x16 | road narrowed on left hand side (road width reduced) (DF4.1) |
| 0x17 | road narrowed on right hand side (road width reduced) (DF4.1) |
| 0x18..0xFE | reserved |
| 0xFF | unknown TMC symbol |

## 30.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 31.) GetNextListPos ( 41, 0x29 )

**Type:** Method

**Description:** This function may be used by the ASG to request any subsequent ID with a certain offset. Deviating from the usual array handling, the ASG requests not the next or previous adjacent list element but the ID of any offset element related to a reference (anchor) element ID (Pos).

## *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| GetNextListPos_Result | Enum | Byte |
| Reserve1 | String | ByteStream |
| currentPos | Number | Word |
| Offset | Number | Word |
| nextPos | Number | Word |
| ListType | Enum | Byte |
| absoluteListPos | Number | Word |
| ErrorCode | Enum | Byte |

## *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** GetNextListPos (41, 0x29 )

| OpCode | Associated Parameters |
|---|---|
| Abort | NoParam, |
| Result | GetNextListPos_Result, currentPos, nextPos, absoluteListPos, |
| Processing | Reserve1, |
| StartResult | currentPos, Offset, ListType, |
| Error | ErrorCode, |

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## *Detailed Parameter Overview*

### 31.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Abort,

**Base Datatype:** Empty

### 31.2.) GetNextListPos_Result [Enum]

**Type:** Enum

**Description:** Indicates the result of the method.

**OpCodes:** Result,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | successful |
| 0x01 | not successful |
| 0x02 | abort successful |
| 0x03 | abort not successful |
| 0x04..0xFF | reserved |

### 31.3.) Reserve1 [String]

**Type:** String

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value.
The FSG shall set this parameter to 0x00 (empty string, refer to [2] for its definition).

**OpCodes:** Processing,

**Base Datatype:** ByteStream

**Length:** 7

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## 31.4.) currentPos [Number]

**Type:** Number

**Description:** This parameter contains the base Pos the request is related to.
The ASG shall always use 16bit-values for "currentPos" even if the specific list requires 8bit-values only.

**OpCodes:** StartResult, Result,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65534

**Step:** 1

**Unit:** unit_none

## 31.5.) Offset [Number]

**Type:** Number

**Description:** This parameter contains the requested offset related to currentPos. The offset is a cardinal offset to support arbitrary and consecutive Pos values. Please consider the sign of this parameter. Negative values are allowed for parameter Offset (indicating reverse direction).
The ASG shall always use 16bit-values for "Offset" even if the specific list requires 8bit-values only.

**OpCodes:** StartResult,

**Base Datatype:** Word

**Range Min:** -32766

**Range Max:** 32767

**Step:** 1

**Unit:** unit_none

## 31.6.) nextPos [Number]

**Type:** Number

**Description:** This parameter contains the requested position information (Pos, element-ID).

The value 0xFFFF indicates, the requested Pos could not be determined by the FSG (negative response). In this case, the FSG shall set parameter GetNextListPos_Result to 0x01 ("not successful").

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**OpCodes:** Result,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 31.7.) ListType [Enum]

**Type:** Enum

**Description:** This parameter contains the list (from LSG "Navigation_SD") the request is related to.

**OpCodes:** StartResult,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | LastDestinations_List |
| 0x01 | FavoriteDestinations_List |
| 0x02 | NavBook |
| 0x03 | Address_List |
| 0x04 | POI_List (DF4.1) |
| 0x05..0xFF | reserved |

## 31.8.) absoluteListPos [Number]

**Type:** Number

**Description:** This parameter contains the absolute (cardinal) requested position information (Pos, element-ID).

The value 0xFFFF indicates, the requested absolute (cardinal) Pos could not be determined by the FSG (negative response). Please be aware, the FSG may set absoluteListPos to 0xFFFF even if the parameter GetNextListPos_Result is set to 0x00 ("successful").

**OpCodes:** Result,

**Base Datatype:** Word

**Range Min:** 0

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**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 31.9.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0x4F | reserved |
| 0x50 | AppErr_MethodAborted |
| 0x51..0xFF | reserved |

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# 32.) NbSpeller ( 42, 0x2A )

**Type:** Method

**Description:** This function realises a speller functionality for the NavBook function (0x20). Its usage depends on the ASG HMI.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Mode | Enum | Byte |
| Reserve1 | Number | Byte |
| NbSpeller_Result | Enum | Byte |
| SearchString | String | ByteStream |
| MatchingEntries | Number | Word |
| Reserve2 | Number | Long |
| Pos | Number | Word |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** NbSpeller (42, 0x2A )

| OpCode | Associated Parameters |
|---|---|
| Abort | NoParam, |
| StartResult | Mode, SearchString, |
| Processing | Reserve1, Reserve2, |
| Result | NbSpeller_Result, MatchingEntries, Pos, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 32.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Abort,

**Base Datatype:** Empty

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## 32.2.) Mode [Enum]

**Type:** Enum

**Description:** Describes the search functionality.
0x00: MatchSpeller;
0x01: search for next character in alphabetical order;
0x02: search for previous character in alphabetical order.

If the FSG receives a reserved value, it shall send an error message with ErrorCode := 0x41 (out of range).

**OpCodes:** StartResult,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | MatchSpeller |
| 0x01 | next character |
| 0x02 | previous character |
| 0x03..0xFF | reserved |

## 32.3.) Reserve1 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value. The FSG shall set this parameter to "0".

**OpCodes:** Processing,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 32.4.) NbSpeller_Result [Enum]

**Type:** Enum

**Description:** Indicates the result of the method.
If the FSG returns 0x01 ("not successful") or 0x02 ("abort successful"), it shall set

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MatchingEntries := 0x0000.
Please note, the FSG is allowed to set "NbSpeller_Result" to 0x00 ("successful") and "MatchingEntries" to 0x0000 in the same message! The ASG shall be able to cope with this value combination.

**OpCodes:** Result,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | successful |
| 0x01 | not successful |
| 0x02 | abort successful |
| 0x03 | abort not successful |
| 0x04..0xFF | reserved |

## 32.5.) SearchString [String]

**Type:** String

**Description:** Contains the UTF-8 encoded search string including one byte length information.

**OpCodes:** StartResult,

**Base Datatype:** ByteStream

**Length:** 51

## 32.6.) MatchingEntries [Number]

**Type:** Number

**Description:** Mode=0: delivers the number of entries exactly matching to SearchString.
0x0000 -> there is no entry stored in NavBook, that matches exactly (but MatchingEntries is supported in general).
0x0001-0xFFFD -> exact number of matching entries.
0xFFFE -> no matching entry while exact number of matching entries is not supported by FSG.
0xFFFF -> at least one matching entry while exact number of matching entries is not supported by FSG. In case of MatchingEntries=0x0000 or 0xFFFE, Pos refers to the next available NavBook entry in alphabetical order (e.g. if SearchString is 'b' and no entry with 'b' is stored in navigation book, Pos will refer to the first entry beginning with 'c').

**OpCodes:** Result,

**Base Datatype:** Word

**Range Min:** 0

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**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 32.7.) Reserve2 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value. The FSG shall set this parameter to "0".

**OpCodes:** Processing,

**Base Datatype:** Long

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 32.8.) Pos [Number]

**Type:** Number

**Description:** Delivers the Pos of the first NavBook entry (in alphabetical order) whose initial letter(s) match to SearchString in chosen direction
(Mode=0x00: forward & matching to SearchString;
Mode=0x01: forward & matching to next character;
Mode=0x02: backward & matching to previous character).

The ASG shall ignore the content of parameter "Pos" if MatchingEntries is set to 0x0000.

**OpCodes:** Result,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

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## 32.9.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0x4F | reserved, refer to annex |
| 0x50 | AppErr_MethodAborted |
| 0x51..0xFF | reserved |

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# 33.) MapColorAndType ( 43, 0x2B )

**Type:** Property

**Description:** This function contains the information of map colour and map type being shown in the instrument cluster.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Colour | Enum | Byte |
| ActiveMapType | Enum | Byte |
| MainMapSetup | Enum | Byte |
| SupportedMapTypes | Bitfield | ByteStream |
| Reserve | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** MapColorAndType (43, 0x2B )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | Colour, ActiveMapType, MainMapSetup, Reserve, |
| Status | Colour, ActiveMapType, MainMapSetup, SupportedMapTypes, |
| HeartbeatStatus | Colour, ActiveMapType, MainMapSetup, SupportedMapTypes, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 33.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 33.2.) Colour [Enum]

**Type:** Enum

**Description:** This parameter contains the current color of map design.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | day |
| 0x01 | night |
| 0x02 | auto |
| 0x03..0xFF | reserved |

## 33.3.) ActiveMapType [Enum]

**Type:** Enum

**Description:** This parameter contains the currently active map type.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | destination |
| 0x01 | position 2D |
| 0x02 | position 3D |
| 0x03 | overview |
| 0x04 | range map |
| 0x05..0xFF | reserved |

## 33.4.) MainMapSetup [Enum]

**Type:** Enum

**Description:** This parameter contains information about the current location of the main map.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | init (no map in ASG) |
| 0x01 | main map in ASG (primary map in ASG / instrument cluster) |
| 0x02 | main map in FSG (secondary map in ASG / instrument cluster) |
| 0x03..0xFE | reserved |
| 0xFF | not supported |

## 33.5.) SupportedMapTypes [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains the supported map types.
Bit0 informs about adjustability of the high resolution map presentation. The high resultion map is refered to as "main map". The "main map" is either presented in BAP ASG or BAP FSG.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | MainMapSetup is supported | MainMapSetup is not supported |
| 1 | destination map is supported | destination map is not supported |
| 2 | position 2D map is supported | position 2D map is not supported |
| 3 | position 3D map is supported | position 3D map is not supported |
| 4 | overview map is supported | overview map is not supported |
| 5 | range map is supported | range map is not supported |
| 6..7 | reserved | reserved |

## 33.6.) Reserve [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value. The FSG shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

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**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 33.7.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 34.) MapViewAndOrientation ( 44, 0x2C )

**Type:** Property

**Description:** This function contains the information of supported map views, the currently active map view and map orientation in the instrument cluster.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| MapView | Enum | Byte |
| SupplementaryMapView | Enum | Byte |
| SupportedMapViews | Bitfield | ByteStream |
| Reserve1 | Number | Byte |
| SupportedSupplementaryMapViews | Bitfield | ByteStream |
| Reserve2 | Number | Byte |
| MapVisibility | Bitfield | ByteStream |
| MapOrientation | Enum | Byte |
| Reserve3 | Number | Byte |
| Modification | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** MapViewAndOrientation (44, 0x2C )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | MapView, SupplementaryMapView, Reserve1, Reserve2, MapVisibility, MapOrientation, Reserve3, |
| Status | MapView, SupplementaryMapView, SupportedMapViews, SupportedSupplementaryMapViews, MapVisibility, MapOrientation, Modification, |
| HeartbeatStatus | MapView, SupplementaryMapView, SupportedMapViews, SupportedSupplementaryMapViews, MapVisibility, MapOrientation, Modification, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

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## 34.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 34.2.) MapView [Enum]

**Type:** Enum

**Description:** This parameter contains the current map view (map being presented in the ASG).

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | standard |
| 0x01 | google earth |
| 0x02 | traffic |
| 0x03..0xFF | reserved |

## 34.3.) SupplementaryMapView [Enum]

**Type:** Enum

**Description:** This parameter contains the type of the supplementary map view

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no supplementary map view |
| 0x01 | Intersection zoom |
| 0x02 | compass |
| 0x03 | 3+1 Box |
| 0x04..0xFF | reserved |

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## 34.4.) SupportedMapViews [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains the supported map views.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | standard map is supported | standard map is not supported |
| 1 | GoogleEarth map is supported | GoogleEarth map is not supported |
| 2 | traffic map is supported | traffic map is not supported |
| 3..7 | reserved | reserved |

## 34.5.) Reserve1 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The FSG shall ignore this value. The ASG shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 34.6.) SupportedSupplementaryMapViews [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains the information about the supported supplementary map views.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

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**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Intersection zoom supported | Intersection zoom not supported |
| 1 | compass supported | compass not supported |
| 2 | 3+1 Box supported | 3+1 Box not supported |
| 3..7 | reserved | reserved |

## 34.7.) Reserve2 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The FSG shall ignore this value. The ASG shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 34.8.) MapVisibility [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains the information about the visibility of the map.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | LVDS map is visible | LVDS map is not visible |
| 1 | supplementary map view is visible | supplementary map view is not visible |
| 2 | standard map is visible | standard map is not visible |
| 3..7 | reserved | reserved |

## 34.9.) MapOrientation [Enum]

**Type:** Enum

**Description:** This parameter contains the current map orientation (map being presented in the ASG).

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | automatic |
| 0x01 | north |
| 0x02 | direction of travel |
| 0x03..0xFF | reserved |

## 34.10.) Reserve3 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The FSG shall ignore this value. The ASG shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

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## 34.11.) Modification [Bitfield]

**Type:** Bitfield

**Description:** The FSG indicates that the functions can temporarily not be changed.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | google map can not be modified | google map can be modified |
| 1..7 | reserved | reserved |

## 34.12.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 35.) MapScale ( 45, 0x2D )

**Type:** Property

**Description:** This function contains the information of current map scale.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Steps | Number | Byte |
| Reserve1 | Number | Byte |
| AutoZoom | Enum | Nibble |
| AutoZoomState | Bitfield | Nibble |
| Scale | Number | Word |
| Reserve2 | Number | Word |
| Unit | Enum | Byte |
| Reserve3 | Number | Byte |
| SupportedAutoZoom | Bitfield | ByteStream |
| Reserve4 | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** MapScale (45, 0x2D )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | Steps, AutoZoom, AutoZoomState, Reserve2, Reserve3, Reserve4, |
| Status | Reserve1, AutoZoom, AutoZoomState, Scale, Unit, SupportedAutoZoom, |
| HeartbeatStatus | Reserve1, AutoZoom, AutoZoomState, Scale, Unit, SupportedAutoZoom, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 35.1.) NoParam [Void]

**Type:** Void

**Description:**

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**OpCodes:** Get,

**Base Datatype:** Empty

## 35.2.) Steps [Number]

**Type:** Number

**Description:** This parameter contains the steps of increment (+) or decrement (-) the map scale.
A positive tick at the operating device (e.g. MFL or rotary encoder) results in an increment by 1; a negative tick results in a decrement by 1.

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** -128

**Range Max:** 127

**Step:** 1

**Unit:** unit_none

## 35.3.) Reserve1 [Number]

**Type:** Number

**Description:** This parameter is just used for symmetric reasons (1 byte dummy parameter).
The receiver shall ignore/skip this parameter. The sender shall set this parameter to "0".

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 35.4.) AutoZoom [Enum]

**Type:** Enum

**Description:** (upper nibble)
Enables and disables the map automatic zoom functionality.
If autozoom is switched ON, the paramter Steps has no effect.

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**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | AutoZoom OFF |
| 0x1 | AutoZoom ON |
| 0x2 | AutoZoom ON for Intersection |
| 0x3..0xE | reserved |
| 0xF | AutoZoom setting not supported |

## 35.5.) AutoZoomState [Bitfield]

**Type:** Bitfield

**Description:** (lower nibble)

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | auto zoom active | auto zoom not active |
| 1..3 | reserved | reserved |

## 35.6.) Scale [Number]

**Type:** Number

**Description:** This parameter contains current map scale.
0x0000: invalid
0x0001..0xFFFD: valid, values: 0.1...6553.3, step: 0.1
0xFFFE: maximum value
0xFFFF: init / unknown

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Word

**Range Min:** 0.1

**Range Max:** 6553.3

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**Step:** 0.1

**Unit:** unit_none

**EnumValues:**

| Code | Description |
|---|---|
| 0x0000 | invalid |
| 0xFFFE | maximum value |
| 0xFFFF | init / unknown |

## 35.7.) Reserve2 [Number]

**Type:** Number

**Description:** This parameter is just used for symmetric reasons (2 byte dummy parameter). The receiver shall ignore/skip this parameter. The sender shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 35.8.) Unit [Enum]

**Type:** Enum

**Description:** This parameter contains the unit of current scale.
In case of Unit is set to 0x06 or 0x07, the ASG shall display the scale including the first decimal place.

The ASG HMI may additionally define further presentation rules for function MapScale.

**Examples**
A scale of 10km is transmitted as: Scale = 100, Unit = 0x01.
A scale of 5mi is transmitted as: Scale = 50, Unit = 0x04.
A scale of 1/4mi is transmitted as: Scale = 10, Unit = 0x05.
A scale of 2.5km is transmitted as: Scale = 25, Unit = 0x06.
A scale of 5.0mi is transmitted as: Scale = 50, Unit = 0x07.

NOTE: Scale = 50 means 50 steps Ã¡ 0.1 -> 50*0.1 = 5.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (UK and US (statute mile)) |
| 0x05 | quarter mile(s) (n* 1/4mile) |
| 0x06 | kilometer (step 0.1) |
| 0x07 | mile (UK and US (statute mile), (step 0.1)) |
| 0x08..0xFE | reserved |
| 0xFF | not supported / no information available |

## 35.9.) Reserve3 [Number]

**Type:** Number

**Description:** This parameter is just used for symmetric reasons (1 byte dummy parameter). The receiver shall ignore/skip this parameter. The sender shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 35.10.) SupportedAutoZoom [Bitfield]

**Type:** Bitfield

**Description:** Setup information for parameter AutoZoom.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | autozoom on for intersection supported | autozoom on for intersection not supported |
| 1..7 | reserved | reserved |

## 35.11.) Reserve4 [Number]

**Type:** Number

**Description:** This parameter is just used for symmetric reasons (1 byte dummy parameter). The receiver shall ignore/skip this parameter. The sender shall set this parameter to "0".

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 35.12.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 36.) DestinationInfo ( 46, 0x2E )

**Type:** Property

**Description:** This function contains the information about the next destination.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Position | Record | Empty |
| TotalNumOfStopovers | Number | Byte |
| Stopover_SN | Number | Byte |
| POI_Type | Enum | Byte |
| POI_Description | String | ByteStream |
| Street | String | ByteStream |
| Town | String | ByteStream |
| State | String | ByteStream |
| PostalCode | String | ByteStream |
| Country | String | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** DestinationInfo (46, 0x2E )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | Position, TotalNumOfStopovers, Stopover_SN, POI_Type, POI_Description, Street, Town, State, PostalCode, Country, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 36.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

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**Base Datatype:** Empty

## 36.2.) Position [Record]

**Type:** Record

**Description:** The destinations nautical coordinates.

**OpCodes:** Status,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_36_2

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Latitude | Number | Long |
| Longitude | Number | Long |

### *Detailed Sub Parameter Overview*

## 36.2.1.) Latitude [Number]

**Type:** Number

**Description:** This parameter contains the destinations latitude (German: Breitengrad).

Positive values: northern hemisphere (German: noerdliche Breite), range 0..+90Â°.
Negative values: southern hemisphere (German: suedliche Breite), range -90Â°..0Â°.

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** -90

**Range Max:** 90

**Step:** 1E-06

**Unit:** unit_none

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## 36.2.2.) Longitude [Number]

**Type:** Number

**Description:** This parameter contains the destinations longitude (German: Laengengrad).

Negative values: western longitude (German: westliche Laenge), range -180Â°..0Â°
Positive values: eastern longitude (German: oestliche Laenge), range 0Â°..180Â°

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** -180

**Range Max:** 180

**Step:** 1E-06

**Unit:** unit_none

## 36.3.) TotalNumOfStopovers [Number]

**Type:** Number

**Description:** This parameter contains the number of stopovers in the active tour.
If route guidance is not active, then TotalNumOfStopover := 0.

**OpCodes:** Status,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 36.4.) Stopover_SN [Number]

**Type:** Number

**Description:** SN:.. sequence number

This parameter contains the stopover sequence number.
E.g., second stopover => Stopover_SN := 2.
If route guidance is not active, then Stopover_SN := 0.

Special values:

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0x00:.. final destination (destination does not belong to a tour)
0xFF:.. final destination (destination belongs to a tour)

**OpCodes:** Status,

**Base Datatype:** Byte

**Range Min:** 1

**Range Max:** 254

**Step:** 1

**Unit:** unit_none

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | final destination (destination does not belong to a tour) |
| 0xFF | final destination (destination belongs to a tour) |

## 36.5.) POI_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates, if the destination is a POI (Point Of Interest) and the POI type.
If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Status,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no POI |
| 0x01 | Accommodation |
| 0x02 | Airport |
| 0x03 | all restaurants |
| 0x04 | Amusement Park |
| 0x05 | Apparel |
| 0x06 | ATM (cashpoint) |
| 0x07 | Attraction |
| 0x08 | Automobile Club |
| 0x09 | Banking and Shopping |

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| | |
|---|---|
| 0x0A | Banking |
| 0x0B | Banquet-Halls |
| 0x0C | Bars and Lounge |
| 0x0D | Boating |
| 0x0E | Bookstore |
| 0x0F | Border Cross |
| 0x10 | Bowling |
| 0x11 | Broadcasting |
| 0x12 | Business and Service |
| 0x13 | Business |
| 0x14 | Busstation |
| 0x15 | Camping |
| 0x16 | Car and Travel |
| 0x17 | Car Track |
| 0x18 | Car Train |
| 0x19 | Car Service |
| 0x1A | Car Wash |
| 0x1B | Casino |
| 0x1C | Cemetry |
| 0x1D | Cityhall |
| 0x1E | coffee |
| 0x1F | Commuter Railstaion |
| 0x20 | Cleaning Launch |
| 0x21 | Clothing Store |
| 0x22 | Communication |
| 0x23 | Community Center |
| 0x24 | Computer Hardware-Software |
| 0x25 | Convenience Store |
| 0x26 | Convention Center |
| 0x27 | Corporations |
| 0x28 | Corporations Public Places |
| 0x29 | Culture Facility |
| 0x2A | Court |
| 0x2B | Department Store |
| 0x2C | Dining and Entertainment |
| 0x2D | Dinning Shopping |
| 0x2E | Discount Store |
| 0x2F | Education |
| 0x30 | Electronics |

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| | |
|---|---|
| 0x31 | Embassy |
| 0x32 | Emergency |
| 0x33 | Emergency and Public |
| 0x34 | Entertainment |
| 0x35 | Fast Food |
| 0x36 | Ferry |
| 0x37 | Fire Department |
| 0x38 | Flooring and Carpeting |
| 0x39 | Flower and Jewelry |
| 0x3A | Food and Beverage |
| 0x3B | Funeral Director |
| 0x3C | Furniture |
| 0x3D | Games, Music and Video |
| 0x3E | Garden and other |
| 0x3F | Gasstation |
| 0x40 | Gasstation DieselGasstation LPG |
| 0x41 | General |
| 0x42 | Gifts and Leisure |
| 0x43 | Gifts- Antiques and Arts |
| 0x44 | Glass and Window |
| 0x45 | Golfing |
| 0x46 | Goverment |
| 0x47 | Grocery Stores |
| 0x48 | Guesthouse |
| 0x49 | Hair and Beauty |
| 0x4A | Harbor |
| 0x4B | Hardware and Lumber |
| 0x4C | Health and Fitness |
| 0x4D | Health- Care |
| 0x4E | Highrise- BLDG |
| 0x4F | Home and Improvment |
| 0x50 | Home Center |
| 0x51 | Home Service |
| 0x52 | Hospital |
| 0x53 | Hotel / Motel |
| 0x54 | Iceskating |
| 0x55 | Insurance |
| 0x56 | Karaokay |
| 0x57 | Library |

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| | |
|---|---|
| 0x58 | LIQ-Gas-Station |
| 0x59 | Major-Appliance |
| 0x5A | Mens-Apparel |
| 0x5B | Monument |
| 0x5C | Motorway Exit |
| 0x5D | Motorway Cross |
| 0x5E | Motorway Service |
| 0x5F | Movie-Theater |
| 0x60 | Moving and Storage |
| 0x61 | Museum |
| 0x62 | Nightclubs |
| 0x63 | Optical |
| 0x64 | Restaurant |
| 0x65 | Park and Fitness |
| 0x66 | Park and Ride |
| 0x67 | Parking |
| 0x68 | Parking Garage |
| 0x69 | Parks |
| 0x6A | Performing-Arts |
| 0x6B | Personal- Service |
| 0x6C | Pharmacies |
| 0x6D | Photographer |
| 0x6E | Place or workship |
| 0x6F | Police Station |
| 0x70 | Post Office |
| 0x71 | Public Places |
| 0x72 | Publisher |
| 0x73 | Railway Station |
| 0x74 | Recreation |
| 0x75 | Rentel Service |
| 0x76 | Repair |
| 0x77 | Rest-Area |
| 0x78 | Retirement Nursing |
| 0x79 | Search |
| 0x7A | Service |
| 0x7B | Shoe Stores |
| 0x7C | Shopping Center |
| 0x7D | Skiing |
| 0x7E | Social Service |

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| | |
|---|---|
| 0x7F | Speciality-Clothing |
| 0x80 | Speciality-Store |
| 0x81 | Speciality-Food |
| 0x82 | Sport-Goods |
| 0x83 | Sports |
| 0x84 | Sport Airport |
| 0x85 | Sports-Complex |
| 0x86 | Tailor |
| 0x87 | Tax-Service |
| 0x88 | Tool-Booth |
| 0x89 | Tourist Attraction |
| 0x8A | Tourist Information |
| 0x8B | Train Station |
| 0x8C | Transportation |
| 0x8D | Travel |
| 0x8E | Travel-Agent |
| 0x8F | University |
| 0x90 | Utilities |
| 0x91 | Variety-Store |
| 0x92 | Wine and Liquor |
| 0x93 | Womens Apparel |
| 0x94 | Winery |
| 0x95 | car repair shop - VW |
| 0x96 | car repair shop - Audi |
| 0x97 | car repair shop - Seat |
| 0x98 | car repair shop - Skoda |
| 0x99 | car repair shop - VW Nutzfahrzeuge |
| 0x9A | car repair shop - Bentley |
| 0x9B | car repair shop - Bugatti |
| 0x9C | car repair shop - Lamborghini |
| 0x9D | car repair shop - Scania |
| 0x9E | car repair shop - MAN |
| 0x9F | car repair shop - Porsche |
| 0xA0 | electrical service station |
| 0xA1 | Diesel service station |
| 0xA2 | CNG service station |
| 0xA3 | rest area ("Autohof") |
| 0xA4 | rest area with WC |
| 0xA5 | motorway service area ("Raststaette") |

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| | |
|---|---|
| 0xA6..0xAF | reserved |
| 0xB0 | Predictive NAV (DF4.6) |
| 0xB1..0xFE | reserved |
| 0xFF | unknown |

## 36.6.) POI_Description [String]

**Type:** String

**Description:** String contains the POI descriptive string (textual description of the POI).

The string may contain up to 20 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 61 bytes. The size of the length identifier shall always be 1 byte.

If this parameter is not available, the FSG shall set it to empty string. For the definition of "empty string" please refer to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 61

## 36.7.) Street [String]

**Type:** String

**Description:** This parameter contains the UTF-8 encoded destinations street and house number.

The maximum number of characters is limited by the string length only (limited size of send buffer). The encoded string must not exceed 127 bytes plus an additional byte for the string length information.

If this parameter is not available, the FSG shall set it to empty string. For the definition of "empty string" please refer to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 128

## 36.8.) Town [String]

**Type:** String

**Description:** This parameter contains the UTF-8 encoded destinations town/city.

The maximum number of characters is limited by the string length only (limited size of send buffer). The encoded string must not exceed 60 bytes plus an additional byte for the string

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length information.

If this parameter is not available, the FSG shall set it to empty string. For the definition of "empty string" please refer to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 61

## 36.9.) State [String]

**Type:** String

**Description:** This parameter contains the UTF-8 encoded region or state of the currently active destination..

The maximum number of characters is limited by the string length only (limited size of send buffer). The encoded string must not exceed 60 bytes plus an additional byte for the string length information.

If this parameter is not available, the FSG shall set it to empty string. For the definition of "empty string" please refer to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 61

## 36.10.) PostalCode [String]

**Type:** String

**Description:** String contains the postal code of the currently active destination.

String may contain up to 7 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 22 bytes. The size of the length identifier shall always be 1 byte.
If this parameter is not availble, the FSG shall set it to empty string. For the definition of empty string please refer to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 22

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## 36.11.) Country [String]

**Type:** String

**Description:** This parameter contains the UTF-8 encoded current destinations country.

The maximum number of characters is limited by the string length only (limited size of send buffer). The encoded string must not exceed 60 bytes plus an additional byte for the string length information.

If this parameter is not available, the FSG shall set it to empty string. For the definition of "empty string" please refer to [2].

**OpCodes:** Status,

**Base Datatype:** ByteStream

**Length:** 61

## 36.12.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 37.) Altitude ( 47, 0x2F )

**Type:** Property

**Description:** This function contains current altitude.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Altitude | Number | Word |
| Unit | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** Altitude (47, 0x2F )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | Altitude, Unit, |
| HeartbeatStatus | Altitude, Unit, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 37.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 37.2.) Altitude [Number]

**Type:** Number

**Description:** This parameter contains current altitude.

0x0000..0x7FFE: positive values

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0x7FFF: init/unknown
0x8000..0xFFFF: negative values

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Word

**Range Min:** -32768

**Range Max:** 32767

**Step:** 1

**Unit:** unit_none

## 37.3.) Unit [Enum]

**Type:** Enum

**Description:** This parameter contains unit of altitude information.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | feet |
| 0x02..0xFF | reserved |

## 37.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 38.) OnlineNavigationState ( 48, 0x30 )

**Type:** Property

**Description:** This function contains current state of the online navigation application.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| State | Enum | Byte |
| Progress | Number | Byte |
| OnlineNavigationSystem | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** OnlineNavigationState (48, 0x30 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | State, Progress, OnlineNavigationSystem, |
| HeartbeatStatus | State, Progress, OnlineNavigationSystem, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 38.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 38.2.) State [Enum]

**Type:** Enum

**Description:** This parameter contains state of online navigation application.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | off |
| 0x01 | init/loading |
| 0x02 | active |
| 0x03 | no data connection |
| 0x04..0xFF | reserved |

## 38.3.) Progress [Number]

**Type:** Number

**Description:** This parameter contains (buffering) progress info.
Progress contains valid values if parameter State is set to 0x01 (init/loading).
0x00..0x64: 0..100%
0x65..0xFE: reserved
0xFF: init/unknown

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 100

**Step:** 1

**Unit:** unit_percent

**EnumValues:**

| Code | Description |
|---|---|
| 0x65..0xFE | reserved |
| 0xFF | init/unknown |

## 38.4.) OnlineNavigationSystem [Enum]

**Type:** Enum

**Description:** Contains information about technical system used for online navigation.

Reserved values shall be handled as 0xFF by the ASG.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | online navigation not built-in |
| 0x01 | GOOGLE navigation |
| 0x02..0xFE | reserved |
| 0xFF | unknown technical system for online navigation |

## 38.5.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 39.) Exitview ( 49, 0x31 )

**Type:** Property

**Description:** This function contains the exit view information.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Variant | Enum | Byte |
| Exitview_ID | Number | Word |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** Exitview (49, 0x31 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | Variant, Exitview_ID, |
| HeartbeatStatus | Variant, Exitview_ID, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 39.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 39.2.) Variant [Enum]

**Type:** Enum

**Description:** This parameter contains state of online navigation application.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | EU |
| 0x01 | NAR |
| 0x02 | ROW |
| 0x03 | ASIA |
| 0x04..0xFE | reserved |
| 0xFF | unknown |

## 39.3.) Exitview_ID [Number]

**Type:** Number

**Description:** This parameter contains the ID of current exit view.
Special value: 0x0 (no info, no exit view available).

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 39.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software.
Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 40.) SemidynamicRouteGuidance ( 50, 0x32 )

**Type:** Property

**Description:** This function returns information about delay on the current route.
If no route guidance is active (refer to function RG_Status), the ASG shall ignore the content of this function.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| TrafficImpact | Bitfield | ByteStream |
| Delay | Record | Empty |
| NewDistanceToDestination | Record | Empty |
| NewTimeToDestination | Record | Empty |
| ValidityInformation | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** SemidynamicRouteGuidance (50, 0x32 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | TrafficImpact, Delay, NewDistanceToDestination, NewTimeToDestination, ValidityInformation, |
| HeartbeatStatus | TrafficImpact, Delay, NewDistanceToDestination, NewTimeToDestination, ValidityInformation, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 40.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 40.2.) TrafficImpact [Bitfield]

**Type:** Bitfield

**Description:** Contains the impact on the traffic

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | traffic impact on current route | no traffic impact on current route or unknown |
| 1 | alternative route available | no alternative route available |
| 2..7 | reserved | reserved |

## 40.3.) Delay [Record]

**Type:** Record

**Description:** Provides the estimated delay on the current route.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_40_3

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Minute | Number | Byte |
| Hour | Number | Byte |

### *Detailed Sub Parameter Overview*

## 40.3.1.) Minute [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit0 in parameter TrafficImpact is set to 0.

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If the route contains any time offset, the parameter Minute contains the number of minutes of delay caused by traffic congestion.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 59

**Step:** 1

**Unit:** unit_min

### 40.3.2.) Hour [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit0 in parameter TrafficImpact is set to 0.
If the route contains any time offset, the parameter Hour contains the number of hours of delay caused by traffic congestion.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_h

## 40.4.) NewDistanceToDestination [Record]

**Type:** Record

**Description:** This parameter contains the new distance to destination if the driver will follow the alternative/semidynamic route.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_40_4

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### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Distance | Number | Long |
| Unit | Enum | Byte |
| TypeOfDistance | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 40.4.1.) Distance [Number]

**Type:** Number

**Description:** [transmitted distance value]=[displayed distance value]*10 (e.g. [displayed distance value]=0.9 -> [transmitted distance value]=9).

In some situations there may be no distance available. Therefore the following coding is defined. If the distance is not known to a specific moment, the bit0 in parameter ValidityInformation is set to 0.

If the Unit == 0x05 (quarter mile), then Distance shall be always an integer, e.g. 30*0.1=3. In that case, the ASG may display 3/4mile (displayed value := Distance * 1/4mi, e.g. 3.0*1/4 = 3/4mi.)

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** 0

**Range Max:** 429496729

**Step:** 0.1

**Unit:** unit_none

## 40.4.2.) Unit [Enum]

**Type:** Enum

**Description:** Unit belonging to the distance value.

**OpCodes:** Empty,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (UK and US (statute mile)) |
| 0x05 | quarter mile(s) (n* 1/4mile) |
| 0x06..0xFE | reserved |
| 0xFF | not supported / no information about unit available |

## 40.4.3.) TypeOfDistance [Enum]

**Type:** Enum

**Description:** Information about the type of the Distance.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | road distance |
| 0x01 | linear distance |
| 0x02..0xFE | reserved |
| 0xFF | distance invalid |

## 40.5.) NewTimeToDestination [Record]

**Type:** Record

**Description:** This parameter contains the new time to destination / arrival time if the driver will follow the alternative/semidynamic route.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_40_5

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### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| TimeInfoType | Enum | Nibble |
| NavigationTimeFormat | Enum | Nibble |
| Minute | Number | Byte |
| Hour | Number | Byte |
| Day | Number | Byte |
| Month | Number | Byte |
| Year | Number | Byte |

### *Detailed Sub Parameter Overview*

## 40.5.1.) TimeInfoType [Enum]

**Type:** Enum

**Description:** (upper nibble)

Type of information is given by the following parameters. The user setting within the navigation unit decides which type is used.

TimeToDestination: Estimated driving time to the next or to the final destination.
ArrivalTime: Estimated arrival time for the next or the final destination.

If not specified in another requirement document, the ASG shall handle a reserved value as value 0xF.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | TimeToDestination |
| 0x1 | ArrivalTime |
| 0x2 | ArrivalTime is local time in another time zone |
| 0x3..0xE | reserved |
| 0xF | not supported (= any time info type) |

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## 40.5.2.) NavigationTimeFormat [Enum]

**Type:** Enum

**Description:** (lower nibble)

Information about the used format for the arrival time in the navigation unit. The Secondary Display may or may not use the same time format, depending on the HMI definitions.
The format of the transmitted time value (Hour, Minute) is always the 24h mode, independent from the parameter NavigationTimeFormat.
If the ASG receives a reserved value, it shall apply the ASG-internal default value for the NavigationTimeFormat.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | 24 hour format |
| 0x1 | 12 hour format |
| 0x2..0xF | reserved |

## 40.5.3.) Minute [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit1 in parameter ValidityInformation is set to 0.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 59

**Step:** 1

**Unit:** unit_min

## 40.5.4.) Hour [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit2 in parameter ValidityInformation is set to FALSE (0).

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**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_h

## 40.5.5.) Day [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit3 in parameter ValidityInformation is set to FALSE (0).
This parameter is only be used if TimeInfoType=0x1 "ArrivalTime" or 0x2 "ArrivalTime in local time another time zone".

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 31

**Step:** 1

**Unit:** unit_d

## 40.5.6.) Month [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit4 in parameter ValidityInformation is set to FALSE (0).
This parameter is only be used if TimeInfoType=0x1 "ArrivalTime" or 0x2 "ArrivalTime in local time another time zone".

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 12

**Step:** 1

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**Unit:** unit_mon

## 40.5.7.) Year [Number]

**Type:** Number

**Description:** If information is not available or shall not be displayed, bit5 in parameter ValidityInformation is set to FALSE (0).
Parameter "Year" is only be used if TimeInfoType=0x1 "ArrivalTime" or 0x2 "ArrivalTime in local time another time zone".
Offset is 2000. I.e. for Year=2014 the value 14 is to be sent.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 2000

**Range Max:** 2099

**Step:** 1

**Unit:** unit_a

## 40.6.) ValidityInformation [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains information about the validity of parameter(s).
Reserved bits shall be set to FALSE (0) by the sender. The receiver shall ignore them.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | NewDistanceToDestination is valid | NewDistanceToDestination is not valid |
| 1 | NewTimeToDestination.Minute is valid | NewTimeToDestination.Minute is not valid |
| 2 | NewTimeToDestination.Hour is valid | NewTimeToDestination.Hour is not valid |
| 3 | NewTimeToDestination.Day is available / to be displayed | NewTimeToDestination.Day is not available / not to be displayed |
| 4 | NewTimeToDestination.Month is available / to be displayed | NewTimeToDestination.Month is not available / not to be displayed |

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| | | |
|---|---|---|
| 5 | NewTimeToDestination.Year is available / to be displayed | NewTimeToDestination.Year is not available / not to be displayed |
| 6..7 | reserved | reserved |

## 40.7.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardized error codes (Standard Protokoll Fehlerwerte) see the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 41.) POI_Search ( 51, 0x33 )

**Type:** Method

**Description:** This function is used to start and abort a search for a specific POI.
The result list of the POI-search is contained in function POI_List.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| SearchType | Enum | Byte |
| AmountOfFoundEntries | Number | Byte |
| POI_Type | Enum | Byte |
| ProgressIndicator | Number | Byte |
| POI_Search_Result | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** POI_Search (51, 0x33 )

| OpCode | Associated Parameters |
|---|---|
| Abort | NoParam, |
| StartResult | SearchType, POI_Type, |
| Processing | AmountOfFoundEntries, ProgressIndicator, |
| Result | AmountOfFoundEntries, POI_Search_Result, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 41.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Abort,

**Base Datatype:** Empty

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## 41.2.) SearchType [Enum]

**Type:** Enum

**Description:** Indicates the type of search for the given POI.

**OpCodes:** StartResult,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | Search along the current route (in case of active RG) |
| 0x01 | Search in the circumference to the current location |
| 0x02..0xFF | reserved |

## 41.3.) AmountOfFoundEntries [Number]

**Type:** Number

**Description:** This parameter indicates the amount of found entries for the current search.

**OpCodes:** Processing, Result,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 41.4.) POI_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates, the POI (Point Of Interest) to be searched.

**OpCodes:** StartResult,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00..0x3E | reserved |
| 0x3F | Gasstation (general) |
| 0x40..0x66 | reserved |
| 0x67 | Parking (general) |
| 0x68..0xFF | reserved |

## 41.5.) ProgressIndicator [Number]

**Type:** Number

**Description:** This parameter indicates the current progress of the search.
The range is: 0% till 100%

**OpCodes:** Processing,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 100

**Step:** 1

**Unit:** unit_percent

## 41.6.) POI_Search_Result [Enum]

**Type:** Enum

**Description:** Indicates the result of the method.
If the POI search cannot be started because the POI-Type does not match to a supported POI in the FSG, the method ends immediately. (FSG sends result message.)

If the POI_Type is set to a reserved value in the StartResult message, the FSG shall send an Error message with ErrorCode set to 0x41 (out of range).

**OpCodes:** Result,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | successful |
| 0x01 | not successful |
| 0x02 | abort successful |
| 0x03 | abort not successful |
| 0x04 | not successful - Entry not found |
| 0x05 | not successful - POI_Type does not match to FSG internal value |
| 0x06..0xFF | reserved |

## 41.7.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0x4F | reserved |
| 0x50 | AppErr_MethodAborted |
| 0x51..0xFF | reserved |

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# 42.) POI_List ( 52, 0x34 )

**Type:** Array

**Description:** This function contains the list of hits of the POI search.

The list of hits shall be cleared, when
- a new route guidance is started by the user
- a new search is started;

For further requirements for clearing the list of hits please refer to the navigation specification.

This list is referred to as "arbitrary list". For the meaning of "arbitrary IDs" the reader is referred to [2].

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| TotalNumListElements | Number | Byte |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** POI_List (52, 0x34 )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 42.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

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If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaneous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | reserved |
| 0x3 | operating unit rear (DF4.2) |
| 0x4..0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | reserved |
| 0xB | operating unit rear, to be evaluated by all ASGs (DF4.2) |
| 0xC..0xF | reserved |

## 42.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

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**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 42.3.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of list elements. In most cases this number is greater than the number of elements stored in the ASG.
This information may be used for dimensioning of scroll bars etc.

Value 0xFF means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 42.4.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_42_4

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### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

### *Detailed Sub Parameter Overview*

## 42.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
Both, the ASG and the FSG shall always use 8-bit addressing for Pos. This means, bit3 shall always be set to 0 by the ASG and the FSG.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Direction = backward | Direction = forward |
| 2 | Transmit_Pos = Pos is transmitted | Transmit_Pos = Pos is not transmitted |
| 3 | reserved | IndexSize = 8 Bit for Start, Elements, Pos |

## 42.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is part of the ModeByte of the ArrayHeader (see also "BAP Datentypen" [2]!).
The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records.

**OpCodes:** Empty,

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**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | Direction_Symbolic, POI_Type, DistanceType, Distance, Unit, MainDescription, Street, City, PostalCode |
| 0x1 | Distance, Unit, MainDescription, Street, City, PostalCode |
| 0x2 | Distance, Unit |
| 0x3 | Direction_Symbolic |
| 0x4 | MainDescription, Street, City, PostalCode |
| 0x5 | Direction_Symbolic, POI_Type, DistanceType, Distance, Unit, MainDescription |
| 0x6..0xE | reserved |
| 0xF | Pos |

## 42.4.3.) Start [Number]

**Type:** Number

**Description:** Parameter Start is the first array element the messages relates to. For usage of this parameter the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 42.4.4.) Elements [Number]

**Type:** Number

**Description:** Parameter Elements contains the number of elements the message relates to. For usage of this parameter the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

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**Step:** 1

**Unit:** unit_none

## 42.5.) Data [Record]

**Type:** Record

**Description:** "POI" contains a list with the requested POI data.
MaxElements = 6. This means, a StatusArray message is limited to 6 complete elements.
Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 254 elements. The FSG is not allowed to generate 0x00 as a valid Pos value.

The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 254

**Subparameter von:** 1_42_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| Direction_Symbolic | Enum | Byte |
| POI_Type | Enum | Byte |
| DistanceType | Enum | Byte |
| Distance | Number | Long |
| Unit | Enum | Byte |
| MainDescription | String | ByteStream |
| Street | String | ByteStream |
| City | String | ByteStream |
| PostalCode | String | ByteStream |

### *Detailed Sub Parameter Overview*

## 42.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos

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**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 42.5.2.) Direction_Symbolic [Enum]

**Type:** Enum

**Description:** The Direction_Symbolic shall be the same as shown on the head unit HMI (navigation device primary display).

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | N |
| 0x01 | NNO |
| 0x02 | NO |
| 0x03 | ONO |
| 0x04 | O |
| 0x05 | OSO |
| 0x06 | SO |
| 0x07 | ssO |
| 0x08 | S |
| 0x09 | ssW |
| 0x0A | SW |
| 0x0B | WSW |
| 0x0C | W |
| 0x0D | WNW |
| 0x0E | NW |
| 0x0F | NNW |
| 0x10..0xFE | reserved |
| 0xFF | not supported |

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## 42.5.3.) POI_Type [Enum]

**Type:** Enum

**Description:** This parameter indicates, if the destination is a POI (Point Of Interest) and the POI type.
If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no POI |
| 0x01 | Accommodation |
| 0x02 | Airport |
| 0x03 | all restaurants |
| 0x04 | Amusement Park |
| 0x05 | Apparel |
| 0x06 | ATM (cashpoint) |
| 0x07 | Attraction |
| 0x08 | Automobile Club |
| 0x09 | Banking and Shopping |
| 0x0A | Banking |
| 0x0B | Banquet-Halls |
| 0x0C | Bars and Lounge |
| 0x0D | Boating |
| 0x0E | Bookstore |
| 0x0F | Border Cross |
| 0x10 | Bowling |
| 0x11 | Broadcasting |
| 0x12 | Business and Service |
| 0x13 | Business |
| 0x14 | Busstation |
| 0x15 | Camping |
| 0x16 | Car and Travel |
| 0x17 | Car Track |
| 0x18 | Car Train |
| 0x19 | Car Service |
| 0x1A | Car Wash |

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| | |
|---|---|
| 0x1B | Casino |
| 0x1C | Cemetry |
| 0x1D | Cityhall |
| 0x1E | coffee |
| 0x1F | Commuter Railstaion |
| 0x20 | Cleaning Launch |
| 0x21 | Clothing Store |
| 0x22 | Communication |
| 0x23 | Community Center |
| 0x24 | Computer Hardware-Software |
| 0x25 | Convenience Store |
| 0x26 | Convention Center |
| 0x27 | Corporations |
| 0x28 | Corporations Public Places |
| 0x29 | Culture Facility |
| 0x2A | Court |
| 0x2B | Department Store |
| 0x2C | Dining and Entertainment |
| 0x2D | Dinning Shopping |
| 0x2E | Discount Store |
| 0x2F | Education |
| 0x30 | Electronics |
| 0x31 | Embassy |
| 0x32 | Emergency |
| 0x33 | Emergency and Public |
| 0x34 | Entertainment |
| 0x35 | Fast Food |
| 0x36 | Ferry |
| 0x37 | Fire Department |
| 0x38 | Flooring and Carpeting |
| 0x39 | Flower and Jewelry |
| 0x3A | Food and Beverage |
| 0x3B | Funeral Director |
| 0x3C | Furniture |
| 0x3D | Games, Music and Video |
| 0x3E | Garden and other |
| 0x3F | Gasstation |
| 0x40 | Gasstation DieselGasstation LPG |
| 0x41 | General |

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| | |
|---|---|
| 0x42 | Gifts and Leisure |
| 0x43 | Gifts- Antiques and Arts |
| 0x44 | Glass and Window |
| 0x45 | Golfing |
| 0x46 | Goverment |
| 0x47 | Grocery Stores |
| 0x48 | Guesthouse |
| 0x49 | Hair and Beauty |
| 0x4A | Harbor |
| 0x4B | Hardware and Lumber |
| 0x4C | Health and Fitness |
| 0x4D | Health- Care |
| 0x4E | Highrise- BLDG |
| 0x4F | Home and Improvment |
| 0x50 | Home Center |
| 0x51 | Home Service |
| 0x52 | Hospital |
| 0x53 | Hotel / Motel |
| 0x54 | Iceskating |
| 0x55 | Insurance |
| 0x56 | Karaokay |
| 0x57 | Library |
| 0x58 | LIQ-Gas-Station |
| 0x59 | Major-Appliance |
| 0x5A | Mens-Apparel |
| 0x5B | Monument |
| 0x5C | Motorway Exit |
| 0x5D | Motorway Cross |
| 0x5E | Motorway Service |
| 0x5F | Movie-Theater |
| 0x60 | Moving and Storage |
| 0x61 | Museum |
| 0x62 | Nightclubs |
| 0x63 | Optical |
| 0x64 | Restaurant |
| 0x65 | Park and Fitness |
| 0x66 | Park and Ride |
| 0x67 | Parking |
| 0x68 | Parking Garage |

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| | |
|---|---|
| 0x69 | Parks |
| 0x6A | Performing-Arts |
| 0x6B | Personal- Service |
| 0x6C | Pharmacies |
| 0x6D | Photographer |
| 0x6E | Place or workship |
| 0x6F | Police Station |
| 0x70 | Post Office |
| 0x71 | Public Places |
| 0x72 | Publisher |
| 0x73 | Railway Station |
| 0x74 | Recreation |
| 0x75 | Rentel Service |
| 0x76 | Repair |
| 0x77 | Rest-Area |
| 0x78 | Retirement Nursing |
| 0x79 | Search |
| 0x7A | Service |
| 0x7B | Shoe Stores |
| 0x7C | Shopping Center |
| 0x7D | Skiing |
| 0x7E | Social Service |
| 0x7F | Speciality-Clothing |
| 0x80 | Speciality-Store |
| 0x81 | Speciality-Food |
| 0x82 | Sport-Goods |
| 0x83 | Sports |
| 0x84 | Sport Airport |
| 0x85 | Sports-Complex |
| 0x86 | Tailor |
| 0x87 | Tax-Service |
| 0x88 | Tool-Booth |
| 0x89 | Tourist Attraction |
| 0x8A | Tourist Information |
| 0x8B | Train Station |
| 0x8C | Transportation |
| 0x8D | Travel |
| 0x8E | Travel-Agent |
| 0x8F | University |

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| | |
|---|---|
| 0x90 | Utilities |
| 0x91 | Variety-Store |
| 0x92 | Wine and Liquor |
| 0x93 | Womens Apparel |
| 0x94 | Winery |
| 0x95 | car repair shop - VW |
| 0x96 | car repair shop - Audi |
| 0x97 | car repair shop - Seat |
| 0x98 | car repair shop - Skoda |
| 0x99 | car repair shop - VW Nutzfahrzeuge |
| 0x9A | car repair shop - Bentley |
| 0x9B | car repair shop - Bugatti |
| 0x9C | car repair shop - Lamborghini |
| 0x9D | car repair shop - Scania |
| 0x9E | car repair shop - MAN |
| 0x9F | car repair shop - Porsche |
| 0xA0 | electrical service station |
| 0xA1 | Diesel fuel station |
| 0xA2 | CNG fuel station |
| 0xA3 | rest area ("Autohof") |
| 0xA4 | rest area with WC |
| 0xA5 | motorway service area ("Raststaette") |
| 0xA6..0xAF | reserved |
| 0xB0 | Predictive NAV (DF4.6) |
| 0xB1..0xFE | reserved |
| 0xFF | unknown |

## 42.5.4.) DistanceType [Enum]

**Type:** Enum

**Description:** This parameter indicates, which type of distance to the requested POI is used.

**OpCodes:** Empty,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no distance available |
| 0x01 | linear distance (air distance) |
| 0x02 | distance along the street |
| 0x03..0xFF | reserved |

## 42.5.5.) Distance [Number]

**Type:** Number

**Description:** [transmitted distance value]=[displayed distance value]*10 (e.g. [displayed distance value]=0.9 -> [transmitted distance value]=9).

If the Unit == 0x05 (quarter mile), then Distance shall always be an integer, e.g. 30*0.1=3. In that case, the ASG may display 3/4mile (displayed value := Distance * 1/4mi, e.g. 3.0*1/4 = 3/4mi.)

**OpCodes:** Empty,

**Base Datatype:** Long

**Range Min:** 0

**Range Max:** 429496729

**Step:** 0.1

**Unit:** unit_none

## 42.5.6.) Unit [Enum]

**Type:** Enum

**Description:** Unit belonging to the distance value.

**OpCodes:** Empty,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (UK and US (statute mile)) |
| 0x05 | quarter mile(s) (n* 1/4mile) |
| 0x06..0xFE | reserved |
| 0xFF | not supported / no information about unit available |

## 42.5.7.) MainDescription [String]

**Type:** String

**Description:** String contains the displayed main description.

String may contain up to 30 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 91 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 91

## 42.5.8.) Street [String]

**Type:** String

**Description:** String contains the displayed street name.

String may contain up to 30 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 91 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 91

## 42.5.9.) City [String]

**Type:** String

**Description:** String contains the displayed city name.

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String may contain up to 30 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 91 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 91

## 42.5.10.) PostalCode [String]

**Type:** String

**Description:** String contains the displayed ZIP code.

String may contain up to 7 UTF-8 encoded Unicode3.0 characters. This results in a maximum length of 22 bytes. The size of the length identifier shall always be 1 byte.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 22

## 42.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved |

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# 43.) FSG_Setup ( 53, 0x35 )

**Type:** Property

**Description:** Controls availability of sub functions for different Fct-ID.
Information is set by ECU coding.
Can be setup only by functional ECU.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| VoiceGuidance | Bitfield | ByteStream |
| Supported_POI_Types | Bitfield | ByteStream |
| FunctionSupport | Bitfield | ByteStream |
| Dummy2 | Number | Byte |
| Dummy3 | Number | Byte |
| Dummy4 | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** FSG_Setup (53, 0x35 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | VoiceGuidance, Supported_POI_Types, FunctionSupport, Dummy2, Dummy3, Dummy4, |
| HeartbeatStatus | VoiceGuidance, Supported_POI_Types, FunctionSupport, Dummy2, Dummy3, Dummy4, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 43.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

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## 43.2.) VoiceGuidance [Bitfield]

**Type:** Bitfield

**Description:** Setup information for the Function 0x24 VoiceGuidance.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | VoiceGuidance ON (full / complete announcements) available | VoiceGuidance ON (full / complete announcements) not available |
| 1 | VoiceGuidance ON reduced (reduced announcements / reduced mode) available | VoiceGuidance ON reduced (reduced announcements / reduced mode) not available |
| 2 | VoiceGuidance ON traffic (reduced announcement traffic) available | VoiceGuidance ON traffic (reduced announcement traffic) (DF4.1) not available |
| 3..7 | reserved | reserved |

## 43.3.) Supported_POI_Types [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains information about the supported POI Types for a POI Search using function POI_Search.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | fuel station and parking area supported | fuel station parking area not supported |
| 1..7 | reserved | reserved |

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## 43.4.) FunctionSupport [Bitfield]

**Type:** Bitfield

**Description:** Information about further FSG functionality.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | activation/deactivation of the video navigation supported (DF4.4) | activation/deactivation of the video navigation not supported (DF4.4) |
| 1 | home address available (DF4.5) | home address not available (DF4.5) |
| 2 | business address available (DF4.5) | business address not available (DF4.5) |
| 3 | Route guidance by mobile device supported (DF4.7) | Route guidance by mobile device not supported (DF4.7) |
| 4 | extended ManeuverDescriptor direction and sidestreets supported (DF4.7) | extended ManeuverDescriptor direction and sidestreets not supported (DF4.7) |
| 5 | Maneuver list supported (DF4.7) | Maneuver list not supported (DF4.7) |
| 6..7 | reserved | reserved |

## 43.5.) Dummy2 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

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## 43.6.) Dummy3 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 43.7.) Dummy4 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 43.8.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes (standard protocol error, standard application error) refer to the annex.

Standard protocol error values will be sent automatically by the software.
Application errors (standard application error, function specific error) have to be set/evaluated by the application.

Standard application error values and standard protocol error values are mandatory.

**OpCodes:** Error,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to annex |
| 0x40..0xFF | reserved, refer to annex |

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# 44.) Map_Presentation ( 54, 0x36 )

**Type:** Property

**Description:** Controls the presentation of the map depending on the HMI state of the ASG.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| ASG_HMI_State | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** Map_Presentation (54, 0x36 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | ASG_HMI_State, |
| Status | ASG_HMI_State, |
| HeartbeatStatus | ASG_HMI_State, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 44.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 44.2.) ASG_HMI_State [Bitfield]

**Type:** Bitfield

**Description:** Contains information of the ASG HMI state being of interesst for the BAP FSG (needed for map rendering).

**OpCodes:** SetGet, Status, HeartbeatStatus,

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**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | large map view | small map view |
| 1 | left side menue open | left sidemenue closed |
| 2 | right side menue open | right side menue closed |
| 3..7 | reserved | reserved |

## 44.3.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 45.) ManeuverState ( 55, 0x37 )

**Type:** Property

**Description:** This function returns the state of the current valid maneuver.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| State | Enum | Byte |
| Dummy1 | Number | Byte |
| Dummy2 | Number | Byte |
| Dummy3 | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** ManeuverState (55, 0x37 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | State, Dummy1, Dummy2, Dummy3, |
| HeartbeatStatus | State, Dummy1, Dummy2, Dummy3, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 45.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 45.2.) State [Enum]

**Type:** Enum

**Description:** Contains the state of the current valid maneuver.

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**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | init / unknown |
| 0x01 | Follow |
| 0x02 | Prepare |
| 0x03 | Distance |
| 0x04 | Call for Action |
| 0x05..0xFF | reserved |

## 45.3.) Dummy1 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 45.4.) Dummy2 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

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**Unit:** unit_none

## 45.5.) Dummy3 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 45.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software.
Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 46.) ETC_Status ( 56, 0x38 )

**Type:** Property

**Description:** This function returns the current ETC status.
ETC: Electronic toll collect

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| CardStatus | Enum | Byte |
| Extension1 | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** ETC_Status (56, 0x38 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | CardStatus, Extension1, |
| HeartbeatStatus | CardStatus, Extension1, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 46.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 46.2.) CardStatus [Enum]

**Type:** Enum

**Description:** Contains the status of the ETC card reader.

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | init / unknown |
| 0x01 | ETC card inserted |
| 0x02 | ETC card not inserted |
| 0x03 | ETC card reader is not connected |
| 0x04..0xFF | reserved |

## 46.3.) Extension1 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 46.4.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 47.) MapContentSettings ( 57, 0x39 )

**Type:** Property

**Description:** Function to report and configure the settings of the map content.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| Reserve1 | Number | Byte |
| AvailableContents | Bitfield | ByteStream |
| ContentStatus | Bitfield | ByteStream |
| Extension1 | Number | Byte |
| Extension2 | Number | Byte |
| Extension3 | Number | Byte |
| Extension4 | Number | Byte |
| ErrorCode | Enum | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** MapContentSettings (57, 0x39 )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| SetGet | Reserve1, ContentStatus, Extension1, Extension2, Extension3, Extension4, |
| Status | AvailableContents, ContentStatus, Extension1, Extension2, Extension3, Extension4, |
| HeartbeatStatus | AvailableContents, ContentStatus, Extension1, Extension2, Extension3, Extension4, |
| Error | ErrorCode, ErrorCode, |

### *Detailed Parameter Overview*

## 47.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Get,

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**Base Datatype:** Empty

## 47.2.) Reserve1 [Number]

**Type:** Number

**Description:** Parameter for symmetry reason. Shall be set to 0 by the sender and not be evaluated by the receiver.

**OpCodes:** SetGet,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 47.3.) AvailableContents [Bitfield]

**Type:** Bitfield

**Description:** Available map contents

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | GoogleInfo available | GoogleInfo not available |
| 1 | OnlineInfo available | OnlineInfo not available |
| 2 | POIInfo available | POIInfo not available |
| 3 | TrafficInfo available | TrafficInfo not available |
| 4..7 | reserved | reserved |

## 47.4.) ContentStatus [Bitfield]

**Type:** Bitfield

**Description:** Status of the content

**OpCodes:** SetGet, Status, HeartbeatStatus,

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**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | GoogleInfo on | GoogleInfo off |
| 1 | OnlineInfo on | OnlineInfo off |
| 2 | POIInfo on | POIInfo off |
| 3 | TrafficInfo on | TrafficInfo off |
| 4..7 | reserved | reserved |

## 47.5.) Extension1 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 47.6.) Extension2 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

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**Unit:** unit_const

## 47.7.) Extension3 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 47.8.) Extension4 [Number]

**Type:** Number

**Description:** This parameter is reserved for future extensions.
The FSG shall set this parameter to "0". The ASG shall ignore this parameter.

**OpCodes:** SetGet, Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 47.9.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software.
Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

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**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

## 47.10.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0xFF | reserved |

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# 48.) VideoStreams ( 58, 0x3A )

**Type:** Array

**Description:** This function contains a list of all currently active RTP Navigation videostreams.

This list is referred to as "arbitrary list". For the meaning of "arbitrary IDs" the reader is referred to [2].

If the ASG requests an invalid array element in the Start parameter, the FSG returns the required START element, ELEMENTS = 0 and no DATA because an invalid element has no successor or predecessor. Please note the special rules for START=0, refer to [2].
If the list contains fewer elements in the given direction than required, the FSG returns the existing elements.
The value of the START byte in a StatusArray message has to be equal to the one from the GetArray message. The value of the Transmit_Pos bit has to be taken into account.

For the general specification of the array function class please refer to [2].

NOTE: In case of a ChangedArray message the parameter Data is either empty or contains only Pos information.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| TotalNumListElements | Number | Byte |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** VideoStreams (58, 0x3A )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

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## 48.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaneous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB..0xF | reserved |

## 48.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

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Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 48.3.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of list elements. In most cases this number is greater than the number of elements stored in the ASG.

Value 0xFF means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 48.4.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_48_4

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### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

### *Detailed Sub Parameter Overview*

## 48.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
The IndexSize shall be set to 8bit always (Bit3 := FALSE). I.e., only 8-bit values are allowed for Start, Elements and Pos.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Direction = backward | Direction = forward |
| 2 | Transmit_Pos = Pos is transmitted | Transmit_Pos = Pos is not transmitted |
| 3 | reserved | IndexSize = 8 Bit for Start, Elements, Pos |

## 48.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is part of the ModeByte of the ArrayHeader (see also "BAP Datentypen" [2]!).
The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records.
Special values:
0x00 may be used by ChangedArray for full range update indication

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**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | reserved |
| 0x1 | StreamType, IPv6_Address, UDP_Port, URI |
| 0x2 | StreamType, IPv6_Address, UDP_Port |
| 0x3..0xE | reserved |
| 0xF | Pos |

## 48.4.3.) Start [Number]

**Type:** Number

**Description:** Parameter Start is the first array element the messages relates to. For usage of this parameter the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 48.4.4.) Elements [Number]

**Type:** Number

**Description:** Parameter Elements contains the number of elements the message relates to. For usage of this parameter the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

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**Unit:** unit_none

## 48.5.) Data [Record]

**Type:** Record

**Description:** This list contains information about all currently existing navigation video streams, which could be displayed in ASG.

MaxElements = 3 if RecordAddres 0x1 is used. This means, a StatusArray message is limited to 3 complete elements.

NOTE! The ASG is allowed to request more elements than given in MaxElements as long as the amount of data to be transferred in bytes does not exceed the data amount needed for transfer of MaxElements using RecordAddress 0x1.
E.g. the ASG is allowed to request more than MaxElements == 3 when using RecordAddress 0xF (POS-only). Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 254 elements. The FSG is not allowed to generate 0x00 as a valid Pos value.

The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 254

**Subparameter von:** 1_48_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| StreamType | Enum | Byte |
| IPv6_Address | String | ByteStream |
| UDP_Port | Number | Word |
| URI | String | ByteStream |

### *Detailed Sub Parameter Overview*

## 48.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos

**OpCodes:** Empty,

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**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 48.5.2.) StreamType [Enum]

**Type:** Enum

**Description:** This parameter indicates the type of navigation video stream. If the FSG cannot determine the type of navigation video stream, it shall be set to unknown (0xFF).
If the ASG receives a reserved value, it shall act as if 0xFF has been received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | main map |
| 0x01 | LaneGuidance |
| 0x02 | ExitView/KDK |
| 0x03 | GoogleEarth map |
| 0x04..0xFE | reserved |
| 0xFF | unknown |

## 48.5.3.) IPv6_Address [String]

**Type:** String

**Description:** This string parameter contains the IPv6 address (destination multicast address) of the video stream as data stream (string). The data stream doesn't contain any UTF8 encoded Unicode3.0 characters.
The IPv6 address is transmitted without colon (":") for seperating the 2Byte blocks.

The first byte provides information about the string length. An IPv6 address always consists of 16 byte. That means if an IPv6 address is transmitted, the string lenght is 16.

For example if the IPv6 address 2001:0DB8:0000:0000:F054:00FF:0000:02EB shall be transmitted, the FSG set this parameter (including length information):
10 20 01 0D B8 00 00 00 00 F0 54 00 FF 00 00 02 EB

**OpCodes:** Empty,

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**Base Datatype:** ByteStream

**Length:** 17

## 48.5.4.) UDP_Port [Number]

**Type:** Number

**Description:** This parameter contains the UDP destination port of the RTP video stream.

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

## 48.5.5.) URI [String]

**Type:** String

**Description:** This string parameter contains the Uniform Resource Identifier (URI) of the navigation video stream. The string contains UTF8 encoded unicode3.0 characters. If the FSG has no information about the URI of the navigation video stream, an empty string shall be transmitted.

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 600 bytes plus two additional byte for the string length information.
The size of the string length identifier shall always be 2 byte.

example for an URI:
*/maprenderer/renderers/bb3372f0-b527-11e3-a5e2-0800200c9a66*

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 602

## 48.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software.

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Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved |

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# 49.) RequestSync_Video ( 59, 0x3B )

**Type:** Method

**Description:** This function is used to request a resynchronisation of navigation video streams in FSG by ASG.
If the ASG is no longer able to decode the H.264 video stream because of an error in video sink (located in FSG), this function can be used.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| ASG_ID | Enum | Nibble |
| Extension1 | Number | Nibble |
| VideoStreamID | Number | Byte |
| RequestSync_Result | Enum | Byte |
| Extension2 | Number | Byte |
| Reserve1 | Number | Byte |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** RequestSync_Video (59, 0x3B )

| OpCode | Associated Parameters |
|---|---|
| Abort | NoParam, |
| StartResult | ASG_ID, Extension1, VideoStreamID, Extension2, |
| Result | ASG_ID, Extension1, RequestSync_Result, |
| Processing | ASG_ID, Extension1, Reserve1, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

## 49.1.) NoParam [Void]

**Type:** Void

**Description:**

**OpCodes:** Abort,

**Base Datatype:** Empty

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## 49.2.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Parameter ASG_ID is used to identify which display ecu request the method in FSG.
Further details about "multi ASG handling" for methods refer [2].
The multi ASG rules shall be applied even if the ASG_ID in the StartResult message is set to a reserved value!

Values > 0x7 shall not be sent in a StartResult message.

**OpCodes:** StartResult, Result, Processing,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaenous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3..0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB..0xF | reserved |

## 49.3.) Extension1 [Number]

**Type:** Number

**Description:** (lower nibble)
Unused filler nibble. Reserved for further extensions.

**OpCodes:** StartResult, Result, Processing,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

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## 49.4.) VideoStreamID [Number]

**Type:** Number

**Description:** This parameter contains the video stream ID of the corrupted video. The VideoStreamID corresponds to the POS value in function VideoStreams (0x3A).

**OpCodes:** StartResult,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 49.5.) RequestSync_Result [Enum]

**Type:** Enum

**Description:** Indicates the result of the method.

**OpCodes:** Result,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | successful |
| 0x01 | not successful |
| 0x02 | abort successful |
| 0x03 | abort not successful |
| 0x04..0xFF | reserved |

## 49.6.) Extension2 [Number]

**Type:** Number

**Description:** Used for further extensions

**OpCodes:** StartResult,

**Base Datatype:** Byte

**Range Min:** 0

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**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 49.7.) Reserve1 [Number]

**Type:** Number

**Description:** This is just used for symmetric reasons. The ASG shall ignore this value. The FSG shall set this parameter to "0".

**OpCodes:** Processing,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 1

**Unit:** unit_const

## 49.8.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40 | reserved |
| 0x41 | AppErr_OutOfRange |
| 0x42..0x45 | reserved, refer to annex |
| 0x46 | AppErr_InvalidArgument |
| 0x47..0x4F | reserved |
| 0x50 | AppErr_MethodAborted |
| 0x51..0xFF | reserved |

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# 50.) DistanceToDestinationExtended ( 60, 0x3C )

**Type:** Property

**Description:** This function is an extension of function DistanceToDestination (0x15).
It returns for example the total distance and current percentage distance to the final destination or stopover from the beginning of starting the route guidance.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| NoParam | Void | Empty |
| TotalDistanceToDestination | Record | Empty |
| DistanceToDestination_percentage | Number | Byte |
| ERangeWarning | Enum | Nibble |
| Extension1 | Number | Nibble |
| Extension2 | Number | Byte |
| Extension3 | Number | Byte |
| Extension4 | Number | Byte |
| ValidityInformation | Bitfield | ByteStream |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** DistanceToDestinationExtended (60, 0x3C )

| OpCode | Associated Parameters |
|---|---|
| Get | NoParam, |
| Status | TotalDistanceToDestination, DistanceToDestination_percentage, ERangeWarning, Extension1, Extension2, Extension3, Extension4, ValidityInformation, |
| HeartbeatStatus | TotalDistanceToDestination, DistanceToDestination_percentage, ERangeWarning, Extension1, Extension2, Extension3, Extension4, ValidityInformation, |
| Error | ErrorCode, |

### *Detailed Parameter Overview*

#### 50.1.) NoParam [Void]

**Type:** Void

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**Description:**

**OpCodes:** Get,

**Base Datatype:** Empty

## 50.2.) TotalDistanceToDestination [Record]

**Type:** Record

**Description:** This parameter contains the total distance to the final destination or stopover along the planned route since start of the route guidance. The content of this parameter (total distance to final destination or stopover) depends also on Fct. DistanceToDestination.DistanceToDestinationType.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_50_2

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Distance | Number | Long |
| Unit | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 50.2.1.) Distance [Number]

**Type:** Number

**Description:** [transmitted distance value]=[displayed distance value]*10 (e.g. [displayed distance value]=0.9 -> [transmitted distance value]=9).

In some situations there may be no distance available. Therefore the following coding is defined. If the distance is not known to a specific moment, the bit0 in parameter ValidityInformation is set to 0.

If the Unit == 0x05 (quarter mile), then Distance shall be always an integer, e.g. 30*0.1=3. In that case, the ASG may display 3/4mile (displayed value := Distance * 1/4mi, e.g. 3.0*1/4 = 3/4mi.)

**OpCodes:** Empty,

**Base Datatype:** Long

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**Range Min:** 0

**Range Max:** 429496729

**Step:** 0.1

**Unit:** unit_none

### 50.2.2.) Unit [Enum]

**Type:** Enum

**Description:** Unit belonging to the distance value.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | meter |
| 0x01 | kilometer |
| 0x02 | yard |
| 0x03 | feet |
| 0x04 | mile (UK and US (statute mile)) |
| 0x05 | quarter mile(s) (n* 1/4mile) |
| 0x06..0xFE | reserved |
| 0xFF | not supported / no information about unit available |

## 50.3.) DistanceToDestination_percentage [Number]

**Type:** Number

**Description:** This parameter contains the current percentage distance to the final destination or stopover along the planned route since start of the route guidance. The content of this parameter (total distance to final destination or stopover) depends also on Fct. DistanceToDestination.DistanceToDestinationType.
Relative fill of DistanceToDestination_percentage: 0%..100%.
The ASG shall map reserved values to 100%.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 100

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**Step:** 1

**Unit:** unit_percent

**EnumValues:**

| Code | Description |
|---|---|
| 0x00..0x64 | valid values |
| 0x65..0xFE | reserved |
| 0xFF | init |

## 50.4.) ERangeWarning [Enum]

**Type:** Enum

**Description:** Value allows to trigger warning popups due to missing electrical range for current navigation.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no warning / init |
| 0x1 | warning level 1 (low range after arrival) |
| 0x2 | warning level 2 (very low range after arrival) |
| 0x3 | warning level 3 (arrival endangered with current usage level) |
| 0x4 | warning level 4 (arrival very endangered with current usage level) |
| 0x5 | warning level 5 (arrival not possible, charge must be planned) |
| 0x6 | warning level 6 (arrival not possible, charge must be done) |
| 0x7..0x8 | reserved |
| 0x9 | info level 1 (RangeExtender will be used) |
| 0xA | info level 2 (RangeExtender usage imminent) |
| 0xB | info level 3 (RangeExtener in use) |
| 0xC..0xF | reserved |

## 50.5.) Extension1 [Number]

**Type:** Number

**Description:** Used for further extensions

**OpCodes:** Status, HeartbeatStatus,

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**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 50.6.) Extension2 [Number]

**Type:** Number

**Description:** Used for further extensions

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 50.7.) Extension3 [Number]

**Type:** Number

**Description:** Used for further extensions

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 50.8.) Extension4 [Number]

**Type:** Number

**Description:** Used for further extensions

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**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 0

**Step:** 0

**Unit:** unit_const

## 50.9.) ValidityInformation [Bitfield]

**Type:** Bitfield

**Description:** This paramters contains information about the validity of parameter(s). The ASG shall ignore reserved values. The FSG shall set reserved bits of the bitfield to FALSE.

**OpCodes:** Status, HeartbeatStatus,

**Base Datatype:** ByteStream

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | TotalDistanceToDestination valid | TotalDistanceToDestination invalid |
| 1 | DistanceToDestination_percentage valid | DistanceToDestination_percentage invalid |
| 2..7 | reserved | reserved |

## 50.10.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

**OpCodes:** Error,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0xFF | reserved, refer to annex |

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# 51.) LaneGuidance2 ( 61, 0x3D )

**Type:** Array

**Description:** Function provides information needed for lane guidance. Lane guidance may be displayed additionally to the maneuver of the route guidance.

Lane guidance is modelled as array of consecutive IDs. Each array entry corresponds to a single lane. For the definition of "consecutive IDs" and the usage for BAP array functions the reader is referred to [2].

NOTE: In case of a ChangedArray message the parameter LaneGuidance is either empty (not sent) or contains only Pos information.

## *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| LaneGuidanceOnOff | Enum | Byte |
| CurrentCarPosition | Number | Byte |
| RecommendedCarPosition | Number | Byte |
| ArrayHeader | Record | Empty |
| LaneGuidance | Record | Empty |
| ErrorCode | Enum | Byte |

## *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** LaneGuidance2 (61, 0x3D )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, LaneGuidanceOnOff, CurrentCarPosition, RecommendedCarPosition, ArrayHeader, LaneGuidance, |
| ChangedArray | LaneGuidanceOnOff, CurrentCarPosition, RecommendedCarPosition, ArrayHeader, LaneGuidance, |
| Error | ErrorCode, |

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## *Detailed Parameter Overview*

### 51.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

If the StatusArray message is a response to a SetGetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the SetGetArray message. If the content of the array element(s) has been changed due to the SetGetArray message, the FSG shall set the most significant bit within ASG_ID to 1 in addition.
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaenous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear |
| 0x4..0x7 | reserved |
| 0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB | operating unit rear, to be evaluated by all ASGs |
| 0xC..0xF | reserved |

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## 51.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with the parameter ASG_ID.

Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceeding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 51.3.) LaneGuidanceOnOff [Enum]

**Type:** Enum

**Description:** Contains on / off trigger for lane guidance.

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | lane guidance shall not be displayed |
| 0x01 | display lane guidance |
| 0x02..0xFE | reserved |
| 0xFF | not supported |

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## 51.4.) CurrentCarPosition [Number]

**Type:** Number

**Description:** Contains the current car position/lane.
The POS value of this function (LaneGuidance array) is used to mark the lane with the current car position.

If the FSG has no information about the current car position, this parameter shall be set to 0x00.

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 51.5.) RecommendedCarPosition [Number]

**Type:** Number

**Description:** Contains the recommended car position/lane.
The POS value of this function (LaneGuidance array) is used to mark the lane for the recommended car position.

If the FSG has no information about the recommended car position, this parameter shall be set to 0x00.

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

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## 51.6.) ArrayHeader [Record]

**Type:** Record

**Description:** ArrayHeader

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_51_6

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

### *Detailed Sub Parameter Overview*

## 51.6.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
The IndexSize shall be set to 8bit always (Bit3 := FALSE). I.e., only 8-bit values are allowed for Start, Elements and Pos.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Shift = 1, refer to [2] | Shift = 0, refer to [2] |
| 1 | Array direction is backward | Array direction is forward |
| 2 | Array position is transmitted | Array position is not transmitted |
| 3 | reserved | IndexSize 8 Bit for Start/Elements |

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## 51.6.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

The RecordAddress is used for array elements consisting of a record. It serves to address a single parameter or groups of parameters of these records [2].

RecordAddress 0x0 delivers all record elements.
RecordAddress 0xF delivers Pos only.
Special values:
0x00 may be used by ChangedArray for full range update indication

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | reserved |
| 0x1 | LaneFocus, LaneDirection1, LaneDirection2, GuidanceInfo1, GuidanceInfo2, LaneSidestreets, LaneType, LaneMarking_left, LaneMarking_right, LaneAttributes, LaneDescription |
| 0x2..0xE | reserved |
| 0xF | Pos |

## 51.6.3.) Start [Number]

**Type:** Number

**Description:** For detailed specification refer to [2].
Parameter Start is the first required array element.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 16

**Step:** 1

**Unit:** unit_none

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## 51.6.4.) Elements [Number]

**Type:** Number

**Description:** For detailed specification refer to [2].
Parameter Elements contains the number of required array elements.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 51.7.) LaneGuidance [Record]

**Type:** Record

**Description:** LaneGuidance may contain up to 8 lanes together with the respective lane and guidance information for each lane focus. That means up to 16 lanes can be transmitted.
The array shall start with the lane focus "current lane" beginning from the most left side.

MaxElements = 16. Please refer to [2], Â§3.10 for the meaning of MaxElements.

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 255

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**Subparameter von:** 1_51_7

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| LaneFocus | Enum | Byte |
| LaneDirection1 | Number | Byte |
| LaneDirection2 | Number | Byte |
| GuidanceInfo1 | Enum | Nibble |
| GuidanceInfo2 | Enum | Nibble |
| LaneSidestreets | String | ByteStream |
| LaneType | Enum | Byte |
| LaneMarking_left | Enum | Nibble |
| LaneMarking_right | Enum | Nibble |
| LaneAttributes | Bitfield | ByteStream |
| LaneDescription | Enum | Byte |

### *Detailed Sub Parameter Overview*

## 51.7.1.) Pos [Number]

**Type:** Number

**Description:** Pos = 0 denotes the left most lane.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 16

**Step:** 1

**Unit:** unit_none

## 51.7.2.) LaneFocus [Enum]

**Type:** Enum

**Description:** This parameter contains the focus of the lane.
If the FSG cannot determine the focus of lane, it shall be set to unknown (0x00).

**OpCodes:** Empty,

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**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | unknown lane focus |
| 0x01 | current lane focus |
| 0x02 | successor lane focus |
| 0x03..0xFF | reserved |

## 51.7.3.) LaneDirection1 [Number]

**Type:** Number

**Description:** The encoded lane direction represents the traffic direction the lane will most likely be used for, in the context of the upcoming maneuver.
A lane can include up to two directions (parameter "LaneDirection1" and "LaneDirection2").
The ASG HMI can combine these directions for example:

The encoding of the LaneDirection is the same as for parameter "Maneuver_< x > .Direction" of function "ManeuverDescriptor" (0x17). The encoding and meaning of "LaneDirection1" is similar to the encoding of "Direction" in maneuver "turn on mainroad". The details are given the functional requirement specification.
The combination of LaneDirection1/2 with LaneSidestreets indicates which of the sidestreets of the lane the driver shall take (in the sense of a reduced RGI). Usually, the direction (sidestreet) contained in LaneDirection is not contained additionally in parameter LaneSidestreets. Further details are given in the functional requirement specification.

Parameter LaneDirection1 is not intended to provide a faithful representation of the symbol painted on the road surface.
NOTE: It makes no sense to combine an inner lane of a street with sidestreets; only outer lanes may be combined with LaneSidestreets.

The default value of LaneDirection1 is "straight" (value: 0x00).

LaneDirection = 0xFF denotes an invalid LaneDirection.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 254

**Step:** 1

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**Unit:** unit_none

**EnumValues:**

| Code | Description |
|---|---|
| 0xFF | invalid LaneDirection |

## 51.7.4.) LaneDirection2 [Number]

**Type:** Number

**Description:** The encoded lane direction represents the traffic direction the lane will most likely be used for, in the context of the upcoming maneuver.
A lane can include up to two directions (parameter 'LaneDirection1' and 'LaneDirection2').
The ASG HMI can combine these directions for example:

The encoding of the LaneDirection is the same as for parameter "Maneuver_< x > .Direction" of function "ManeuverDescriptor" (0x17). The encoding and meaning of "LaneDirection2" is similar to the encoding of "Direction" in maneuver "turn on mainroad". The details are given the functional requirement specification.
The combination of LaneDirection2 with LaneSidestreets indicates which of the sidestreets of the lane the driver shall take (in the sense of a reduced RGI). Usually, the direction (sidestreet) contained in LaneDirection is not contained additionally in parameter LaneSidestreets. Further details are given in the functional requirement specification.

Parameter LaneDirection2 is not intended to provide a faithful representation of the symbol painted on the road surface.
NOTE: It makes no sense to combine an inner lane of a street with sidestreets; only outer lanes may be combined with LaneSidestreets.

The default value of LaneDirection is "straight" (value: 0x00).

If the lane includes only one direction, LaneDirection2 is invalid (value: 0xFF).

LaneDirection2 = 0xFF denotes an invalid LaneDirection.

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 254

**Step:** 1

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**Unit:** unit_none

**EnumValues:**

| Code | Description |
|---|---|
| 0xFF | invalid LaneDirection |

## 51.7.5.) GuidanceInfo1 [Enum]

**Type:** Enum

**Description:** (lower nibble)

This parameter describes the lane direction recommendation related to the parameter "LaneDirection1".

If the ASG receives a reserved value, it shall act if the value 0x0 ("no direction recommended") is received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no direction recommended |
| 0x1 | recommended direction |
| 0x2 | best recommended direction |
| 0x3..0xF | reserved |

## 51.7.6.) GuidanceInfo2 [Enum]

**Type:** Enum

**Description:** (upper nibble)

This parameter describes the lane direction recommendation related to the parameter "LaneDirection2".

If no LaneDirection2 exists "GuidanceInfo2"is not recommended (value: 0x0).

If the ASG receives a reserved value, it shall act if the value 0x0 ("not recommended lane") is received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

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**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no direction recommended |
| 0x1 | recommended direction |
| 0x2 | best recommended direction |
| 0x3..0xF | reserved |

## 51.7.7.) LaneSidestreets [String]

**Type:** String

**Description:** The string contains the complete set of available sidestreets of the lane as data stream (string). The data stream doesn't contain any UTF8 encoded Unicode3.0 characters.
This string contains information about the (existing) sidestreets, which shall be combined with the current lane.

The first byte provides information about the string length.
The encoding and meaning is similar to the encoding of "sidestreets" in maneuver "turn on mainroad". Further details are given in functional requirement specification.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 17

## 51.7.8.) LaneType [Enum]

**Type:** Enum

**Description:** This parameter contains the lane type information, e.g. bus lane, taxi lane, bicycle lane.

If the ASG receives a reserved value, it shall act as if value 0x00 ("unspecific lane type") had been received.

Value 0x17 ("further lanes on the left hand side") is applicable for the left most lane only.
Value 0x18 ("further lanes on the right hand side") is applicable for the rightmost lane only.
The leftmost lane is the one with the smallest numeric Pos value.

**OpCodes:** Empty,

**Base Datatype:** Byte

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**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | unspecific lane type |
| 0x01 | normal lane |
| 0x02 | lane blocked |
| 0x03 | lane forbidden |
| 0x04 | lane restricted |
| 0x05 | breakdown lane |
| 0x06 | dynamically controlled |
| 0x07 | car pools |
| 0x08 | bus lane |
| 0x09 | bicycle lane |
| 0x0A | taxi lane |
| 0x0B | truck lane |
| 0x0C | bus & bicycle lane |
| 0x0D | bus & taxi lane |
| 0x0E | taxi & bicycle lane |
| 0x0F | bus, taxi & bicycle lane |
| 0x10 | green strip |
| 0x11 | motorway/highway lane |
| 0x12 | tramway on normal road |
| 0x13 | separated tramway |
| 0x14 | exit lane |
| 0x15 | enter lane |
| 0x16 | toll lane |
| 0x17 | hide lane |
| 0x18 | ETC / non ETC mixed lane |
| 0x19..0xFF | reserved |

## 51.7.9.) LaneMarking_left [Enum]

**Type:** Enum

**Description:** (upper nibble)

This parameter contains the lane marking (at the left border of the lane). Lane markings of adjacent lanes may be combined. E.g., if the lane marking is a double solid line, lane "n-1" reports LaneMarking_right = 0x1 and lane "n" reports LaneMarking_left = 0x1.
If the ASG receives a reserved value, it shall act as if value 0x0 ("no line") had been received.

**OpCodes:** Empty,

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**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no lane marking at the left border of the lane (no line) |
| 0x1 | solid divider line |
| 0x2 | dashed divider line |
| 0x3 | dotted divider line |
| 0x4..0xF | reserved |

## 51.7.10.) LaneMarking_right [Enum]

**Type:** Enum

**Description:** (lower nibble)

This parameter contains the lane marking (at the right border of the lane). Lane markings of adjacent lanes may be combined. E.g., if the lane marking is a double solid line, lane "n-1" reports LaneMarking_right = 0x1 and lane "n" reports LaneMarking_left = 0x1.
If the ASG receives a reserved value, it shall act as if value 0x0 ("no line") had been received.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | no lane marking at the right border of the lane (no line) |
| 0x1 | solid divider line |
| 0x2 | dashed divider line |
| 0x3 | dotted divider line |
| 0x4..0xF | reserved |

## 51.7.11.) LaneAttributes [Bitfield]

**Type:** Bitfield

**Description:** This parameter contains additional attributes of the specific lane.

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:**1

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**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | Further lanes on the left hand side (not shown) | No further lanes on the left hand side |
| 1 | Further lanes on the right hand side (not shown) | No further lanes on the right hand side |
| 2..7 | reserved | reserved |

## 51.7.12.) LaneDescription [Enum]

**Type:** Enum

**Description:** Provides information of lane availability further down the route. This information can be used to provide suitable graphical representations for additional lanes, which are not yet available at the present vehicle position (expanding lanes) or lanes which will end/merge ahead of the present vehicle position (merging lanes).

If the ASG receives a reserved value, it shall act as if 0x00 ("lane available") is received.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | lane available |
| 0x01 | lane merging to left |
| 0x02 | lane merging to right |
| 0x03 | lane ending later |
| 0x04 | forbidden lane |
| 0x05 | lane expanding to left |
| 0x06 | lane expanding to right |
| 0x07 | lane available later |
| 0x08 | multiple additional lane begin |
| 0x09 | multiple continuation with lane ends |
| 0x0A | lane begins in the middle |
| 0x0B..0xFE | reserved |
| 0xFF | not supported - no description available |

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## 51.8.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software. Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |
| 0x46..0xFF | reserved |

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# 52.) Picture ( 62, 0x3E )

**Type:** Array

**Description:** This function is used to transfer a link (URL) for navigation pictures, e.g. photorealistic "ExitView/JunctionView" or "kanban" located in FSG ( e.g. in a cdn/images service). The transmission of the picture data will be done by HTTP afterwards.

This list is referred to as "arbitrary list". For the meaning of "arbitrary IDs" the reader is referred to [2].

If the ASG requests an invalid array element in the Start parameter, the FSG returns the required START element, ELEMENTS = 0 and no DATA because an invalid element has no successor or predecessor. Please note the special rules for START=0, refer to [2].
If the list contains fewer elements in the given direction than required, the FSG returns the existing elements.
The value of the START byte in a StatusArray message has to be equal to the one from the GetArray message. The value of the Transmit_Pos bit has to be taken into account.

For the general specification of the array function class please refer to [2].

NOTE: In case of a ChangedArray message the parameter Data is either empty or contains only Pos information.

### *Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| ASG_ID | Enum | Nibble |
| TAID | Number | Nibble |
| TotalNumListElements | Number | Byte |
| ArrayHeader | Record | Empty |
| Data | Record | Empty |
| ErrorCode | Enum | Byte |

### *OpCode List*

**LCU:** Navigation_SD (50, 0x32 ) , **Function:** Picture (62, 0x3E )

| OpCode | Associated Parameters |
|---|---|
| GetArray | ASG_ID, TAID, ArrayHeader, |
| StatusArray | ASG_ID, TAID, TotalNumListElements, ArrayHeader, Data, |
| ChangedArray | ArrayHeader, Data, |
| Error | ErrorCode, |

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## *Detailed Parameter Overview*

### 52.1.) ASG_ID [Enum]

**Type:** Enum

**Description:** (upper nibble)

Identifies the requesting ASG.
If the FSG sends a StatusArray message without preceeding GetArray message, the FSG shall set this parameter to 0x0.

If the StatusArray message is a response to a GetArray message, the FSG shall set the parameters ASG_ID and TAID to the values received in the GetArray message. This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!
In that case (if the most significant bit within ASG_ID is set to 1), then all ASGs have to evaluate that StatusArray message.
This rule shall be applied even if the ASG_ID in the GetArray message is set to a reserved value!

Values > 0x7 shall not be sent in a GetArray message.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | default ASG (any ASG, spontaenous FSG message) |
| 0x1 | instrument cluster |
| 0x2 | head-up display |
| 0x3 | operating unit rear |
| 0x4..0x8 | reserved |
| 0x9 | instrument cluster, to be evaluated by all ASGs |
| 0xA | head-up display, to be evaluated by all ASGs |
| 0xB | operating unit rear, to be evaluated by all ASGs |
| 0xC..0xF | reserved |

### 52.2.) TAID [Number]

**Type:** Number

**Description:** (lower nibble)

TAID:.. transaction ID
Identifies the transaction (pair of GetArray and StatusArray). The transaction ID (TAID) is unique in combination with parameter ASG_ID.

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Special values:
0: Is to be generated/used by the FSG only if the FSG sends a StatusArray without preceeding GetArray. This means, the ASG is not allowed to generate TAID = 0.

**OpCodes:** GetArray, StatusArray,

**Base Datatype:** Nibble

**Range Min:** 0

**Range Max:** 15

**Step:** 1

**Unit:** unit_none

## 52.3.) TotalNumListElements [Number]

**Type:** Number

**Description:** Contains the total number of current list elements. In most cases this number is greater than the number of elements stored in the ASG.
This information may be used for dimensioning of scroll bars etc.

Value 0xFF means, the total number of list elements is unknown.

**OpCodes:** StatusArray,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 52.4.) ArrayHeader [Record]

**Type:** Record

**Description:** For meaning, encoding and decoding of the ArrayHeader the reader is referred to [2].

**OpCodes:** GetArray, StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 1

**Subparameter von:** 1_52_4

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### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Mode | Bitfield | Nibble |
| RecordAddress | Enum | Nibble |
| Start | Number | Byte |
| Elements | Number | Byte |

### *Detailed Sub Parameter Overview*

## 52.4.1.) Mode [Bitfield]

**Type:** Bitfield

**Description:** (upper nibble)

For detailed specification of "Mode" refer to [2].
The IndexSize shall be set to 8bit always (Bit3 := FALSE). I.e., only 8-bit values are allowed for Start, Elements and Pos.

**OpCodes:** Empty,

**Base Datatype:** Nibble

**Length:**1

**Boolfields:**

| Bit Position | True Description | False Description |
|---|---|---|
| 0 | shift=1, refer to [2] | shift=0, refer to [2] |
| 1 | Array direction is backward | Array direction is forward |
| 2 | Array position is transmitted | Array position is not transmitted |
| 3 | reserved | IndexSize = 8 Bit for Start, Elements, Pos |

## 52.4.2.) RecordAddress [Enum]

**Type:** Enum

**Description:** (lower nibble)

For the definition and usage of RecordAddress please refer to [2]. Special values:
0x00 may be used by ChangedArray for full range update indication

**OpCodes:** Empty,

**Base Datatype:** Nibble

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**EnumValues:**

| Code | Description |
|---|---|
| 0x0 | reserved |
| 0x1 | PictureType, IPv6_Address, TCP_Port, URI |
| 0x2..0xE | reserved |
| 0xF | Pos |

## 52.4.3.) Start [Number]

**Type:** Number

**Description:** Parameter Start is the first array element the message relates to. For usage of this parameter, the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 52.4.4.) Elements [Number]

**Type:** Number

**Description:** Parameter Elements contains the number of array elements the message relates to. For the definition and usage of "Elements", the reader is referred to [2].

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

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## 52.5.) Data [Record]

**Type:** Record

**Description:** This list contains information about all navigation pictures.

If ASG shall display a navigation picture like 'ExitView', the FSG will insert it as a new list element in array. If the ASG shall no longer display this picture, the FSG delete the related list element in array and ASG shall no longer display it.

MaxElements = 1 if RecordAddres 0x1 is used. This means, a StatusArray message is limited to 1 complete elements.

NOTE! The ASG is allowed to request more elements than given in MaxElements as long as the amount of data to be transferred in bytes does not exceed the data amount needed for transfer of MaxElements using RecordAddress 0x1.
E.g. the ASG is allowed to request more than MaxElements == 1 when using RecordAddress 0xF (POS-only). Please refer to [2], Â§3.10 for the meaning of MaxElements.

The logical length of this list (array) is limited to 254 elements. The FSG is not allowed to generate 0x00 as a valid Pos value.

The FSG is responsible for the logical order of list elements (array elements).

**OpCodes:** StatusArray, ChangedArray,

**Base Datatype:** Empty

**Count:** 254

**Subparameter von:** 1_52_5

### *Sub Parameter List*

| Name | Type | BaseDataType |
|---|---|---|
| Pos | Number | Byte |
| PictureType | Enum | Byte |
| IPv6_Address | String | ByteStream |
| TCP_Port | Number | Word |
| URI | String | ByteStream |

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### *Detailed Sub Parameter Overview*

## 52.5.1.) Pos [Number]

**Type:** Number

**Description:** Pos

**OpCodes:** Empty,

**Base Datatype:** Byte

**Range Min:** 0

**Range Max:** 255

**Step:** 1

**Unit:** unit_none

## 52.5.2.) PictureType [Enum]

**Type:** Enum

**Description:** This parameter contains the type of picture.

**OpCodes:** Empty,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | no picture |
| 0x01 | Sign |
| 0x02 | GreenKanban |
| 0x03 | BlueKanban |
| 0x04 | JunctionView |
| 0x05 | IntersectionView |
| 0x06 | ExitView |
| 0x07 | TollGateView |
| 0x08 | MotorwayInfo |
| 0x09 | ServiceParkingArea |
| 0x0A | TrafficDetails |
| 0x0B | LocalHazardInformation |
| 0x0C | GhostDriver |

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| | |
|---|---|
| 0x0D | VICSLevel1 |
| 0x0E | HotspotMap |
| 0x0F..0xFE | reserved |
| 0xFF | unknown picture |

## 52.5.3.) IPv6_Address [String]

**Type:** String

**Description:** This string parameter contains the IPv6 address (destination address) of the TCP connection for the specific picture service (e.g. CDN) as data stream (string). The data stream doesn't contain any UTF8 encoded Unicode3.0 characters.
The IPv6 address is transmitted without colon (":") for seperating the 2Byte blocks.

The first byte provides information about the string length. An IPv6 address always consists of 16 byte. That means if an IPv6 address is transmitted, the string lenght is 16.

For example if the IPv6 address 2001:0DB8:0000:0000:F054:00FF:0000:02EB shall be transmitted, the FSG set this parameter (including length information):
10 20 01 0D B8 00 00 00 00 F0 54 00 FF 00 00 02 EB

The FSG shall transmit an empty string, if the PictureType is "no picture" (0x00).

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 17

## 52.5.4.) TCP_Port [Number]

**Type:** Number

**Description:** This parameter contains the destination port of the TCP connection for the specific picture service (e.g. CDN).
For example the destination port for the call pictures could be 29735.

The FSG shall set the TCP port = 0, if the PictureType is "no picture" (0x00).

**OpCodes:** Empty,

**Base Datatype:** Word

**Range Min:** 0

**Range Max:** 65535

**Step:** 1

**Unit:** unit_none

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## 52.5.5.) URI [String]

**Type:** String

**Description:** This string parameter contains the Uniform Resource Identifier (URI) of the navigation pictures (e.g. in a CDN image service). The string contains UTF8 encoded unicode3.0 characters.
If the FSG cannot provide an image, the URI shall be transmitted as an empty string.

The FSG shall transmit an empty string, if the PictureType is "no picture" (0x00).

The maximum number of characters in this string is limited by the string length only.
The encoded length must not exceed 600 bytes plus an additional byte for the string length information.
The size of the string length identifier shall always be 2 byte.

example for an URI:
*/cdn/images/navigation_picture21458746217514*

**OpCodes:** Empty,

**Base Datatype:** ByteStream

**Length:** 602

## 52.6.) ErrorCode [Enum]

**Type:** Enum

**Description:** For standardised error codes refer to the BAP specification [1].

Standard protocol error values will be automatically sent by the BAP standard software.
Application errors (standard application error, function specific error) have to be set/evaluated by the application if defined for this function.

Standard application error values are mandatory.

**OpCodes:** Error,

**Base Datatype:** Byte

**EnumValues:**

| Code | Description |
|---|---|
| 0x00 | AppErr_NoError |
| 0x01..0x3F | standard BAP error codes, refer to [1] |
| 0x40..0x42 | reserved, refer to annex |
| 0x43 | AppErr_MaxDataLengthExceeded |
| 0x44 | reserved, refer to annex |
| 0x45 | AppErr_ParameterMismatch |

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0x46..0xFF reserved

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